ARITHMETIC VOLUMES OF QUASISPLIT SHTUKAS IN ARBITRARY COWEIGHT TYPE

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ABSTRACT. Let $X/\mathbf{F}_q$ be a smooth projective geometrically connected curve and let G/X be an everywhere-reductive quasisplit connected semisimple group scheme, split by a finite étale Galois cover $X' \to X$. We combine Feng–Yun–Zhang's quasisplit multi-graph descent with Wang–Wei's arbitrary-coweight local-volume-data formalism. For representative-leg data we prove an exact relative Borel–Moore product formula and obtain, on the Ran and augmentation associated graded,

$$\kappa(X')b_v \,\mathrm{id} + \nabla_{A_v}, \qquad \kappa(X') = -\int_{X'_{\overline{\mathbf{F}}_q}} c_1(TX').$$

We also introduce point-decorated degree-zero data, which supply the independent scalar needed to recover the Feng–Yun–Zhang minuscule formula, while the split undecorated specialization recovers Wang–Wei. Applying the result to determinant intersection-complex data gives an arbitrary-coweight quasisplit determinant-volume formula. Purity and an analytic superdeterminant argument give absolute convergence of the regularized trace after fixing an embedding $\overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C}$.

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1. INTRODUCTION

1.1. **Motivation.** Feng–Yun–Zhang define arithmetic volumes of moduli stacks of shtukas and relate them to higher derivatives of Artin L-functions. Their nonsplit theorem is formulated for quasisplit group schemes presented by a faithful pinned finite étale Galois splitting cover, but the modification coweights are assumed minuscule. Wang–Wei remove the minuscule restriction for split connected semisimple groups. Their method replaces the smooth Schubert-cell calculation by a formalism of local volume data supported on arbitrary affine-Grassmannian Schubert closures, with intersection complexes as coefficients.

The purpose of this article is to prove the arbitrary-coweight quasisplit determinant-volume formula and to place both source formulas inside one scalar-plus-derivation trace principle. We retain the full-X'-leg convention and the graph-transfer formalism of Feng–Yun–Zhang, and we use the arbitrary-coweight local volume data and relative Borel–Moore operator calculus of Wang–Wei. The key point is that the quasisplit modification becomes, after pullback to the splitting cover,

a split modification along a disjoint union of conjugate graph divisors. This permits the one-leg Wang–Wei identity to be telescoped over the graphs.

There are two logically different extensions being combined. Feng–Yun–Zhang supply the descent geometry: a single quasisplit modification becomes a family of split modifications along the conjugate graphs on X'. Wang–Wei supply the singular local calculation: arbitrary Schubert closures are handled by intersection complexes and relative Borel–Moore correspondences. The theorem below amounts to showing that these two operations commute. In particular, one must not merely sum coefficient-level identities; one must prove that the complete local volume datum, including both correspondence maps, globalizes through the multi-graph geometry.

1.2. **Main result.** For a dominant coweight $\lambda \in X_*(T_0)^+$, the Wang–Wei local-volume-data formalism attaches to the determinant datum fixed below a scalar b_λ and an endomorphism

$$E_\lambda \in \mathrm{End}(V_0).$$

The pinned descent produces a degree-preserving endomorphism

$$A_\lambda \in \mathrm{End}\, H_{>0}(X, \mathbb{V}),$$

called the graph-transfer of E_λ. It is defined intrinsically in Definition 4.7; in the representative notation of [FYZ, §7.4.8], its action on a Künneth generator is represented by

$$v^z \longmapsto \sum_{\gamma\in\Gamma} \big(E_\lambda(\gamma v)\big)^{\gamma^{-1}z}. \tag{1.1}$$

Equation (1.1) is the representative expression of the descended graph-indexed construction. Concretely, the construction has three stages: the graph D_γ contributes the local operator of type $\gamma\lambda$; pinned equivariance transports that operator to the fixed representative chart; and the resulting invariant graph sum is passed from invariants to descent coinvariants. The displayed formula records the outcome of these stages, not a separate definition on naive coinvariants.

Let $\lambda_I = (\lambda_1, \dots, \lambda_r)$ be arbitrary dominant coweights such that the total component shift is zero. Assume that the adjoints $A^\vee_{\lambda_i}$ commute on

$$M := H^{<0}(X, \mathbb{V}^\vee).$$

After a finite extension of K_ℓ, decompose M into simultaneous generalized eigenspaces

$$M = \bigoplus_{\varepsilon} M[\varepsilon], \qquad \varepsilon = (\varepsilon_1, \dots, \varepsilon_r), \tag{1.2}$$

where ε_i is the actual generalized eigenvalue of $A^\vee_{\lambda_i}$ on $M[\varepsilon]$. In the complex formulas below, b_{λ_i}, ε_i, and every Frobenius determinant are viewed through the fixed embedding ι of (2.1); this scalar extension is suppressed from the notation. The two hypotheses used here have different purposes. The zero component-shift condition makes the iterated correspondence return to the component on which the trace is taken. Commutativity of the $A^\vee_{\lambda_i}$ is used only at the final linear-algebra stage, where it permits the simultaneous generalized-eigenspace decomposition and hence a multivariable L-function.

Define

$$L^*_X(\mathbf{s}, \mathbb{V}^\vee) = \prod_{\varepsilon} \prod_{m<0} \det\big(1 - q^{-s_\varepsilon}\,\mathrm{Frob} \mid H^m(X, \mathbb{V}^\vee)[\varepsilon]\big)^{(-1)^{m-1}} \tag{1.3}$$

and

$$D_i = \kappa(X')b_{\lambda_i} - (\log q)^{-1} \sum_{\varepsilon} \varepsilon_i\, \partial_{s_\varepsilon}. \tag{1.4}$$

Theorem 1.1 (Arbitrary-coweight quasisplit determinant volume)**.** *Fix the faithful pinned splitting datum $(X'/X, \Gamma, G_0, \tau)$ used in* (2.2)*, and let G/X be its quasisplit descent. Let λ_I be a sequence of arbitrary dominant coweights satisfying the zero component-shift condition. Equip each local Schubert Hecke stack with the factorwise-normalized determinant IC volume datum defined in Equation* (7.3)*, and globalize it in the full-X'-leg convention of Feng–Yun–Zhang. Assume the commutativity hypothesis above. The determinant family carries canonical coherent pinned equivariance and compatible Weil structures by Proposition 7.2; no common number-field model is assumed. When G_0 is almost simple, this is Wang–Wei's determinant datum; for a general semisimple product it is the factorwise-normalized datum specified here. Then, for every Frobenius-stable geometrically connected component Bun_G^{ω}, the arithmetic volume satisfies*

$$\mathrm{vol}\Big({}^{\omega}\,\mathrm{Sht}_G^{\leq \lambda_I}, v_{\det,I}^{\mathrm{rep}}\Big) = q^{\dim \mathrm{Bun}_G} \left(\prod_{i=1}^{r} D_i\right) L_X^*(\mathbf{s}, \mathbb{V}^\vee)\Big|_{\mathbf{s}=0}. \tag{1.5}$$

The product of the D_i may be taken in the order of the iterated Hecke correspondence; under the commutativity hypothesis the order is immaterial.

Relation with the source theorems. When $X' = X$ and $\Gamma = 1$, the general local-volume-data formula specializes to the operator-invariant content of Wang–Wei's Theorem 2.13; Remark 6.14 records the sign translation required by their later ordered eigenweight symbols. For their determinant data it gives their headline theorem after summing the split semisimple components; in the almost-simple case the determinant line used here is literally theirs. For a general semisimple product, the determinant theorem concerns the factorwise-normalized datum defined in this article.

Feng–Yun–Zhang allow a minuscule class $\eta + \xi'\eta'$. Its two summands have different geometric roles: η produces the scalar-plus-derivation operator, while the η'-term becomes degree preserving only after multiplication by the degree-two point class ξ'. Section 5 isolates this point-decoration mechanism and proves that its non-unit commutator terms lie two Ran steps lower. Arbitrary FYZ classes need not be invariant under the stabilizer of the coweight. The graph-labelled orbit construction of Definition 3.3 and Proposition 3.4 retains the graph label and therefore transports these classes without imposing stabilizer invariance. The resulting master formula recovers all terms of their theorem for a geometrically connected splitting cover and gives the ordinary-total-pushforward correction for a disconnected cover. The embedding ι is fixed exactly as in Feng–Yun–Zhang and is suppressed from the name and notation of the theorem.

The proof occupies Sections 3, 4.1, 4.2, 5 and 6. Our coefficients ε_i are the actual eigenvalues of the operators occurring in the derivations, so the differential sign is fixed by the identity $t\partial_t = -(\log q)^{-1}\partial_s$.

How to read the formula. The four ingredients of (1.5) come from different stages of the proof. The scalar $\kappa(X') = -\int_{X'} c_1(TX')$ records integration along the full representative-leg curve. The number b_{λ_i} is the value of the corresponding local degree-two volume datum on the unit. The operator A_{λ_i} is obtained by summing the conjugate one-leg operators after transport to the representative chart; its derivation is the non-scalar part of the associated-graded global Hecke operator. Finally, L_X^* is the Frobenius superdeterminant on the dual negative-degree Gross-motive cohomology. Thus the theorem is not obtained by guessing a differential operator: the scalar and derivation are first calculated geometrically, and only then converted into derivatives of the L-function.

1.3. **Strategy of proof.** The proof follows the organization of the two source papers. We first set up the quasisplit Hecke correspondence, tautological Künneth classes, and Wang–Wei's local volume data with their target/source arrow convention. The representative-leg global datum is constructed from a Cartesian local-model diagram. The stack-level Cartesian assertion is

proved by Beauville–Laszlo gluing, including the semilinear Γ-descent cocycle, while the required Beck–Chevalley and relative-duality exchanges are verified in the bounded six-functor formalism.

Next we pull the universal modification to X'. The support is the disjoint union of the graphs D_γ, and the local type at D_γ is $\gamma\lambda$. Telescoping the one-leg Wang–Wei identity gives a multi-graph commutator. This ordinary cohomology identity is then applied to the relative Borel–Moore class $H^*(c)(1)$, after which $H^*(d)$ is applied. The resulting exact product formula is the quasisplit analogue of Wang–Wei's formula (3.8).

The zero-bracket term gives $\delta^{\omega}_{\lambda,v} + \kappa(X')b_v$; in the standing semisimple determinant setting the horizontal term $\delta^{\omega}_{\lambda,v}$ vanishes. The one-bracket terms give the graph-transfer derivation ∇_{A_v}, and terms with at least two brackets lower the Ran degree. Thus the associated-graded one-leg operator is scalar plus derivation. Finally, degreewise trace additivity reduces the regularized trace to a finite-dimensional symmetric algebra. Purity gives exponential Frobenius decay, while the derivations have polynomial growth. An analytic superdeterminant identity then converts the trace into the differential operator appearing in the theorem.

1.4. **Organization.** Section 2 fixes notation, the quasisplit splitting datum, the full-leg Hecke stacks, and tautological classes. Section 3 develops the general formalism of equivariant local volume data and their representative-leg globalisation. Section 4 calculates the global relative Hecke operators and proves the scalar-plus-derivation formula. Section 5 introduces globally point-decorated degree-zero data and proves their Ran-filtration effect. Section 6 proves the regularized trace theorem, the scalar-plus-derivation master formula, the connected-reductive trace package, the Wang–Wei specialization, and the precise Feng–Yun–Zhang recovery together with the disconnected-cover normalization correction. Section 7 treats determinant intersection-complex data. Appendix A states the precise foundational theorems used, verifies their hypotheses for the local and global Hecke stacks, and proves the application-specific Beauville–Laszlo and relative Borel–Moore mate calculations.

2. Notation and quasisplit Hecke stacks

2.1. **Coefficients and trace convention.** Let $k = \mathbf{F}_q$ and fix an algebraic closure $\bar{k}$. Fix a prime $\ell \neq \operatorname{char} k$ and a finite extension $K_\ell/\mathbf{Q}_\ell$ inside $\overline{\mathbf{Q}}_\ell$ containing a chosen square root $q^{1/2}$. For every k-stack Y, the notation $\mathrm{Shv}(Y)$ means the geometric sheaf category on $Y_{\bar{k}}$, and $H^*(Y)$ means $H^*(Y_{\bar{k}}, K_\ell)$. Weil structures on objects and morphisms defined over k are recorded separately. Thus all cohomology is geometric ℓ-adic cohomology with coefficients in K_ℓ, and Frob denotes geometric Frobenius. We fix the rank-one half-Tate Weil object $K_\ell(1/2)$ on which geometric Frobenius acts by $q^{-1/2}$; it defines the normalization $\langle n\rangle = [n](n/2)$ for every integer n. We use the homological grading on curve classes adopted in [FYZ, Remark 7.2.2]. Cohomological shifts and Tate twists are retained in the definitions of volume data, but are suppressed in identities in ordinary and relative Borel–Moore cohomology when they are forced by degree. All cited results stated with $\mathbf{Q}_\ell$-coefficients are used after scalar extension to K_ℓ.

Fix once and for all an abstract field embedding

$$\iota : \overline{\mathbf{Q}}_\ell \hookrightarrow \mathbf{C} \tag{2.1}$$

whose restriction to K_ℓ sends the chosen $q^{1/2}$ to the positive real square root. Every finite scalar extension, generalized eigenvalue, degreewise trace, determinant, norm, and absolute value below is viewed through ι; we usually suppress ι from the notation. Thus the regularized trace is the complex series

$$\mathrm{Tr}_\iota(\mathrm{Frob}^{-1}\, T \mid H^*) := \sum_n (-1)^n\, \iota\big(\mathrm{Tr}(\mathrm{Frob}^{-1}\, T \mid H^n)\big)$$

whenever it is absolutely convergent. As in Feng–Yun–Zhang, we fix this embedding once and suppress it from the notation; we write vol for the resulting arithmetic volume. This is the

convention of [FYZ, Proposition 5.5.3]. We require compatible Weil structures, but no number-field form. The Frobenius eigenvalues to which weight estimates are applied are algebraic Weil numbers, so their complex absolute values are controlled under the restriction of ι to the algebraic closure of $\mathbf{Q}$. The scalars and finite-dimensional local operators supplied by a local volume datum may be arbitrary K_ℓ-linear data: after applying ι they are fixed complex scalars and matrices, and the convergence proof below uses only polynomial bounds in the symmetric degree.

We work in the constructible sheaf and cohomological-correspondence formalism used in [WW, §1.4 and §2.1], whose underlying definitions are given in [LW, §4.1–§4.2]. Every Hecke stack below is already bounded by a fixed affine-Grassmannian Schubert closure; its two horizontal maps are representable and proper of finite presentation by Proposition 3.8. All pushforwards, Beck–Chevalley maps, and relative-duality exchanges are taken in this bounded setting.

2.2. **Quasisplit group schemes and Atiyah–Bott generators.** Let X/k be smooth, projective, and geometrically connected, and let

$$\nu : X' \longrightarrow X$$

be a finite étale Galois cover with deck group Γ. The cover X' may be geometrically disconnected. Let G_0/k be a split connected semisimple group with a pinning

$$T_0 \subset B_0 \subset G_0$$

and a faithful pinned action

$$\tau : \Gamma \hookrightarrow \mathrm{Aut}(G_0, B_0, T_0).$$

Put

$$G = \big(\mathrm{Res}_{X'/X}(G_0 \times X')\big)^{\Gamma}. \tag{2.2}$$

This is the setup of [FYZ, §7.1]. After choosing a faithful pinned Galois splitting datum, every everywhere-reductive quasisplit connected semisimple group scheme over X has such a presentation [FYZ, Remark 7.1.1]. The chosen cover is part of the full-leg presentation used throughout the paper. A point $x' \in X'$ is the representative leg, while the pulled-back modification is supported simultaneously at all of its Γ-translates. Keeping these two viewpoints separate is essential: the coefficient object comes from one representative local model, whereas the commutator with tautological classes sees the entire orbit of graph divisors.

Pullback by the pinned automorphisms is contravariant, so we regard the coefficient factor as a right Γ-module. Thus, for $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$ and $v \in V_0$, we put

$$\alpha \cdot \gamma := a_\gamma^* \alpha, \qquad v \cdot \gamma := a_\gamma^* v,$$

and, to agree with the notation of [FYZ, §7.4.8], we abbreviate $v \cdot \gamma$ to γv. Deck transformations act on $H_*(X')$ on the left; the right diagonal descent action on $H_*(X') \otimes V_0$ is therefore

$$(z \otimes v) \cdot \gamma := (\gamma^{-1} z) \otimes (\gamma v). \tag{2.3}$$

All invariants and coinvariants below refer to this right action. With these conventions Poincaré duality gives $\gamma^* \,\mathrm{PD}(z) = \mathrm{PD}(\gamma^{-1} z)$.

Let

$$V_0 = \mathrm{Gr}^1_{\mathrm{aug}} H^*(BG_0, K_\ell)$$

be the Gross-motive space and let

$$\mathbb{V} = (V_0 \times X')^{\Gamma}$$

be the descended graded local system on X. We use the homological grading convention of [FYZ, Remark 7.2.2], so that

$$H_*(X, \mathbb{V}) \cong \big(H_*(X') \otimes V_0\big)_{\Gamma}. \tag{2.4}$$

For each geometrically connected component Bun_G^{ω}, Feng–Yun–Zhang give a doubly filtered Atiyah–Bott isomorphism

$$\mathrm{Gr}^{\bullet}_{\mathrm{aug}}\,\mathrm{Gr}^{\bullet}_{\mathrm{Ran}}\,H^*(\mathrm{Bun}_G^{\omega}) \cong \mathrm{Sym}^{\bullet}\big(H_{>0}(X,\mathbb{V})\big). \tag{2.5}$$

We use the componentwise identifications induced by the same tautological map. They are compatible with the source and target tautological classes in the Hecke correspondences, which is the component-identification convention used in [FYZ, §7.5.2].

Lemma 2.1 (Componentwise Atiyah–Bott compatibility)**.** *The isomorphisms* (2.5) *for the geometric components of* Bun_G *are induced by the same tautological map. Under the resulting canonical identifications of the associated gradeds of two components, tautological classes with the same name correspond. Frobenius carries the map for* ω *to the map for* $\mathrm{Frob}(\omega)$*. For a one-leg correspondence from* ω *to* $\omega+\bar{\lambda}$*, the diagrams*

(2.6)

$$\begin{array}{ccc} H_{>0}(X,\mathbb{V}) & \xrightarrow{\tau_{\omega+\bar{\lambda}}} & \mathrm{Gr}^1_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(\mathrm{Bun}_G^{\omega+\bar{\lambda}}) \\ {\scriptstyle\mathrm{id}}\downarrow & & \downarrow{\scriptstyle h_1^*} \\ H_{>0}(X,\mathbb{V}) & \xrightarrow{\tau_{S,1}} & \mathrm{Gr}^1_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(S_\lambda) \end{array} \qquad \begin{array}{ccc} H_{>0}(X,\mathbb{V}) & \xrightarrow{\tau_{\omega}} & \mathrm{Gr}^1_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(\mathrm{Bun}_G^{\omega}) \\ {\scriptstyle\mathrm{id}}\downarrow & & \downarrow{\scriptstyle h_0^*} \\ H_{>0}(X,\mathbb{V}) & \xrightarrow{\tau_{S,0}} & \mathrm{Gr}^1_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(S_\lambda) \end{array}$$

commute. The difference $\tau_{S,1}(\eta)-\tau_{S,0}(\eta)$ *is the typed graph commutator of Corollary 4.3.*

Proof. For every component ω, let

$$\mathrm{AB}_{\omega} : \mathrm{Sym}\big(H_{>0}(X,\mathbb{V})\big) \xrightarrow{\sim} \mathrm{Gr}_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(\mathrm{Bun}_G^{\omega})$$

be the map constructed from the Künneth components of the universal characteristic classes in [FYZ, Theorem 4.2.8]. We define the comparison between the associated gradeds of two components ω,ω' to be

$$\mathrm{AB}_{\omega'}\circ\mathrm{AB}_{\omega}^{-1}\,.$$

This definition is canonical because the source symmetric algebra and the tautological construction are the same for every component. It sends every named tautological class to the class with the same name. Frobenius compatibility follows from [FYZ, Remark 4.2.10], since Frobenius carries the universal bundle on the ω-component to the universal bundle on the $\mathrm{Frob}(\omega)$-component.

The upper horizontal map in the first square of (2.6) is formed from the universal target bundle, while the lower horizontal map is formed from its pullback to the Hecke stack. The latter is exactly the pullback along h_1, so functoriality of evaluation and Künneth decomposition proves commutativity before passing to either filtration. The source square is identical with h_0 in place of h_1. Their difference is therefore the target-minus-source universal-bundle class computed in Corollary 4.3. Hence every associated-graded one-leg operator is canonically a map between two copies of $\mathrm{Sym}(H_{>0}(X,\mathbb{V}))$. □

2.3. **The Euler scalar of the splitting cover.** The global Wang–Wei operator includes ordinary proper pushforward along the complete leg curve. Accordingly the scalar is the total Euler characteristic of X', not the Euler characteristic of one geometric component.

Definition 2.2. Define

$$\kappa(X') = -\int_{X'_{\bar{k}}} c_1(TX'). \tag{2.7}$$

Lemma 2.3. *One has*

$$\kappa(X') = \deg(\nu)(2g(X)-2). \tag{2.8}$$

If X' is geometrically connected, then $\kappa(X') = 2g(X') - 2$. If $X'_{\overline{k}} = \coprod_{a=1}^{N} X'_a$ and all X'_a have genus g_a, then

$$\kappa(X') = \sum_{a=1}^{N} (2g_a - 2).$$

Equivalently, if $g_{\mathrm{a}}(X') := 1 - \chi(X', \mathcal{O}_{X'})$ denotes the arithmetic genus of the possibly disconnected curve, then $\kappa(X') = 2g_{\mathrm{a}}(X') - 2$.

Proof. Because ν is étale, $TX' \cong \nu^* TX$. Proper push-pull gives

$$\int_{X'} c_1(TX') = \int_X \nu_* \nu^* c_1(TX) = \deg(\nu) \int_X c_1(TX) = \deg(\nu)(2 - 2g(X)).$$

The remaining assertions are the componentwise Gauss–Bonnet formula and, in the geometrically connected case, Riemann–Hurwitz for an étale cover. □

2.4. **Full-leg Hecke and shtuka stacks.** For $\lambda \in X_*(T_0)^+$, let

$$\mathrm{Hk}_{\widetilde{G}}^{\leq \lambda}$$

be the full-X'-leg quasisplit Hecke stack of [FYZ, §7.3]. An S-point is a tuple

$$(x', \mathcal{F}_0, \mathcal{F}_1, \alpha)$$

with $x' \in X'(S)$, G-bundles $\mathcal{F}_0, \mathcal{F}_1$ on $X \times S$, and an isomorphism away from the graph of $x = \nu(x')$, such that after pullback to $X' \times S$ the modification has relative position $\leq \gamma\lambda$ along the graph of $\gamma x'$ for every $\gamma \in \Gamma$. Write

$$p : \mathrm{Hk}_{\widetilde{G}}^{\leq \lambda} \to X', \qquad h_0, h_1 : \mathrm{Hk}_{\widetilde{G}}^{\leq \lambda} \to \mathrm{Bun}_G\,.$$

For a source component ω, write ${}^{\omega}\mathrm{Hk}_{\widetilde{G}}^{\leq \lambda}$ for the corresponding open-and-closed substack. The target component is $\omega + \bar{\lambda}$, using the component map of [FYZ, §7.2].

Although the stack has a single leg valued in X', its pullback to the splitting cover has one support graph for each $\gamma \in \Gamma$. This is why the later non-scalar operator is a graph sum. It does not mean that the coefficient sheaf is a tensor product over those graphs: the other local modifications are determined by descent from the representative one.

For an r-tuple λ_I, the iterated Hecke stack is the usual fiber product over the intermediate copies of Bun_G. We impose

$$\sum_{i=1}^{r} \bar{\lambda}_i = 0 \quad \text{in } \pi_0(\mathrm{Bun}_G). \tag{2.9}$$

The condition (2.9) is global bookkeeping rather than a restriction on any one local Schubert variety: it says that the ordered composition of one-leg correspondences starts and ends on the same connected component. The associated shtuka stack is the Frobenius-twisted fixed-point stack. We use the componentwise volume convention of [FYZ, Definition 7.4.2]; a literal trace on one component is taken only when that component is Frobenius-stable. For a sequence of local volume data, the notation used below means the iterated composition of the one-leg cohomological correspondences along the intermediate components.

Definition 2.4 (Componentwise arithmetic volume)**.** Assume (2.9) and $\mathrm{Frob}(\omega) = \omega$. The componentwise arithmetic volume

$$\mathrm{vol}\left({}^{\omega}\mathrm{Sht}_{G,I}, v_I^{\mathrm{rep}}\right)$$

is the regularized compactly supported Lefschetz supertrace of the Frobenius-twisted iterated cohomological correspondence, formed using the convention of Section 2.1: one applies ι to each finite-dimensional degreewise trace and then sums the resulting complex series. The shifts and twists are those of [FYZ, Definition 7.4.2] and [WW, §2.4.1]. Absolute convergence is proved in

Proposition 6.4; thus the definition does not assume a number-field model. Under Poincaré duality it is related to the ordinary-cohomology operator by (6.16) below. The stack dimension $\dim \mathrm{Bun}_G$ always means its geometric stack dimension over $\overline{k}$.

2.5. **Tautological classes.** Let

$$\mathrm{ev}_\omega : X' \times \mathrm{Bun}_G^\omega \longrightarrow BG_0$$

classify the pullback of the universal G-bundle. The characteristic classes used as multipliers below are Tate-normalized. Thus, for a homogeneous class of degree m, we use the notation

$$H^m_{\mathrm{Tate}}(Y) := \mathrm{Hom}^0_{\mathrm{Shv}(Y_{\bar{k}})}\big(\mathbf{1}, \mathbf{1}\langle m\rangle\big) = H^m(Y_{\bar{k}}, K_\ell(m/2)).$$

For example, the canonical degree-$2r$ characteristic classes of BG_0 lie in $H^{2r}(BG_0, K_\ell(r)) = H^{2r}_{\mathrm{Tate}}(BG_0)$. We suppress the coefficient twist in their names. This convention is used only for cohomology classes acting as multipliers; the untwisted Frobenius module $H^*(BG_0, K_\ell)$ used to define V_0 is unchanged.

For $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$ and $z \in H_*(X')$, define

$$\alpha^z_\omega := \mathrm{pr}_{2,!}\big(\mathrm{ev}^*_\omega \alpha \cup \mathrm{pr}_1^* \,\mathrm{PD}(z)\big). \tag{2.10}$$

The construction is K_ℓ-linear in z and is natural for Frobenius. On the direct sum over all geometric components it assembles to

$$H^*_{\mathrm{Tate}}(BG_0) \otimes H_*(X') \longrightarrow \bigoplus_\omega H^*(\mathrm{Bun}_G^\omega), \qquad \alpha \otimes z \longmapsto (\alpha^z_\omega)_\omega. \tag{2.11}$$

Frobenius carries the ω-component of this map to the $\mathrm{Frob}(\omega)$-component. When ω is Frobenius-stable, as in the componentwise trace theorem, equivariance reads

$$\mathrm{Frob}(\alpha^z_\omega) = (\mathrm{Frob}\,\alpha)^{\mathrm{Frob}\, z}_\omega. \tag{2.12}$$

The normalized characteristic classes used below are Frobenius-stable, so on a Frobenius-stable component the right side is $\alpha^{\mathrm{Frob}\, z}_\omega$. We do not assign a rank-one Weil object to an arbitrary vector z: a vector of $H_j(X')$ need neither be Frobenius-fixed nor span a Frobenius-stable line. The normalized shift belongs to the multiplier α, while the curve input is retained as a vector in the full Frobenius module $H_*(X')$. After forgetting Weil structure, (2.10) is the usual untwisted tautological class. The map (2.11) is invariant under the diagonal descent action and factors through coinvariants. Its augmentation-degree-one part identifies with $H_{>0}(X, \mathbb{V})$ under (2.5).

We shall also need a version allowing the class $\hbar$ of the tangent line of the formal disk. Put

$$\mathcal{B}_0 = B(L^+G_0 \rtimes \mathrm{Aut}\,\mathcal{D}).$$

The standard evaluation map

$$f_\omega : \mathrm{Bun}_G^\omega \times X' \longrightarrow \mathcal{B}_0$$

restricts the pulled-back G_0-bundle to the formal disk at the selected point. For $\beta \in H^*_{\mathrm{Tate}}(\mathcal{B}_0)$ and $z \in H_*(X')$, define

$$\mathsf{Glob}_\omega(\beta; z) := \mathrm{pr}_{1,!}\big(f^*_\omega \beta \cup \mathrm{pr}_2^* \,\mathrm{PD}(z)\big). \tag{2.13}$$

Thus $\mathsf{Glob}_\omega(\alpha; z) = \alpha^z_\omega$ for $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$. If $[X'] \in H_2(X')$ is the total fundamental class, then

$$\mathsf{Glob}_\omega(\hbar; [X']) = \int_{X'} c_1(TX') = -\kappa(X'). \tag{2.14}$$

Here we used the local-to-global identity $f^*_\omega \hbar = c_1(TX')$, the analogue of [WW, (3.3)] with $C = X'$.

The notation in (2.13) suppresses the harmless flip between $X' \times \mathrm{Bun}_G^\omega$ and $\mathrm{Bun}_G^\omega \times X'$. The class $\hbar$ is the first Chern class of the tangent line to the moving formal disk. Under the evaluation map, moving the chosen formal disk with the point of X' identifies this tangent line with TX'.

This explains both the identity $f_\omega^*\hbar = c_1(TX')$ and why the unit contribution is governed by the total Euler class of the entire leg curve rather than by one chosen geometric component.

3. General formalism

In this section we formulate the local and global relative Hecke operators used throughout the article. The setup is the quasisplit analogue of Wang–Wei's general formalism. A representative local datum is transported to the conjugate graphs by its graph-labelled pinned orbit; coherent coweight-indexed equivariance is a useful special case, not a necessary hypothesis.

3.1. **Local volume data and local relative Hecke operators.** Let $\mathcal{H}^\ell_{G_0}$ be the full local Hecke stack. We follow the arrow convention of Wang–Wei:

$$\mathcal{B}_0 \xleftarrow{\overleftarrow{h}^\ell} \mathcal{H}^\ell_{G_0} \xrightarrow{\overrightarrow{h}^\ell} \mathcal{B}_0,$$

where $\overrightarrow{h}^\ell$ records the bundle *before* the modification and $\overleftarrow{h}^\ell$ records the bundle *after* the modification. Thus, for a modification $\mathcal{E}_0 \dashrightarrow \mathcal{E}_1$, the right arrow records $\mathcal{E}_0$ and the left arrow records $\mathcal{E}_1$. This convention is forced by [WW, (2.9)–(2.10) and (3.7)] and will be used without further comment.

For a dominant coweight λ, let

$$\iota_\lambda : \mathcal{H}^\ell_{\le\lambda} \hookrightarrow \mathcal{H}^\ell_{G_0}$$

be the closed local Schubert Hecke stack. A local volume datum supported on it is a triple

$$v_\lambda = (K_\lambda, c^\ell_\lambda, d^\ell_\lambda)$$

with

$$c^\ell_\lambda \in \mathrm{Hom}^0\big(\overleftarrow{h}^\ell_{\lambda,!}K_\lambda\langle -d_c\rangle, \mathbf{1}_{\mathcal{B}_0}\big), \qquad d^\ell_\lambda \in \mathrm{Hom}^0\big(\mathbf{1}_{\mathcal{B}_0}, \overrightarrow{h}^\ell_{\lambda,!}K_\lambda\langle d_d\rangle\big).$$

Its degree is $d_v = d_c + d_d$ [WW, Definition 2.1]. The principal theorem concerns degree-two data.

The two correspondence morphisms play different roles. By adjunction, c^ℓ_λ produces a relative Borel–Moore class $u^\ell_\lambda = H^*(c^\ell_\lambda)(1)$ on the local Hecke stack. Ordinary cohomology acts on this class through the source and target pullbacks. The morphism d^ℓ_λ then evaluates the resulting Borel–Moore class and returns a class on the base. Schematically, the local operator is

$$H^*(\mathcal{B}_0) \xrightarrow{H^*(c)} H^{\mathrm{BM}}_{-*}(\mathcal{H}^\ell/\mathcal{B}_0, K) \xrightarrow{H^*(d)} H^{*+d_v}(\mathcal{B}_0).$$

This order is essential later: the graph commutator acts on $H^*(c)(1)$ before $H^*(d)$ is applied.

Only Tate-normalized cohomology classes are used to multiply local volume data. For a homogeneous class

$$\theta \in H^m_{\mathrm{Tate}}(\mathcal{H}^\ell_{\le\lambda}) = \mathrm{Hom}^0\big(\mathbf{1}, \mathbf{1}\langle m\rangle\big),$$

define

$$\theta \cdot v_\lambda := (K_\lambda, \theta \cup c^\ell_\lambda, d^\ell_\lambda). \tag{3.1}$$

It has

$$d_c(\theta\cdot v_\lambda) = d_c(v_\lambda) + m, \qquad d_d(\theta\cdot v_\lambda) = d_d(v_\lambda), \qquad d_{\theta\cdot v_\lambda} = d_v + m.$$

Equivalently, its first correspondence is the composition

$$\overleftarrow{h}^\ell_{\lambda,!}K_\lambda\langle -(d_c+m)\rangle \xrightarrow{\cup\theta} \overleftarrow{h}^\ell_{\lambda,!}K_\lambda\langle -d_c\rangle \xrightarrow{c^\ell_\lambda} \mathbf{1}_{\mathcal{B}_0}.$$

This is the Tate-normalized form of [WW, Example 2.3]. The restriction is essential: an arbitrary class in untwisted H^m supplies the cohomological shift $[m]$, but not the half-Tate twist $(m/2)$. All multipliers occurring below are products of normalized characteristic classes and their local brackets, so they lie in this normalized cohomology.

The local relative operator is homogeneous for normalized Tate degree:

$$\Gamma^{\ell}_{v_\lambda} : H^m_{\mathrm{Tate}}(\mathcal{B}_0) \longrightarrow H^{m+d_v}_{\mathrm{Tate}}(\mathcal{B}_0).$$

Lemma 3.1 (Degree two on the local base)**.** *For a split connected reductive group G_0, geometric cohomology admits a canonical graded-algebra identification*

$$H^*_{\mathrm{Tate}}(\mathcal{B}_0) \simeq H^*_{\mathrm{Tate}}(BG_0)[\hbar],$$

where $\hbar = c_1(T_{\mathcal{D}})$ is the normalized tangent-line class. Consequently

$$H^2_{\mathrm{Tate}}(\mathcal{B}_0) = H^2_{\mathrm{Tate}}(BG_0) \oplus K_\ell\,\hbar. \tag{3.2}$$

Under the standing semisimple hypothesis, $H^2_{\mathrm{Tate}}(BG_0) = 0$ and the first summand vanishes.

Proof. This is the cohomology calculation of [WW, §2.3.1], after scalar extension to K_ℓ: evaluation at the closed point identifies the positive-loop factor with BG_0 in cohomology, the pro-unipotent congruence kernels contribute no cohomology, and the quotient of $\mathrm{Aut}(\mathcal{D})$ by its pro-unipotent kernel is $\mathbf{G}_m$, whose degree-two generator is $\hbar$. Equivalently, $H^*(B(L^+G_0 \rtimes \mathrm{Aut}\,\mathcal{D})) = H^*(BG_0)[\hbar]$, which gives (3.2). The degree-two part of $H^*(BG_0)$ vanishes for connected semisimple G_0 because there are no Weyl-invariant linear forms. □

For any degree-two datum, Lemma 3.1 gives a unique decomposition

$$\Gamma^{\ell}_{v_\lambda}(1) = \chi_{\lambda,v} - b_{\lambda,v}\hbar, \qquad \chi_{\lambda,v} \in H^2_{\mathrm{Tate}}(BG_0), \quad b_{\lambda,v} \in K_\ell. \tag{3.3}$$

Define the horizontal component scalar

$$\delta^{\omega}_{\lambda,v} := \mathsf{Glob}_{\omega}(\chi_{\lambda,v}; [X']) \in H^0(\mathrm{Bun}^{\omega}_G) = K_\ell. \tag{3.4}$$

Under the standing semisimple hypothesis, $\chi_{\lambda,v} = 0$ and $\delta^{\omega}_{\lambda,v} = 0$. For the determinant datum we abbreviate $b_{\lambda,v}$ to b_λ.

The local bracket is first defined on the *full* local Hecke stack. For $\alpha \in H^m_{\mathrm{Tate}}(\mathcal{B}_0)$, let $[\alpha]$ be the class introduced in Wang–Wei's equation [WW, (2.15)], characterized by

$$\hbar[\alpha] = \overleftarrow{h}^{\ell *}\alpha - \overrightarrow{h}^{\ell *}\alpha.$$

Existence is part of their construction of the bracket. Their $K_\ell[\hbar]$-flatness result makes multiplication by $\hbar$ injective and therefore makes this class unique. We write

$$[\alpha] := \hbar^{-1}\big(\overleftarrow{h}^{\ell *}\alpha - \overrightarrow{h}^{\ell *}\alpha\big) \in H^{m-2}_{\mathrm{Tate}}(\mathcal{H}^{\ell}_{G_0}). \tag{3.5}$$

We then set

$$[\alpha]_\lambda := \iota^*_\lambda[\alpha] \in H^{m-2}_{\mathrm{Tate}}(\mathcal{H}^{\ell}_{\leq\lambda}). \tag{3.6}$$

Thus existence is the Wang–Wei bracket construction, while flatness is used only for uniqueness; no flatness assertion for the cohomology of the closed Schubert substack is needed.

For a degree-two datum put

$$\widetilde{\nabla}^{\ell}_{v_\lambda}(\alpha) := \Gamma^{\ell}_{[\alpha]_\lambda \cdot v_\lambda}(1). \tag{3.7}$$

Because $[\alpha]_\lambda \cdot v_\lambda$ has normalized degree $|\alpha|$, the right side lies in $H^{|\alpha|}_{\mathrm{Tate}}(\mathcal{B}_0)$. After forgetting the displayed Tate normalization and reducing modulo $\hbar$, one obtains Wang–Wei's derivation on untwisted cohomology,

$$\nabla^{\ell}_{v_\lambda} : H^*(BG_0, K_\ell) \longrightarrow H^*(BG_0, K_\ell),$$

which preserves the augmentation filtration. Its augmentation-degree-one part is

$$E_\lambda = \mathrm{Gr}^1_{\mathrm{aug}}\,\nabla^{\ell}_{v_\lambda} \in \mathrm{End}(V_0). \tag{3.8}$$

For the determinant datum in the almost-simple case, the symbols b_λ and E_λ are Wang–Wei's invariants. Under the standing semisimple hypothesis, these are the only local numerical data

that survive on the top associated graded: b_λ is the tangent-line unit coefficient, while E_λ is the linear response to commuting the operator past one augmentation-degree-one generator. For a reductive group there is in addition the horizontal class $\chi_{\lambda,v}$ and its component scalar $\delta^{\omega}_{\lambda,v}$. The global calculation below sends these three pieces to the horizontal scalar, the Euler-scaled tangent scalar, and the graph-transfer derivation.

3.2. **Pinned transport and graph-labelled orbits.** For $\gamma \in \Gamma$, let a_γ denote the pinned automorphism of G_0 and also its induced maps on $\mathcal{B}_0$ and the local Hecke stacks.

Definition 3.2. A family $\{v_\lambda\}$ is *coherently pinned-equivariant in the Weil category* if it is equipped with isomorphisms of local volume data

$$\theta_{\gamma,\lambda} : a_\gamma^* v_{\gamma\lambda} \xrightarrow{\sim} v_\lambda \tag{3.9}$$

that commute with the chosen Weil structures and satisfy, for all $\gamma, \delta \in \Gamma$,

$$\theta_{\gamma\delta,\lambda} = \theta_{\delta,\lambda} \circ a_\delta^*(\theta_{\gamma,\delta\lambda}). \tag{3.10}$$

Here $a_{\gamma\delta} = a_\gamma \circ a_\delta$, so the source on the right is $a_\delta^* a_\gamma^* v_{\gamma\delta\lambda} = a_{\gamma\delta}^* v_{\gamma\delta\lambda}$. No rationality condition on a number field is part of this definition.

Functoriality immediately gives

$$a_\gamma^* \chi_{\gamma\lambda,v} = \chi_{\lambda,v}, \qquad b_{\gamma\lambda,v} = b_{\lambda,v}, \qquad a_\gamma^* E_{\gamma\lambda} = E_\lambda a_\gamma^*. \tag{3.11}$$

In particular, the horizontal unit classes and their componentwise globalizations are compatible with pinned descent. Moreover,

$$a_\gamma^*[\alpha]_{\gamma\lambda} = [a_\gamma^*\alpha]_\lambda. \tag{3.12}$$

Definition 3.3 (Graph-labelled pinned orbit)**.** Fix a dominant coweight λ and a local volume datum v_λ on $\mathcal{H}^{\ell}_{\le\lambda}$. Its *graph-labelled pinned orbit* is the family

$$v_\lambda^{[\gamma]} := (a_{\gamma^{-1}})^* v_\lambda \qquad (\gamma \in \Gamma), \tag{3.13}$$

where $v_\lambda^{[\gamma]}$ is regarded as a datum on $\mathcal{H}^{\ell}_{\le\gamma\lambda}$. The label γ is retained even when $\gamma\lambda = \gamma'\lambda$. For $\delta, \gamma \in \Gamma$, contravariant functoriality gives canonical isomorphisms

$$\vartheta_{\delta,\gamma} : a_\delta^* v_\lambda^{[\delta\gamma]} \xrightarrow{\sim} v_\lambda^{[\gamma]}, \tag{3.14}$$

which satisfy the labelled cocycle condition. If v_λ has a Weil structure, all the data in (3.13)–(3.14) are taken in the Weil category.

The distinction between the graph label and the coweight is essential. If $h \in \Gamma$ fixes λ, the datum on the graph D_h is $(a_{h^{-1}})^* v_\lambda$; it is not identified with v_λ unless such an identification is part of the input. Thus the graph-labelled orbit of an arbitrary datum imposes no invariance under the stabilizer of λ. A coherently coweight-indexed family in the sense of Definition 3.2 identifies its graph-labelled orbit with the data $v_{\gamma\lambda}$, but the labelled construction is more general.

Proposition 3.4 (Labelled extension principle)**.** *The representative-leg globalisation, the multi-graph commutator, the exact product formula, the graph-transfer, the associated-graded theorem, the point-decoration theorem, and the scalar-plus-derivation trace formula remain valid for a local datum equipped with its graph-labelled pinned orbit. In all of these statements the datum on D_γ is $v_\lambda^{[\gamma]}$, and after transport to the representative chart the formulas are exactly the formulas written for v_λ in Sections 4.1, 4.2, 5 and 6.*

Proof. Write

$$v_\lambda^{[\gamma]} = (K^{[\gamma]}, c^{[\gamma]}, d^{[\gamma]})$$

with degree pair (d_c, d_d), independent of the graph label. The representative-leg coefficient object is still $f_{\mathrm{Hk}}^* K_\lambda$. On the graph D_γ, the local-model map satisfies $f_\gamma \simeq a_\gamma \circ f_{\mathrm{Hk}}$, and hence

$$f_\gamma^* K^{[\gamma]} = f_\gamma^* (a_{\gamma^{-1}})^* K_\lambda \xrightarrow{\sim} f_{\mathrm{Hk}}^* K_\lambda.$$

For $\delta, \gamma \in \Gamma$, the labelled cocycle $\vartheta_{\delta,\gamma}$ is an isomorphism of complete local volume data, not merely of coefficient objects. Thus the following two diagrams commute; the vertical arrows are the proper Beck–Chevalley identifications together with $\vartheta_{\delta,\gamma}$:

$$\begin{array}{ccc} a_\delta^*(\overleftarrow{h}^{\ell}_{\delta\gamma\lambda,!} K^{[\delta\gamma]}\langle -d_c\rangle) & \xrightarrow{a_\delta^* c^{[\delta\gamma]}} & a_\delta^* \mathbf{1} \\ \sim\downarrow & & \downarrow\sim \\ \overleftarrow{h}^{\ell}_{\gamma\lambda,!} K^{[\gamma]}\langle -d_c\rangle & \xrightarrow{c^{[\gamma]}} & \mathbf{1}, \end{array} \qquad \begin{array}{ccc} a_\delta^* \mathbf{1} & \xrightarrow{a_\delta^* d^{[\delta\gamma]}} & a_\delta^*(\overrightarrow{h}^{\ell}_{\delta\gamma\lambda,!} K^{[\delta\gamma]}\langle d_d\rangle) \\ \sim\downarrow & & \downarrow\sim \\ \mathbf{1} & \xrightarrow{d^{[\gamma]}} & \overrightarrow{h}^{\ell}_{\gamma\lambda,!} K^{[\gamma]}\langle d_d\rangle. \end{array}$$

The same diagrams remain commutative after pullback by the graph-local model maps. Indeed, the proper Beck–Chevalley maps, the exceptional-pullback comparisons, and their mate/counit identifications are natural in 2-Cartesian squares. Hence the global c- and d-morphisms attached to $D_{\delta\gamma}$ transport to those attached to D_γ.

Under a deck transformation δ, the graph D_γ is sent to $D_{\delta\gamma}$, and the preceding diagrams are precisely the compatibility needed to reindex every graph-labelled relative Borel–Moore term. The same argument applies to every bracket multiplier, since the bracket is functorial for pinned automorphisms. Thus every use of coherent pinned equivariance in the multi-graph and relative Borel–Moore arguments is replaced by the labelled cocycle without imposing stabilizer invariance on the representative datum.

Transporting the D_γ contribution back to the representative chart turns the local bracket into $[\alpha \cdot \gamma]_\lambda$ and the linear local operator into $E_\lambda(\gamma v)$. Therefore the exact product formula, the representative graph-transfer formula, and all Ran-filtration calculations are unchanged. The unit term uses only the representative local datum, while the point-decorated construction is linear in the same one-leg correspondence. Finally, the trace proof uses only the resulting scalar-plus-derivation operator and therefore applies without change. □

3.3. **Representative-leg globalisation.** Set

$$S_\lambda = \mathrm{Hk}_G^{\leq\lambda}, \qquad \overleftarrow{h} = (h_1, p), \qquad \overrightarrow{h} = (h_0, p) : S_\lambda \longrightarrow \mathrm{Bun}_G \times X'.$$

Thus $\overleftarrow{h}$ records the target bundle and $\overrightarrow{h}$ records the source bundle, in agreement with the Wang–Wei convention fixed above. All constructions are made on S_λ and then restricted to the open-and-closed source-component substack ${}^{\omega}S_\lambda$.

Let

$$f : \mathrm{Bun}_G \times X' \longrightarrow \mathcal{B}_0$$

be the formal restriction of the pulled-back G_0-bundle at the selected point, and let

$$f_{\mathrm{Hk}} : S_\lambda \longrightarrow \mathcal{H}^{\ell}_{\leq\lambda}$$

be formal restriction of the modification at the representative graph. These maps are maps to quotient prestacks; no coordinate is part of their definition.

Lemma 3.5 (Finite jet groups)**.** *Put*

$$H = L^+ G_0 \rtimes \mathrm{Aut}\, \mathcal{D}, \qquad \mathcal{B}_0 = BH.$$

For $n \geq 1$, put $A_n = k[t]/(t^{n+1})$, let

$$L_n^+ G_0 := \mathrm{Res}_{A_n/k}(G_0 \times_k A_n)$$

be the n-jet group, let $\mathrm{Aut}_n(\mathcal{D})$ be the group of automorphisms of A_n preserving the ideal (t), and put

$$H_n = L_n^+ G_0 \rtimes \mathrm{Aut}_n(\mathcal{D}).$$

Then

$$H \simeq \varprojlim_n H_n, \tag{3.15}$$

where every H_n is a smooth affine algebraic group and every transition $H_{n+1} \to H_n$ is smooth, affine, and surjective. In particular, $H \to \operatorname{Spec} k$ is elementary pro-smooth and $* \to BH$ is a representable pro-smooth surjection.

Proof. A complete finite-level proof is given in Section A.3. There the transition kernels are identified explicitly and the representable pro-smooth atlas $* \to BH$ is verified on arbitrary test schemes. □

Lemma 3.6 (Coordinate-and-trivialization torsors)**.** *There is a filtered exhaustion*

$$\mathrm{Bun}_G \times X' = \bigcup_\beta Y_\beta$$

by quasi-compact quasi-separated open algebraic substacks. The stack $\mathrm{Bun}_G \times X'$ is smooth over k: for a G-bundle $\mathcal{F}$ the obstruction space is $H^2(X, \mathrm{Ad}(\mathcal{F})) = 0$, since X is a curve. Hence every nonempty Y_β admits a finite-type scheme U_β, smooth over k, together with a smooth surjection $U_\beta \to Y_\beta$. Let $x'_\beta : U_\beta \to X'$ be the selected leg, let $\mathcal{E}_\beta$ be the pullback to $X' \times U_\beta$ of the universal G-bundle, viewed as a G_0-bundle with its semilinear Γ-descent datum, and let D_β be the graph of x'_β.

With H_n as in Lemma 3.5, let $P_{\beta,n}$ classify pairs

$$(t_n, \tau_n),$$

where t_n is a coordinate on the nth infinitesimal neighbourhood of D_β and τ_n is a trivialization of $\mathcal{E}_\beta$ there. Then $P_{\beta,n} \to U_\beta$ is an H_n-torsor, represented by an affine scheme smooth and surjective over U_β. The forgetful maps $P_{\beta,n+1} \to P_{\beta,n}$ are smooth, affine, and surjective, and

$$P_\beta := \varprojlim_n P_{\beta,n} \tag{3.16}$$

is an affine qcqs scheme over U_β. It represents simultaneous formal coordinates and a formal trivialization, and its moduli interpretation gives a canonical identification

$$P_\beta \xrightarrow{\sim} U_\beta \times_{BH} *. \tag{3.17}$$

Both maps

$$P_\beta \longrightarrow Y_\beta, \qquad P_\beta \longrightarrow BH$$

are pro-smooth surjections. Hence $f|_{Y_\beta} : Y_\beta \to BH$ satisfies the covering hypothesis of [LW, Proposition 4.10]*. The same is true after base change to either bounded global Hecke projection.*

Proof. See Section A.4. The appendix constructs the coordinate and trivialization torsors at every jet level, proves effectivity of their inverse limit, and checks both pro-smooth surjectivity conditions required by locally pro-smooth base change. □

Lemma 3.7 (Bounded local quotient and properness)**.** *Let*

$$Z_\lambda := \mathcal{H}^\ell_{\le\lambda} \times_{\mathcal{B}_0, \overleftarrow{h}^\ell_\lambda} *$$

be the target-trivialized bounded local Hecke space. Target trivialization identifies Z_λ with the projective Schubert variety $\overline{\mathrm{Gr}}_\lambda$, and changing the target coordinate and trivialization gives an H-action with

$$\mathcal{H}^\ell_{\le\lambda} \simeq [Z_\lambda/H]. \tag{3.18}$$

The two local projections

$$\overleftarrow{h}^\ell_\lambda, \overrightarrow{h}^\ell_\lambda : \mathcal{H}^\ell_{\le\lambda} \longrightarrow BH$$

are representable and proper of finite presentation.

Proof. See Section A.5. The quotient description is fpqc descent along $* \to BH$. Properness is checked after an arbitrary base change to an H-torsor, where the target projection becomes the associated bundle with fiber Z_λ; the source projection is treated by inversion of local modifications. □

Proposition 3.8 (Local-model squares and base change)**.** *Fix a local volume datum $v_\lambda = (K_\lambda, c^\ell_\lambda, d^\ell_\lambda)$ supported on $\mathcal{H}^\ell_{\le\lambda}$. Both squares in*

$$\begin{array}{ccccc} \mathrm{Bun}_G \times X' & \xleftarrow{\overleftarrow{h}} & S_\lambda & \xrightarrow{\overrightarrow{h}} & \mathrm{Bun}_G \times X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f_{\mathrm{Hk}}} & & \downarrow{\scriptstyle f} \\ \mathcal{B}_0 & \xleftarrow{\overleftarrow{h}^\ell_\lambda} & \mathcal{H}^\ell_{\le\lambda} & \xrightarrow{\overrightarrow{h}^\ell_\lambda} & \mathcal{B}_0 \end{array} \tag{3.19}$$

are 2-Cartesian. The four horizontal morphisms are representable and proper of finite presentation. Consequently the proper Beck–Chevalley maps are equivalences:

$$\overleftarrow{h}_! f^*_{\mathrm{Hk}} K_\lambda \xrightarrow{\sim} f^* \overleftarrow{h}^\ell_{\lambda,!} K_\lambda, \qquad f^* \overrightarrow{h}^\ell_{\lambda,!} K_\lambda \xrightarrow{\sim} \overrightarrow{h}_! f^*_{\mathrm{Hk}} K_\lambda. \tag{3.20}$$

Moreover the coordinate-and-trivialization presentations are locally pro-smooth. The locally pro-smooth $f_!$-exchange, together with the constant-proper-factor duality theorem of Proposition A.8, gives canonical exceptional exchange isomorphisms

$$f^*_{\mathrm{Hk}}(\overleftarrow{h}^\ell_\lambda)^! \mathbf{1}_{\mathcal{B}_0} \xrightarrow{\sim} \overleftarrow{h}^! \mathbf{1}_{\mathrm{Bun}_G \times X'}, \qquad f^*_{\mathrm{Hk}}(\overrightarrow{h}^\ell_\lambda)^! \mathbf{1}_{\mathcal{B}_0} \xrightarrow{\sim} \overrightarrow{h}^! \mathbf{1}_{\mathrm{Bun}_G \times X'}. \tag{3.21}$$

The equivalences (3.20) *and* (3.21) *are compatible with mates, the projection formula, multiplication by cohomology classes, and open-and-closed restriction to components.*

Proof. The stack-level equivalences are proved in Section A.6 by multi-disc Beauville–Laszlo gluing, including the semilinear Γ-descent cocycle and the verification on automorphism groupoids. Properness and the two sheaf-theoretic exchange isomorphisms are proved in Sections A.5 and A.7. □

Definition 3.9 (Representative-leg globalisation)**.** The representative-leg globalisation of v_λ is

$$v^{\mathrm{rep}}_\lambda = (K^{\mathrm{rep}}_\lambda, c^{\mathrm{rep}}_\lambda, d^{\mathrm{rep}}_\lambda), \qquad K^{\mathrm{rep}}_\lambda = f^*_{\mathrm{Hk}} K_\lambda,$$

where

$$\overleftarrow{h}_! K^{\mathrm{rep}}_\lambda\langle -d_c\rangle \xrightarrow[\mathrm{BC}]{\sim} f^* \overleftarrow{h}^\ell_{\lambda,!} K_\lambda\langle -d_c\rangle \xrightarrow{f^* c^\ell_\lambda} \mathbf{1}_{\mathrm{Bun}_G \times X'},$$

$$\mathbf{1}_{\mathrm{Bun}_G \times X'} \xrightarrow{f^* d^\ell_\lambda} f^* \overrightarrow{h}^\ell_{\lambda,!} K_\lambda\langle d_d\rangle \xrightarrow[\mathrm{BC}]{\sim} \overrightarrow{h}_! K^{\mathrm{rep}}_\lambda\langle d_d\rangle.$$

This is the construction of [WW, (2.4)–(2.6)] with the leg curve C replaced by X' and with the full-X'-leg quasisplit local-model square supplying the two base changes. The arrow directions explain the component direction of the resulting ordinary operator. The map c begins with a class on the target component and produces a relative Borel–Moore class on the Hecke stack; the map d evaluates that class against the source projection. After integrating over the leg, the ordinary operator therefore runs from the target component $\omega + \bar{\lambda}$ back to the source component ω.

Geometrically, a target bundle together with the representative local modification determines the source lattice at every conjugate graph by pinned transport and semilinear Γ-descent. Consequently K_λ^{rep} is the pullback of a *single* local coefficient object. It is not a tensor product of independent IC complexes indexed by the conjugate graphs. Those graphs enter later when the source and target tautological classes are compared; they do not supply additional coefficient factors. This distinction is what makes the representative-leg scalar a one-leg local scalar integrated over X' while the non-scalar commutator is a graph sum.

Proposition 3.10 (Relative Borel–Moore compatibility)**.** *Use Wang–Wei's canonical identification*

$$\omega_{\mathcal{H}^\ell/\mathcal{B}_0} := \overleftarrow{h}^{\ell!}\mathbf{1}_{\mathcal{B}_0} \xrightarrow{\sim} \overrightarrow{h}^{\ell!}\mathbf{1}_{\mathcal{B}_0}$$

and its base change

$$\omega_{S_\lambda/(\mathrm{Bun}_G \times X')} := \overleftarrow{h}^{!}\mathbf{1} \xrightarrow{\sim} \overrightarrow{h}^{!}\mathbf{1}.$$

For a coefficient object K put

$$H^{r,s}_{\mathrm{BM}}(Y/Z, K) := \mathrm{Hom}^0_{\mathrm{Shv}(Y)}\big(K, \omega_{Y/Z}[r](s)\big) \qquad (r \in \mathbf{Z},\ s \in \tfrac{1}{2}\mathbf{Z}).$$

The notation $H^{\mathrm{BM}}_{-*}(Y/Z, K)$ denotes this bigraded object with the homological reindexing of [WW, (2.8)]*, while suppressing the Tate index. The exchange isomorphism* (3.21) *defines a pullback map on relative Borel–Moore cohomology. The next diagram should be read from left to right: the first arrow forms the Borel–Moore class supplied by c, the middle vertical arrow pulls that class through the local-model square, and the second arrow evaluates it using d. Its commutativity is stronger than ordinary proper base change because it also identifies the two relative dualizing objects. With this interpretation, the diagram*

$$\begin{array}{ccccc} H^*(\mathcal{B}_0) & \xrightarrow{H^*(c_\lambda^\ell)} & H^{\mathrm{BM}}_{-*}(\mathcal{H}^\ell_{\le\lambda}/\mathcal{B}_0, K_\lambda) & \xrightarrow{H^*(d_\lambda^\ell)} & H^{*+d_v}(\mathcal{B}_0) \\ {\scriptstyle f^*}\downarrow & & {\scriptstyle f^*_{\mathrm{Hk}}}\downarrow & & \downarrow{\scriptstyle f^*} \\ H^*(\mathrm{Bun}_G \times X') & \xrightarrow{H^*(c_\lambda^{\mathrm{rep}})} & H^{\mathrm{BM}}_{-*}(S_\lambda/(\mathrm{Bun}_G \times X'), K_\lambda^{\mathrm{rep}}) & \xrightarrow{H^*(d_\lambda^{\mathrm{rep}})} & H^{*+d_v}(\mathrm{Bun}_G \times X') \end{array} \tag{3.22}$$

commutes. In particular, if

$$u_\lambda^\ell := H^*(c_\lambda^\ell)(1) \in H^{d_c, d_c/2}_{\mathrm{BM}}(\mathcal{H}^\ell_{\le\lambda}/\mathcal{B}_0, K_\lambda),$$
$$u_\lambda^{\mathrm{rep}} := H^*(c_\lambda^{\mathrm{rep}})(1) \in H^{d_c, d_c/2}_{\mathrm{BM}}(S_\lambda/(\mathrm{Bun}_G \times X'), K_\lambda^{\mathrm{rep}}).$$

then $u_\lambda^{\mathrm{rep}} = f^*_{\mathrm{Hk}} u_\lambda^\ell$, and for every homogeneous $\theta \in H^*_{\mathrm{Tate}}(\mathcal{H}^\ell_{\le\lambda})$,

$$H^*(d_\lambda^{\mathrm{rep}})\big(f^*_{\mathrm{Hk}}\theta \cdot u_\lambda^{\mathrm{rep}}\big) = f^*\Big(H^*(d_\lambda^\ell)(\theta \cdot u_\lambda^\ell)\Big) = f^*\Gamma^\ell_{\theta \cdot v_\lambda}(1). \tag{3.23}$$

Both sides have normalized degree $d_v + |\theta|$. In (3.22)*, the Tate index is suppressed; the products used in* (3.23) *lie on the Tate-normalized diagonal defined above.*

Proof. The definitions of bigraded relative Borel–Moore cohomology and its cohomology-module structures are given in Section A.8. The mate calculation proving (3.22) and (3.23) is carried out in Section A.9. □

4. Calculating global relative Hecke operators

This section contains the main geometric calculation. The universal quasisplit modification is pulled to the splitting cover, where it is supported on disjoint conjugate graphs. We first prove the corresponding ordinary cohomology commutator, then pass through relative Borel–Moore cohomology to obtain the exact product formula and its associated-graded consequence. There are three levels of identity, and keeping them distinct prevents a common type error. The multi-graph formula first compares ordinary source and target characteristic classes. It is then allowed to act on the relative Borel–Moore class $H^*(c)(1)$. Only after that action is the evaluation map $H^*(d)$ applied to obtain the global Hecke operator.

4.1. **The multi-graph commutator.**

4.1.1. *The graph divisors.* Pull the universal source and target G-bundles on $X \times S_\lambda$ back to $X' \times S_\lambda$. Denote the resulting G_0-bundles by $\mathcal{F}'_0$ and $\mathcal{F}'_1$. They are isomorphic away from

$$D = \coprod_{\gamma \in \Gamma} D_\gamma, \qquad D_\gamma = \Gamma_{\gamma p} \subset X' \times S_\lambda. \tag{4.1}$$

The graphs are pairwise disjoint because $X' \to X$ is a Galois torsor, and the modification has type $\gamma\lambda$ along D_γ. Let

$$f_\gamma : S_\lambda \longrightarrow \mathcal{H}^\ell_{\leq \gamma\lambda}$$

be the local-model map at D_γ. The descent datum gives a canonical 2-isomorphism

$$f_\gamma \simeq a_\gamma \circ f_{\mathrm{Hk}}. \tag{4.2}$$

Let

$$\mathrm{Ev}_0, \mathrm{Ev}_1 : X' \times S_\lambda \longrightarrow BG_0$$

classify $\mathcal{F}'_0$ and $\mathcal{F}'_1$ at an auxiliary point. Thus Ev_0 is the source evaluation and Ev_1 is the target evaluation.

4.1.2. *The ordinary cohomology identity.*

Lemma 4.1 (Family form of Wang–Wei's one-leg identity)**.** *Let $C/\overline{k}$ be a smooth projective curve, not necessarily connected, let S be a prestack for which the sheaf operations in use are defined, and let $\mathcal{E}_0 \dashrightarrow \mathcal{E}_1$ be a family of G_0-modifications of type $\leq \mu$ along the graph of $l : S \to C$. Let*

$$q : S \longrightarrow \mathrm{Hk}^{\leq \mu}_{G_0,C}, \qquad u : S \longrightarrow \mathcal{H}^\ell_{\leq \mu}$$

be the global and local classifying maps. Then, for every $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$,

$$\mathrm{Ev}_1^* \alpha - \mathrm{Ev}_0^* \alpha = \Delta_{l,!}\, u^*[\alpha]_\mu \qquad \text{in the corresponding Tate-normalized cohomology group of } C \times S, \tag{4.3}$$

where $\Delta_l : S \hookrightarrow C \times S$ is the graph of l.

Proof. If C is connected, Wang–Wei's Theorem 3.2 is an equality for the universal one-leg modification, with their left arrow recording the target and their right arrow recording the source. Pull that equality back along the classifying map q. The pullback of the universal diagonal is the graph Δ_l, and proper base change identifies the universal Gysin morphism with $\Delta_{l,!}$. If C is disconnected, restrict to each open-and-closed connected component: the identity is the connected one on the component containing the graph and is the zero identity on every other component. The componentwise identities reassemble to (4.3). No injectivity statement for pullback to an atlas is used. □

Proposition 4.2 (Multi-graph Wang–Wei identity)**.** *For every $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$,*

$$\mathrm{Ev}_1^*\,\alpha - \mathrm{Ev}_0^*\,\alpha = \sum_{\gamma\in\Gamma} \Delta_{\gamma,!}\, f_\gamma^*[\alpha]_{\gamma\lambda} \tag{4.4}$$

inside the corresponding Tate-normalized cohomology group of $X' \times S_\lambda$, where $\Delta_\gamma : S_\lambda \hookrightarrow X' \times S_\lambda$ is the graph of γp.

Proof. Work geometrically over $\overline{k}$ and on an open-and-closed substack on which the selected leg lies in one fixed geometric component of $X'_{\overline{k}}$. Choose an ordering $\Gamma = \{\gamma_1, \dots, \gamma_N\}$. Since the graph divisors are disjoint, Beauville–Laszlo gluing gives functorial intermediate bundles

$$\mathcal{F}'^{(0)} = \mathcal{F}'_0, \mathcal{F}'^{(1)}, \dots, \mathcal{F}'^{(N)} = \mathcal{F}'_1$$

such that $\mathcal{F}'^{(j-1)} \dashrightarrow \mathcal{F}'^{(j)}$ is supported only on D_{γ_j} and has type $\gamma_j\lambda$. Apply Lemma 4.1 to this one-graph modification. If Ev_j classifies $\mathcal{F}'^{(j)}$, it gives

$$\mathrm{Ev}_j^*\,\alpha - \mathrm{Ev}_{j-1}^*\,\alpha = \Delta_{\gamma_j,!} f_{\gamma_j}^*[\alpha]_{\gamma_j\lambda}.$$

On the other geometric components the two intermediate bundles agree, so the identity extends to all of $X'_{\overline{k}} \times S_{\lambda,\overline{k}}$. Summing over j telescopes to (4.4). The construction of the intermediate bundles and all graph Gysin maps is functorial under base change, so the equality descends. Disjointness eliminates excess-intersection terms, and the result is independent of the chosen ordering because the left side is the fixed difference $\mathrm{Ev}_1^*\,\alpha - \mathrm{Ev}_0^*\,\alpha$. □

Pairing with a curve homology class gives the typed tautological commutator. We write α^z_ω for the class on the source component and $\alpha^z_{\omega+\bar\lambda}$ for the corresponding class on the target component.

Corollary 4.3 (Typed tautological commutator)**.** *For $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$ and $z \in H_*(X')$,*

$$h_1^*(\alpha^z_{\omega+\bar\lambda}) - h_0^*(\alpha^z_\omega) = \sum_{\gamma\in\Gamma} f_\gamma^*[\alpha]_{\gamma\lambda} \cup (\gamma p)^*\,\mathrm{PD}(z) \tag{4.5}$$

$$= \sum_{\gamma\in\Gamma} f_{\mathrm{Hk}}^*[\alpha\cdot\gamma]_\lambda \cup p^*\,\mathrm{PD}(\gamma^{-1}z). \tag{4.6}$$

Equivalently, the left side is $\overleftarrow{h}^*(\alpha^z_{\omega+\bar\lambda}) - \overrightarrow{h}^*(\alpha^z_\omega)$.

Proof. Apply

$$\mathrm{pr}_{S,!}\big(- \cup \mathrm{pr}_{X'}^*\,\mathrm{PD}(z)\big)$$

to (4.4). The graph projection formula gives (4.5). Equation (4.6) follows from (4.2), pinned equivariance of the universal bracket, and $(\gamma p)^*\,\mathrm{PD}(z) = p^*\,\mathrm{PD}(\gamma^{-1}z)$. □

4.2. **Relative Borel–Moore module structure.** Let $v_\lambda^{\mathrm{rep}} = (K, c, d)$ be the global datum of Definition 3.9. Its relative operator is

$$\Gamma^{X'}_{v_\lambda} : H^*(\mathrm{Bun}_G \times X') \xrightarrow{H^*(c)} H^{\mathrm{BM}}_{-*}(S_\lambda/(\mathrm{Bun}_G \times X'), K) \xrightarrow{H^*(d)} H^{*+d_v}(\mathrm{Bun}_G \times X').$$

The absolute operator is

$$\Gamma_{\lambda,v} = \mathrm{pr}_{1,!} \circ \Gamma^{X'}_{v_\lambda} \circ \mathrm{pr}_1^*\,. \tag{4.7}$$

For $d_v = 2$ it has degree zero. On components it is

$$\Gamma^{\omega}_{\lambda,v} : H^*(\mathrm{Bun}_G^{\omega+\bar\lambda}) \longrightarrow H^*(\mathrm{Bun}_G^{\omega}). \tag{4.8}$$

Put

$$u = H^*(c)(1) \in H^{d_c, d_c/2}_{\mathrm{BM}}(S_\lambda/(\mathrm{Bun}_G \times X'), K).$$

The module identities of [WW, (2.9)–(2.10)], now with the arrows correctly aligned, are

$$H^*(c)(x) = \overleftarrow{h}^* x \cdot u = h_1^* x \cdot u \tag{4.9}$$

and

$$H^*(d)(\overrightarrow{h}^* y \cdot m) = H^*(d)(h_0^* y \cdot m) = y \cdot H^*(d)(m). \tag{4.10}$$

Thus x in (4.9) is a class on the target component, while y in (4.10) is a class on the source component. By Proposition 3.10, for every homogeneous $\theta \in H^*_{\mathrm{Tate}}(\mathcal{H}^{\ell}_{\leq\lambda})$,

$$H^*(d)\big(f_{\mathrm{Hk}}^*\theta \cdot u\big) = f^*\Big(\Gamma^{\ell}_{\theta\cdot v_\lambda}(1)\Big). \tag{4.11}$$

This is the relative Borel–Moore calculation: the ordinary graph identity acts on $u = H^*(c)(1)$, and only then is $H^*(d)$ applied. The final pushforward $\mathrm{pr}_{1,!}$ in (4.7) then integrates the remaining curve class over the full leg X'. For a degree-two local datum this curve integration lowers the degree by two, so the resulting absolute operator has degree zero and can enter an ordinary Frobenius trace.

4.3. **Exact product formula.** For homogeneous $z_i \in H_*(X')$, define the ordered intersection

$$\bigcap_{i\in A} z_i := \mathrm{PD}^{-1}\left(\prod_{i\in A}^{\longrightarrow} \mathrm{PD}(z_i)\right), \qquad \bigcap_{i\in\varnothing} z_i = [X'],$$

where the product follows the increasing order of the index set. Put

$$x_{i,+} := \alpha_{i,\omega+\bar{\lambda}}^{z_i}, \qquad x_{i,-} := \alpha_{i,\omega}^{z_i}.$$

For $J \subset I = \{1,\dots,N\}$ put $A = I \setminus J$ and

$$\epsilon(J) := \sum_{\substack{i<j \\ i\in A,\ j\in J}} |z_i|\,|z_j| \pmod 2. \tag{4.12}$$

Since $H^*_{\mathrm{Tate}}(BG_0)$ and the local brackets are even, this is exactly the Koszul sign obtained by moving the unchanged source tautological factors past the preceding curve classes and then ordering the source factors, curve classes, and local brackets increasingly by their original indices. It is independent of the graph labels.

The combinatorics of the formula are as follows. The subset J records those factors for which the target pullback is replaced by the unchanged source pullback. Its complement $A = I \setminus J$ records the factors for which a local bracket is inserted. A tuple $\boldsymbol{\gamma}$ chooses the conjugate graph supporting each inserted bracket. The bracket classes are multiplied locally, while their curve inputs are combined by Poincaré intersection before the final pushforward along X'. Thus each summand keeps separate track of the algebraic operation on the local Hecke stack and the geometric contraction of curve classes.

Theorem 4.4 (Exact quasisplit product formula)**.** *Let v_λ be a homogeneous local volume datum of degree d_v, equipped with its graph-labelled pinned orbit, for which the normalized bracket multipliers act as in* (3.1)*, and let $\alpha_i \in H^*_{\mathrm{Tate}}(BG_0)$ and $z_i \in H_*(X')$ be homogeneous. The resulting absolute operator has degree $d_v - 2$. We shall use the formula for $d_v = 2$ and $d_v = 0$. Then*

$$\begin{aligned}
&\Gamma^{\omega}_{\lambda,v}\left(\prod_{i\in I}^{\longrightarrow} x_{i,+}\right) \\
&\quad= \sum_{J\subset I}\left(\prod_{j\in J}^{\longrightarrow} x_{j,-}\right)\sum_{\boldsymbol{\gamma}\in\Gamma^{I\setminus J}} (-1)^{\epsilon(J)}\,\mathsf{Glob}_{\omega}\left(\Gamma^{\ell}_{(\prod^{\longrightarrow}_{i\in I\setminus J}[\alpha_i\cdot\gamma_i]_\lambda)\cdot v_\lambda}(1);\ \bigcap_{i\in I\setminus J}^{\longrightarrow}\gamma_i^{-1} z_i\right).
\end{aligned} \tag{4.13}$$

For $J = I$, the graph-label sum has one empty term and the intersection class is $[X']$. Formula (4.13) *is first an equality in geometric K_ℓ-cohomology. As the z_i vary, all its operations assemble into a Frobenius-equivariant multilinear identity on the full modules $H_{|z_i|}(X')$: deck transport, graph Gysin, Poincaré intersection, cup product, and proper pushforward all commute with Frobenius.*

Proof. By (4.9),

$$H^*(c)\left(\prod_i x_{i,+}\right) = \left(\prod_i \overleftarrow{h}^* x_{i,+}\right) u.$$

For every i, (4.6) gives

$$\overleftarrow{h}^* x_{i,+} = \overrightarrow{h}^* x_{i,-} + \sum_{\gamma_i \in \Gamma} f^*_{\mathrm{Hk}}[\alpha_i \cdot \gamma_i]_\lambda \cup p^* \,\mathrm{PD}(\gamma_i^{-1} z_i). \tag{4.14}$$

Expanding the ordered product, the subset J records the source-pullback choices and $\boldsymbol{\gamma}$ records the graph choices. Reordering into source, curve, and bracket blocks gives precisely the sign (4.12). Every source tautological class and every curve class is a $\overrightarrow{h}$-pullback, with

$$p^* \,\mathrm{PD}(w) = \overrightarrow{h}^*(1 \boxtimes \mathrm{PD}(w)).$$

Target-linearity (4.10) pulls these factors through $H^*(d)$. The relative Borel–Moore identity (4.11) converts the ordered local-bracket product acting on u into the local operator for the multiplied datum. Finally, the ordered curve product is Poincaré dual to the displayed intersection, and applying $\mathrm{pr}_{1,!}$ gives (4.13). □

The case of two generators. For $I = \{1, 2\}$, formula (4.13) has four types of terms:

$$\begin{aligned}
\Gamma^{\omega}_{\lambda,v}(x_{1,+}x_{2,+}) = {} & x_{1,-}x_{2,-}\,\mathsf{Glob}_\omega\big(\Gamma^\ell_v(1); [X']\big) \\
& + x_{1,-} \sum_{\gamma_2} \mathsf{Glob}_\omega\big(\Gamma^\ell_{[\alpha_2\cdot\gamma_2]_\lambda \cdot v}(1); \gamma_2^{-1} z_2\big) \\
& + (-1)^{|z_1||z_2|} x_{2,-} \sum_{\gamma_1} \mathsf{Glob}_\omega\big(\Gamma^\ell_{[\alpha_1\cdot\gamma_1]_\lambda \cdot v}(1); \gamma_1^{-1} z_1\big) \\
& + \sum_{\gamma_1,\gamma_2} \mathsf{Glob}_\omega\big(\Gamma^\ell_{[\alpha_1\cdot\gamma_1]_\lambda[\alpha_2\cdot\gamma_2]_\lambda \cdot v}(1); \gamma_1^{-1} z_1 \cap \gamma_2^{-1} z_2\big).
\end{aligned}$$

The first line is the scalar term, the middle two lines are the two possible one-bracket terms, and the final line is the first term that lowers Ran degree. This example is the complete combinatorial pattern of the general formula.

Lemma 4.5 (Degree and Tate-twist check)**.** *Every summand in* (4.13) *has the same cohomological degree and Tate twist as the left side.*

Proof. Put $A = I \setminus J$ and $k = |A|$. Write $\alpha_i \in H^{2r_i}_{\mathrm{Tate}}(BG_0) = H^{2r_i}(BG_0, K_\ell(r_i))$. By (3.5),

$$[\alpha_i \cdot \gamma_i]_\lambda \in H^{2r_i-2}_{\mathrm{Tate}}(\mathcal{H}^\ell_{\le\lambda}),$$

so the local multiplier

$$\theta_A = \prod_{i\in A} [\alpha_i \cdot \gamma_i]_\lambda$$

has normalized degree $\sum_{i\in A}(|\alpha_i| - 2)$. Consequently $\Gamma^\ell_{\theta_A \cdot v_\lambda}(1)$ has normalized degree

$$d_v + \sum_{i\in A}(|\alpha_i| - 2).$$

The ordered intersection has homological degree

$$\left|\bigcap_{i\in A}\gamma_i^{-1}z_i\right| = \sum_{i\in A}|z_i| - 2(k-1)$$

when it is nonzero. Hence the corresponding Glob-term has cohomological degree

$$d_v + \sum_{i\in A}(|\alpha_i|-2) - \left(\sum_{i\in A}|z_i| - 2(k-1)\right) = d_v - 2 + \sum_{i\in A}(|\alpha_i| - |z_i|).$$

Adding the unchanged outside factors gives the degree of the input monomial plus $d_v - 2$, which is exactly the degree of the absolute operator.

The normalized Tate shifts match at the level of the actual Weil morphisms. For every inserted bracket, the homogeneous identity

$$\overleftarrow{h}^{\ell *}\alpha_i - \overrightarrow{h}^{\ell *}\alpha_i = \hbar[\alpha_i]$$

places $[\alpha_i]$ in normalized degree $|\alpha_i| - 2$. The corresponding graph Gysin morphism is $[2](1) = \langle 2\rangle$, so its normalized shift cancels the loss produced by division by $\hbar$. Thus each bracket–Gysin pair contributes the same normalized shift as the original target-minus-source tautological factor. The final pushforward along the curve has relative dualizing shift $[-2](-1) = \langle -2\rangle$, exactly the shift included in the definition of Glob; together with the local degree d_v, this gives total absolute degree $d_v - 2$. The Poincaré intersection maps

$$\bigotimes_{i\in A} H_{|z_i|}(X') \longrightarrow H_{\sum_{i\in A}|z_i| - 2(|A|-1)}(X')$$

are Frobenius-equivariant multilinear maps. The curve-homology inputs acquire no auxiliary Tate shifts and are not replaced by individual rank-one Weil objects. Hence the geometric sheaf-level expansion is homogeneous term by term for the same total cohomological shift and Tate twist as the input monomial, and the resulting multilinear identity is Frobenius-equivariant. □

Corollary 4.6 (Unit and one-generator formulas)**.** *Assume in this corollary that v_λ has local degree two, and use the decomposition* (3.3)*. Then*

$$\Gamma^\omega_{\lambda,v}(1) = \delta^\omega_{\lambda,v} + \kappa(X')b_{\lambda,v}. \tag{4.15}$$

For homogeneous $\alpha \in H^*_{\mathrm{Tate}}(BG_0)$ and $z \in H_*(X')$,

$$\Gamma^\omega_{\lambda,v}(\alpha^z_{\omega+\bar\lambda}) = \left(\delta^\omega_{\lambda,v} + \kappa(X')b_{\lambda,v}\right)\alpha^z_\omega + \sum_{\gamma\in\Gamma}\mathsf{Glob}_\omega\left(\widetilde\nabla^\ell_{v_\lambda}(\alpha\cdot\gamma);\gamma^{-1}z\right). \tag{4.16}$$

Proof. Take $I = \varnothing$ for the unit. Equations (3.3), (3.4), and (2.14) give

$$\mathsf{Glob}_\omega(\chi_{\lambda,v} - b_{\lambda,v}\hbar;[X']) = \delta^\omega_{\lambda,v} - b_{\lambda,v}\int_{X'} c_1(TX') = \delta^\omega_{\lambda,v} + \kappa(X')b_{\lambda,v}.$$

For one generator, the two subsets of $\{1\}$ give the scalar and one-bracket terms. The input lies on the target component and the output on the source component, in accordance with the arrow convention. □

4.4. **The graph-transfer operator.** Let

$$\tau_\omega : H_{>0}(X,\mathbb{V}) \longrightarrow \mathrm{Gr}^1_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,H^*(\mathrm{Bun}^\omega_G)$$

be the degree-one Atiyah–Bott map, and let $m_\omega(\eta)$ denote multiplication by $\tau_\omega(\eta)$. The typed commutator is

$$[\Gamma^\omega_{\lambda,v}, m(\eta)] := \Gamma^\omega_{\lambda,v}\circ m_{\omega+\bar\lambda}(\eta) - m_\omega(\eta)\circ\Gamma^\omega_{\lambda,v}. \tag{4.17}$$

Definition 4.7. The graph-transfer $A_{\lambda,v} \in \operatorname{End} H_{>0}(X, \mathbb{V})$ is the unique map satisfying

$$\tau_\omega(A_{\lambda,v}\eta) = \operatorname{Gr}^1_{\mathrm{aug}} \operatorname{Gr}_{\mathrm{Ran}}[\Gamma^\omega_{\lambda,v}, m(\eta)](1). \tag{4.18}$$

It is independent of ω by (4.16).

Proposition 4.8 (Representative expression)**.** *Let $E_\lambda = \operatorname{Gr}^1_{\mathrm{aug}} \nabla^\ell_{v_\lambda}$. If $\eta \in H_{>0}(X, \mathbb{V})$ is represented in $(H_*(X') \otimes V_0)_\Gamma$ by v^z, then $A_{\lambda,v}\eta$ is represented by*

$$\sum_{\gamma\in\Gamma} \big(E_\lambda(\gamma v)\big)^{\gamma^{-1}z}. \tag{4.19}$$

This is the representative-coordinate expression of the intrinsic commutator (4.18)*; hence its class in coinvariants is independent of the representative v^z.*

Proof. Subtract the full unit scalar multiple of the output generator from (4.16) and pass to top Ran degree and augmentation degree one. Reducing modulo $\hbar$ replaces $\widetilde{\nabla}^\ell_{v_\lambda}$ by the derivation $\nabla^\ell_{v_\lambda}$, whose action on V_0 is E_λ. At D_γ, the labelled isomorphism (3.14) transports the operator of the datum $v_\lambda^{[\gamma]}$ to $E_\lambda(\gamma v)$ in the representative chart, while the graph pulls the curve class back to $\gamma^{-1}z$. Summing over the graph divisors gives (4.19). By the definition (4.18) through the typed commutator and the componentwise Atiyah–Bott map, the graph-indexed expression annihilates the descent relations and therefore induces the displayed class in the descent coinvariants; no commutation of E_λ with Γ is assumed. □

The order of logic is important. The endomorphism $A_{\lambda,v}$ is defined by the intrinsic typed commutator, which already lives on the descended cohomology. Formula (4.19) is obtained only after choosing a representative in $H_*(X') \otimes V_0$. If that representative is changed by a descent relation, the graph labels are reindexed and the labelled cocycle transports the local operator on D_γ to the corresponding operator on the reindexed graph. The total graph-indexed commutator is therefore unchanged, even though a single displayed summand need not be invariant by itself.

For the determinant datum we abbreviate $A_{\lambda,v}$ to A_λ.

4.5. **Associated graded.** Recall that the Ran filtration is the increasing filtration for which a tautological monomial

$$\prod_i \alpha_i^{z_i}$$

has filtration index at most $\sum_i |z_i|$; this is the filtration of [WW, §3.1.3], transported through the componentwise Atiyah–Bott isomorphism.

Theorem 4.9 (Associated-graded operator)**.** *The operator $\Gamma^\omega_{\lambda,v}$ preserves the Ran filtration and, on its Ran associated graded, preserves the augmentation filtration. Under the componentwise Atiyah–Bott identifications,*

$$\operatorname{Gr}^\bullet_{\mathrm{aug}} \operatorname{Gr}^\bullet_{\mathrm{Ran}} \Gamma^\omega_{\lambda,v} = \big(\delta^\omega_{\lambda,v} + \kappa(X')b_{\lambda,v}\big)\,\mathrm{id} + \nabla_{A_{\lambda,v}} \tag{4.20}$$

on $\operatorname{Sym}(H_{>0}(X, \mathbb{V}))$.

Proof. Apply Theorem 4.4 to a homogeneous tautological monomial and let $A = I \setminus J$ be the set of inserted brackets. Put

$$R_{\mathrm{in}} = \sum_{i\in I} |z_i|.$$

If $A \neq \varnothing$ and the ordered intersection in (4.13) is nonzero, then

$$\left|\bigcap_{i\in A} \gamma_i^{-1} z_i\right| = \sum_{i\in A} |z_i| - 2(|A| - 1).$$

Hence the output Ran index of that summand is

$$R_{\mathrm{out}} = \sum_{j\in J} |z_j| + \left|\bigcap_{i\in A} \gamma_i^{-1} z_i\right| = R_{\mathrm{in}} - 2(|A|-1). \tag{4.21}$$

For $A = \varnothing$ the scalar term preserves the index; for $|A| = 1$ the index is again preserved; and for $|A| \geq 2$ it drops by at least two. This proves preservation of the Ran filtration and shows that only the zero-bracket and one-bracket terms survive on its top associated graded. It is the graph-labelled form of the calculation in [WW, Lemma 3.6].

The zero-bracket term is multiplication by (4.15). In a one-bracket term, the local analogue of passage to the top Ran associated graded is reduction modulo $\hbar$ [WW, §3.4, especially Lemma 3.6]. Indeed, after globalisation a factor $\hbar$ becomes $c_1(TX')$: its linear contribution lowers the curve-homology index by two, while powers at least two vanish on the curve. Thus only the reduction of $\widetilde{\nabla}^{\ell}_{v_\lambda}$ modulo $\hbar$ survives. Wang–Wei's Lemma 2.6 identifies that reduction with the derivation $\nabla^{\ell}_{v_\lambda}$, whose augmentation-degree-one action is E_λ. By Proposition 4.8, the one-bracket contribution on a generator is $A_{\lambda,v}$, and the sum over the chosen generator is the derivation $\nabla_{A_{\lambda,v}}$. This proves (4.20) and augmentation-filtration preservation. □

In particular, the scalar and derivation in (4.20) have independent geometric meanings. The scalar is determined by the value of the one-leg operator on the unit. Its horizontal part $\delta^{\omega}_{\lambda,v}$ comes from $H^2(BG_0)$, while its tangent part $\kappa(X')b_{\lambda,v}$ remembers the Euler class of the full leg curve. The derivation is determined by the commutator with one tautological generator and therefore remembers the sum of the conjugate local operators. Multiple commutators disappear from the top Ran graded because every additional bracket forces one further curve-class contraction.

Proposition 4.10 (Split and minuscule checks)**.** *If $X' = X$ and $\Gamma = 1$, the relative Borel–Moore calculation above specializes term by term to* [WW, (3.7)–(3.8)]*, and* (4.20) *becomes* [WW, Proposition 3.7]*. If the coweight is minuscule, the ordinary commutator has the sign $h_1^* - h_0^*$ and the graph-transfer agrees with* [FYZ, Proposition 7.5.1 and §7.4.8] *for the corresponding local datum.*

Proof. For $\Gamma = 1$, (4.14) is Wang–Wei's replacement of the target pullback by the source pullback plus the local bracket, and (4.9), (4.10), and (4.11) reproduce their calculation without an additional transport or graph label. In the minuscule case, (4.5) has the target-minus-source sign $h_1^* - h_0^*$ of [FYZ, Proposition 7.5.1], and (4.19) is the full graph-transfer of [FYZ, §7.4.8]. □

5. Globally decorated local volume data

The local-volume-data formalism produces degree-two absolute operators by globalizing a degree-two local datum over $\mathrm{Bun}_G \times X'$ and then integrating over the one-dimensional leg. Feng–Yun–Zhang use one additional construction: a degree-zero local correspondence is multiplied by a degree-two class from the leg curve before the final pushforward. This section isolates that operation and proves that it changes only the scalar on the top Ran associated graded.

5.1. **Point-decorated degree-zero data.** Let $w_\lambda = (K^0_\lambda, c^{\ell,0}_\lambda, d^{\ell,0}_\lambda)$ be a local volume datum of degree zero, equipped with its graph-labelled pinned orbit. Its representative-leg globalisation is a relative global volume datum

$$w^{\mathrm{rep}}_\lambda = (K_w, c_w, d_w)$$

on S_λ, with degree pair (d^w_c, d^w_d) satisfying $d^w_c + d^w_d = 0$. Thus

$$c_w : \overleftarrow{h}_! K_w\langle -d^w_c\rangle \longrightarrow \mathbf{1}, \qquad d_w : \mathbf{1} \longrightarrow \overrightarrow{h}_! K_w\langle d^w_d\rangle.$$

Let

$$\zeta \in H^2_{\mathrm{Tate}}(X') = H^2(X'_{\bar{k}}, K_\ell(1))$$

be Γ-invariant and Frobenius-stable. Since $p = \mathrm{pr}_2 \circ \overleftarrow{h} = \mathrm{pr}_2 \circ \overrightarrow{h}$, multiplication by $p^*\zeta$ defines

$$\begin{aligned} c_{\zeta w} : \quad \overleftarrow{h}_! K_w\langle -(d_c^w + 2)\rangle &\xrightarrow{\overleftarrow{h}_!(\cup p^*\zeta)} \overleftarrow{h}_! K_w\langle -d_c^w\rangle \\ &\xrightarrow{c_w} \mathbf{1}, \qquad d_{\zeta w} := d_w. \end{aligned} \tag{5.1}$$

Hence

$$(\zeta \star w_\lambda)^{\mathrm{rep}} := (K_w, c_{\zeta w}, d_w) \tag{5.2}$$

is an actual global volume datum of total degree $(d_c^w + 2) + d_d^w = 2$.

Proposition 5.1 (Cohomological action of the point decoration)**.** *The relative ordinary operator of $(\zeta \star w_\lambda)^{\mathrm{rep}}$ is*

$$\Gamma^{X',\omega}_{\lambda,\zeta\star w}(x) = \mathrm{pr}_2^*\,\zeta \cup \Gamma^{X',\omega}_{w_\lambda}(x). \tag{5.3}$$

Its absolute operator is therefore

$$\Gamma^{\omega}_{\lambda,\zeta\star w}(x) := \mathrm{pr}_{1,!}\Big(\mathrm{pr}_2^*\,\zeta \cup \Gamma^{X',\omega}_{w_\lambda}(\mathrm{pr}_1^*\,x)\Big)\,. \tag{5.4}$$

In particular, the point decoration is represented by a genuine degree-two cohomological correspondence, so its compactly supported operator and its Lefschetz trace are defined by the ordinary local-volume-data formalism.

Proof. Put $u_w = H^*(c_w)(1)$ in the relative Borel–Moore group of the global Hecke stack. By (5.1),

$$H^*(c_{\zeta w})(x) = \overleftarrow{h}^* x \cdot p^*\zeta \cdot u_w.$$

Because $p^*\zeta = \overrightarrow{h}^*\,\mathrm{pr}_2^*\,\zeta$, target-linearity of $H^*(d_w)$ gives

$$H^*(d_w)H^*(c_{\zeta w})(x) = \mathrm{pr}_2^*\,\zeta \cup H^*(d_w)H^*(c_w)(x),$$

which is (5.3). Pulling an absolute input back by pr_1 and applying the proper pushforward $\mathrm{pr}_{1,!}$ gives (5.4). This is exactly the projection-formula realization of the sheaf-theoretic datum (5.2). □

Put

$$e^{\omega}_{\lambda,\zeta,w} := \Gamma^{\omega}_{\lambda,\zeta\star w}(1) \in H^0(\mathrm{Bun}^{\omega}_G) = K_\ell. \tag{5.5}$$

For a homology class $z \in H_*(X')$, write

$$\zeta \cap z := \mathrm{PD}^{-1}(\zeta \cup \mathrm{PD}(z))\,. \tag{5.6}$$

This convention is componentwise when X' is geometrically disconnected and imposes no normalization on ζ.

Definition 5.2 (Globally decorated datum)**.** Let $v_\lambda^{\mathrm{rep}} = (K_v, c_v, d_v)$ be a degree-two global datum of degree pair (d_c^v, d_d^v), and let $(\zeta \star w_\lambda)^{\mathrm{rep}}$ be as above. Set

$$t = d_c^v - (d_c^w + 2) = d_d^w - d_d^v. \tag{5.7}$$

After replacing the point-decorated coefficient by $K_w\langle t\rangle$, its degree pair becomes (d_c^v, d_d^v). More explicitly,

$$\overleftarrow{h}_!(K_w\langle t\rangle)\langle -d_c^v\rangle = \overleftarrow{h}_! K_w\langle -(d_c^w + 2)\rangle, \qquad \overrightarrow{h}_!(K_w\langle t\rangle)\langle d_d^v\rangle = \overrightarrow{h}_! K_w\langle d_d^w\rangle.$$

The symbols $c_{\zeta w}[t]$ and $d_w[t]$ below denote $c_{\zeta w}$ and d_w under these canonical shift identifications. The decorated global datum is the biproduct

$$\mathfrak{v}_\lambda^{\mathrm{rep}} := \Big(K_v \oplus K_w\langle t\rangle, (c_v, c_{\zeta w}[t]), \begin{pmatrix} d_v \\ d_w[t] \end{pmatrix}\Big) \tag{5.8}$$

with common degree pair (d_c^v, d_d^v). Its ordinary operator is

$$\Gamma^\omega_{\lambda,\mathfrak{v}} = \Gamma^\omega_{\lambda,v} + \Gamma^\omega_{\lambda,\zeta\star w}. \tag{5.9}$$

For a sequence $\mathfrak{v}_I$, the compact and ordinary iterated correspondences are the ordered compositions of these biproduct correspondences, expanded distributively.

If $t = 0$, $K_v = K_w$, and $d_v = d_w$, one may equivalently use the *compressed model* with coefficient object K_v, common d-map, and c-map $c_v + c_{\zeta w}$. By linearity in c, the compressed and biproduct models induce the same ordinary and compact cohomological operators. This is the model used in the FYZ specialization below.

Lemma 5.3 (Duality for a point decoration)**.** *The compactly supported operator attached to the degree-two global datum $(\zeta \star w_\lambda)^{\mathrm{rep}}$ is graded-dual to* (5.4)*. Consequently the compact operator of* (5.8) *is graded-dual to the ordinary operator* (5.9)*; the same conclusion holds for the compressed model when it is available.*

Proof. The first assertion follows formally from Poincaré–Verdier duality for the genuine global volume datum (5.2). In cohomological terms, cup product by the even class $\mathrm{pr}_2^*\zeta$ is self-adjoint, while pr_1^* and $\mathrm{pr}_{1,!}$ are adjoint; the projection formula therefore identifies the dual operator with the point-decorated compact relative operator. The normalized shifts cancel because the local datum has degree zero, ζ has degree two, and integration along X' has degree -2. Verdier duality is additive for finite biproducts, proving the second assertion. In the compressed case, both compact and ordinary operators are linear in the summed c-map. □

Proposition 5.4 (Exact point-decorated product formula)**.** *Let $\alpha_i \in H^*_{\mathrm{Tate}}(BG_0)$ and $z_i \in H_*(X')$ be homogeneous, and retain the notation of Theorem 4.4. Then*

$$\begin{aligned} &\Gamma^\omega_{\lambda,\zeta\star w}\left(\overrightarrow{\prod_{i\in I}} x_{i,+}\right) \\ &\quad = \sum_{J\subset I}\left(\overrightarrow{\prod_{j\in J}} x_{j,-}\right)\sum_{\boldsymbol{\gamma}\in\Gamma^{I\setminus J}}(-1)^{\epsilon(J)}\mathsf{Glob}_\omega\left(\Gamma^\ell_{(\prod^{\rightarrow}_{i\in I\setminus J}[\alpha_i\cdot\gamma_i]_\lambda)\cdot w_\lambda}(1); \zeta\cap\bigcap_{i\in I\setminus J}\gamma_i^{-1}z_i\right). \end{aligned} \tag{5.10}$$

For $J = I$, the curve input is $\zeta \cap [X']$ and the local multiplier is empty.

Proof. Repeat the proof of Theorem 4.4 for the degree-zero datum w_λ, and before the final pushforward along X' multiply every expanded summand by $\mathrm{pr}_2^*\zeta$. The source factors still pass through $H^*(d)$ by target-linearity, the local brackets are handled by (4.11), and the curve factors become

$$\zeta \cup \overrightarrow{\prod_{i\in I\setminus J}} \mathrm{PD}(\gamma_i^{-1}z_i).$$

By (5.6), its Poincaré dual is the curve input displayed in (5.10). The Koszul sign is unchanged because ζ has even degree and is inserted after the ordered expansion. The degree check is

$$\sum_{i\in I\setminus J}(|\alpha_i|-2) - \left(\sum_{i\in I\setminus J}|z_i| - 2|I\setminus J|\right) = \sum_{i\in I\setminus J}(|\alpha_i| - |z_i|),$$

so every decorated summand has the same degree as the input monomial. The normalized Tate twists match as well: w_λ has degree zero, multiplication by $\zeta \in H^2_{\mathrm{Tate}}(X')$ contributes $\langle 2\rangle$, and the final curve pushforward contributes $\langle -2\rangle$. Every bracket–Gysin pair has the same cancelling normalized shifts as in Lemma 4.5; curve intersection is a Frobenius-equivariant multilinear operation on the full homology module and introduces no auxiliary rank-one Weil object. □

Proposition 5.5 (Ran effect of a point decoration)**.** *The point-decorated operator preserves the Ran filtration and satisfies*

$$\mathrm{Gr}_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,\Gamma^{\omega}_{\lambda,\zeta\star w} = e^{\omega}_{\lambda,\zeta,w}\ \mathrm{id}\,. \tag{5.11}$$

Consequently, for a decorated datum Definition 5.2,

$$\mathrm{Gr}_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,\Gamma^{\omega}_{\lambda,\mathfrak{v}} = \big(\delta^{\omega}_{\lambda,v} + \kappa(X')b_{\lambda,v} + e^{\omega}_{\lambda,\zeta,w}\big)\,\mathrm{id} + \nabla_{A_{\lambda,v}}. \tag{5.12}$$

Proof. Let $A = I \setminus J$ be the set of inserted brackets in (5.10). If $|A| = k$, then

$$\left|\zeta \cap \bigcap_{i\in A} \gamma_i^{-1} z_i\right| = \sum_{i\in A} |z_i| - 2(k-1) - 2 = \sum_{i\in A} |z_i| - 2k$$

whenever the class is nonzero. Hence the output Ran index is

$$R^{\zeta}_{\mathrm{out}} = R_{\mathrm{in}} - 2|A|. \tag{5.13}$$

For $A = \varnothing$, the summand is the unit scalar $e^{\omega}_{\lambda,\zeta,w}$ times the unchanged monomial. Every summand with at least one bracket lowers the Ran index by at least two. This proves (5.11). Adding Theorem 4.9 proves (5.12). □

The three scalar mechanisms are independent. The horizontal term $\delta^{\omega}_{\lambda,v}$ comes from the $H^2(BG_0)$-part of the local unit and can depend on the component; it vanishes for semisimple G_0. The term $\kappa(X')b_{\lambda,v}$ comes from the formal-disk tangent class $\hbar$ and remembers the Euler class of the full leg curve. The term $e^{\omega}_{\lambda,\zeta,w}$ comes from a separately chosen global class on X'; it need not be divisible by $\kappa(X')$ and may remain nonzero when X' has genus one.

6. Arithmetic volume of quasisplit shtukas

We now pass from the associated-graded operator to the arithmetic-volume formula. As in Feng–Yun–Zhang, the embedding ι was fixed once in Section 2.1 and is suppressed from the notation. The argument does not require a common number-field model. Two separate points must be proved. First, in each cohomological degree, trace additivity identifies the trace of the filtered operator with the trace of its associated-graded scalar-plus-derivation model. Second, the sum over all cohomological degrees must converge absolutely. The first point is finite-dimensional algebra; the second uses purity to dominate polynomial derivation growth by exponential Frobenius decay.

6.1. Frobenius and component shifts.

From this point onward assume that the representative local data and their graph-labelled pinned orbits carry compatible Weil structures. The labelled cocycle globalizes the graph-indexed construction over k, and the fixed embedding (2.1) is used solely to interpret the degreewise ℓ-adic traces as a complex series. No number-field model is assumed.

Lemma 6.1. *The graph-transfer $A_{\lambda,v}$ commutes with geometric Frobenius on $H_*(X,\mathbb{V})$. Its adjoint therefore preserves the Frobenius module $H^{<0}(X,\mathbb{V}^\vee)$.*

Proof. The Hecke stack, graph divisors, deck transformations, pinned automorphisms, and the local correspondence morphisms are all defined over k. Frobenius consequently commutes with every operation in the typed commutator (4.18). Equivalently, it permutes the graph-labelled construction underlying (4.19) and commutes with the sum. □

Lemma 6.2 (Frobenius-stable intermediate components)**.** *The component shift $\bar{\lambda}$ is fixed by Frobenius. Hence, if ω is Frobenius-stable and*

$$\omega_i = \omega + \bar{\lambda}_1 + \cdots + \bar{\lambda}_i,$$

then every ω_i is Frobenius-stable.

Proof. The full-X'-leg Hecke stack and the source and target component maps are defined over k. Hence the translation of the component group determined by this correspondence is defined over k. Its translation element is $\bar{\lambda}$, so $\mathrm{Frob}(\bar{\lambda}) = \bar{\lambda}$. Equivalently, Frobenius carries a modification from ω to $\omega + \bar{\lambda}$ to one from $\mathrm{Frob}(\omega)$ to $\mathrm{Frob}(\omega) + \bar{\lambda}$. Induction on i proves the assertion. □

Let $\lambda_I = (\lambda_1, \ldots, \lambda_r)$ satisfy (2.9), and for the selected degree-two data abbreviate A_{λ_i, v_i} to A_{λ_i}. We impose the same commutativity condition as the two source theorems. By the perfect Poincaré-duality pairing between $H_{>0}(X, \mathbb{V})$ and $H^{<0}(X, \mathbb{V}^\vee)$, this is equivalent to pairwise commutativity of the corresponding operators on $H_{>0}(X, \mathbb{V})$:

$$A^\vee_{\lambda_1}, \ldots, A^\vee_{\lambda_r} \quad \text{pairwise commute on } M = H^{<0}(X, \mathbb{V}^\vee). \tag{6.1}$$

By finite-dimensional duality, the corresponding endomorphisms A_{λ_i} pairwise commute on $H_{>0}(X, \mathbb{V})$; hence their derivations commute on the Atiyah–Bott symmetric algebra. After a finite scalar extension, write

$$M = \bigoplus_{\varepsilon} M[\varepsilon] \tag{6.2}$$

for the simultaneous generalized eigenspace decomposition, where ε_i is the actual generalized eigenvalue of $A^\vee_{\lambda_i}$ on $M[\varepsilon]$. By Lemma 6.1, each $M[\varepsilon]$ is Frobenius-stable. The perfect pairing identifies the dual summand $W[\varepsilon] \subset W := H_{>0}(X, \mathbb{V})$ with the simultaneous generalized eigenspace on which every A_{λ_i} has the same actual eigenvalue ε_i. Thus the index ε used in the symmetric-algebra trace and in the L-function is the same. Define L^*_X and D_i by (1.3) and (1.4).

6.2. **Convergence and trace comparison.** There are two logically separate points. First, in each fixed cohomological degree, ordinary trace additivity allows the operator on $H^*(\mathrm{Bun}^\omega_G)$ to be replaced by its doubly associated-graded scalar-plus-derivation model. Second, the sum over all cohomological degrees must converge absolutely. The first point is finite-dimensional and filtration-theoretic; the second follows from polynomial growth of the symmetric algebra and exponential decay of inverse Frobenius. Keeping these steps separate avoids any need to estimate matrix coefficients of the original IC correspondence.

Lemma 6.3 (Purity in the Atiyah–Bott grading)**.** *Let $W = H_{>0}(X, \mathbb{V})$ with the cohomological grading appearing in* (2.5)*. For every $n > 0$, every eigenvalue of geometric Frobenius on W^n has complex absolute value $q^{n/2}$; equivalently, every eigenvalue of* Frob^{-1} *has absolute value $q^{-n/2}$.*

Proof. It suffices to calculate after pullback to X'. After extending scalars to $\overline{\mathbf{Q}}_\ell$, decompose V_0^{2d} and $H_j(X')$ into generalized Frobenius eigenspaces. If a and b are eigenvalues on such two generalized eigenspaces, then ab is the corresponding eigenvalue on their tensor product; no semisimplicity is required. In the homological convention of [FYZ, Remark 7.2.2], the resulting Künneth summand has Atiyah–Bott cohomological degree

$$|v^z| = 2d - j. \tag{6.3}$$

The graded piece V_0^{2d} is Tate of weight $2d$: it is a direct sum of copies of $K_\ell(-d)$, so every geometric-Frobenius eigenvalue a on it has absolute value q^d. Poincaré duality identifies $H_j(X') = H^{2-j}(X', K_\ell)(1)$; since X' is smooth and proper, every Frobenius eigenvalue b on $H_j(X')$ has

absolute value $q^{-j/2}$. Hence every product eigenvalue ab on the tensor-product summand has absolute value

$$q^d q^{-j/2} = q^{(2d-j)/2} = q^{|v^z|/2}.$$

The action of Γ commutes with Frobenius. Over characteristic-zero coefficients, invariants and coinvariants are direct summands of the finite-monodromy module, so passing from $H_*(X') \otimes V_0$ to $H_*(X, \mathbb{V})$ preserves these absolute values. This proves the assertion. It is the quasisplit form of the degree–weight calculation used in [FYZ, §5.5.18]. □

Proposition 6.4 (Trace comparison and convergence)**.** *Let $T = T_1 \circ \cdots \circ T_r$ be a fixed finite composition of Frobenius-equivariant, filtration-preserving one-leg operators with total component shift zero. Suppose that, under the componentwise Atiyah–Bott identifications,*

$$\mathrm{Gr}_{\mathrm{aug}} \mathrm{Gr}_{\mathrm{Ran}} T_i = c_i \,\mathrm{id} + \nabla_{A_i}$$

for fixed scalars c_i and fixed degree-preserving endomorphisms A_i of $W = H_{>0}(X, \mathbb{V})$ commuting with Frobenius. On a Frobenius-stable component, the complex series

$$\sum_n (-1)^n \, \iota\big(\mathrm{Tr}\big(\mathrm{Frob}^{-1} T \mid H^n(\mathrm{Bun}_G^{\omega})\big)\big)$$

is absolutely convergent. Its sum equals the complex supertrace of the induced operator on the doubly associated graded

$$\mathrm{Gr}_{\mathrm{aug}} \mathrm{Gr}_{\mathrm{Ran}} H^*(\mathrm{Bun}_G^{\omega}).$$

No number-field form of the local data or of the pinned descent maps is required.

Proof. Put $W = H_{>0}(X, \mathbb{V})$. Since W is finite-dimensional and positively graded, every cohomological degree $\mathrm{Sym}(W)^n$ is finite-dimensional. More explicitly, if $d_{\min} > 0$ is the least degree of a nonzero homogeneous generator, then a degree-n monomial has at most $n/d_{\min}$ factors. Its augmentation degree is therefore bounded, and its Ran index is a sum of at most that many curve-homology degrees drawn from the finite set occurring in $H_*(X')$. Thus the Ran and augmentation filtrations are exhaustive and separated and have only finitely many jumps in fixed cohomological degree. The filtered Atiyah–Bott theorem identifies the doubly associated graded of $H^n(\mathrm{Bun}_G^{\omega})$ with the finite-dimensional space $\mathrm{Sym}(W)^n$, so $H^n(\mathrm{Bun}_G^{\omega})$ is finite-dimensional. Frobenius preserves both filtrations. The stated hypotheses give preservation of both filtrations and Frobenius equivariance for every one-leg operator. Before applying ι, finite-dimensional trace additivity first for the Ran filtration and then for the augmentation filtration gives an equality in K_ℓ,

$$\mathrm{Tr}\big(\mathrm{Frob}^{-1} T \mid H^n(\mathrm{Bun}_G^{\omega})\big) = \mathrm{Tr}\big(\mathrm{Frob}^{-1} \mathrm{Gr}\, T \mid \mathrm{Gr}_{\mathrm{aug}} \mathrm{Gr}_{\mathrm{Ran}} H^n(\mathrm{Bun}_G^{\omega})\big)\,. \tag{6.4}$$

Applying ι preserves this equality. From this point onward all absolute values and norms are taken on the resulting complex vector spaces. It remains to prove absolute convergence of the right side.

After extending W through ι, it is a finite-dimensional positively graded complex super vector space, and the scalar extension of the doubly associated graded is $\mathrm{Sym}(W)$. By hypothesis,

$$B := \mathrm{Gr}\, T = \prod_{i=1}^{r} (c_i \,\mathrm{id} + \nabla_{A_i}).$$

Commutativity among the A_i is not needed for the estimate. Choose a homogeneous basis of W and the induced monomial ℓ^1-norms. If $d_{\min} > 0$ is the least degree of a nonzero homogeneous vector, a monomial of cohomological degree n has at most $n/d_{\min}$ factors. Since a derivation applies A_i to one factor at a time, there is a constant C_i such that

$$\left\| \nabla_{A_i} \big|_{\mathrm{Sym}(W)^n} \right\| \le C_i(1+n). \tag{6.5}$$

Consequently

$$\left\| B\big|_{\mathrm{Sym}(W)^n} \right\| \le C(1+n)^r. \tag{6.6}$$

Set

$$F_n = \mathrm{Frob}^{-1}\big|_{\mathrm{Sym}(W)^n}, \qquad B_n = B\big|_{\mathrm{Sym}(W)^n}.$$

Every A_i commutes with Frobenius, hence $F_nB_n = B_nF_n$. Over an algebraic closure the pair (F_n, B_n) is simultaneously upper triangular. By multiplicativity of Frobenius eigenvalues, every monomial of total cohomological degree n in $\mathrm{Sym}(W)$ has Frobenius absolute value $q^{n/2}$; hence every eigenvalue of F_n has absolute value $q^{-n/2}$. Triangularize separately on the even and odd subspaces, which are preserved by both operators. Every diagonal entry of B_n is an eigenvalue of B_n and therefore has absolute value at most the operator norm in (6.6). Thus

$$|\mathrm{STr}(F_nB_n)| \le \dim \mathrm{Sym}(W)^n\, C(1+n)^r q^{-n/2}. \tag{6.7}$$

No estimate on the off-diagonal entries of a Jordan block is involved: only the diagonal of the common triangular form contributes to the trace.

The Hilbert function of a finite-dimensional positively graded super vector space has polynomial growth. Indeed, the odd generators contribute a finite exterior factor, and the coefficient sequence of the product of the finitely many even geometric series is polynomially bounded. Hence

$$\dim \mathrm{Sym}(W)^n \le C'(1+n)^M$$

for some C', M. The right side of (6.7) is therefore bounded by $C''(1+n)^{M+r}q^{-n/2}$, whose sum converges. The constants may depend on T and on ι, but not on n. Equation (6.4) proves both absolute convergence of the original trace and equality with the associated-graded supertrace. □

Corollary 6.5 (Trace convergence without a number-field model)**.** *For the full-X'-leg operators considered here, compatible Weil structures and a scalar-plus-derivation associated-graded formula—in particular* (4.20) *or* (5.12)*—suffice to define* vol *as an absolutely convergent complex trace in the convention of Section 2.1. Wang–Wei's Assumption 2.9 and any additional number-field rationality of the pinned descent maps are not required for this convergence statement.*

Proof. All degreewise traces lie in K_ℓ, since the cohomology groups and correspondence maps are K_ℓ-linear. Compatibility of the Weil structures ensures that the one-leg operators commute with Frobenius. Proposition 6.4 proves absolute convergence after applying ι and identifies the sum with the finite-dimensional scalar-plus-derivation model. No other use of a number-field form occurs in the construction or in the trace identity. □

6.3. **The symmetric-algebra identity.**

Lemma 6.6 (Analyticity of the multivariable supertrace)**.** *Let W be a finite-dimensional super vector space, let F be an invertible parity-preserving endomorphism, and let $W = \bigoplus_\varepsilon W[\varepsilon]$ be an F-stable decomposition. For variables $\mathbf{t} = (t_\varepsilon)$ define $T(\mathbf{t})|_{W[\varepsilon]} = t_\varepsilon\,\mathrm{id}$ and the formal series*

$$\Lambda(\mathbf{t}) := \sum_{(N_\varepsilon)} \mathrm{STr}\left(F^{-1}\Big|_{\bigotimes_\varepsilon \mathrm{Sym}^{N_\varepsilon} W[\varepsilon]} \right) \prod_\varepsilon t_\varepsilon^{N_\varepsilon}. \tag{6.8}$$

Then

$$\Lambda(\mathbf{t}) = \mathrm{sdet}\big(1 - F^{-1}T(\mathbf{t}) \mid W\big)^{-1} \tag{6.9}$$

as a formal power series. If every eigenvalue of F^{-1} on W has complex absolute value < 1, then (6.8) *converges absolutely and locally uniformly on a polydisc containing $\mathbf{t} = \mathbf{1}$. The same is true after applying any fixed finite product of the Euler operators $t_\varepsilon\partial_{t_\varepsilon}$.*

Proof. After extending scalars, put F^{-1} into upper-triangular form on each $W[\varepsilon]$ and let $a_{\varepsilon,j}$ denote its diagonal eigenvalues. The trace of every symmetric or exterior power depends only on these diagonal eigenvalues. Since the graded-symmetric algebra is the tensor product of the ordinary symmetric algebra on the even part and the exterior algebra on the odd part, one obtains

$$\Lambda(\mathbf{t}) = \prod_{\varepsilon} \prod_{j \text{ even}} (1 - a_{\varepsilon,j} t_\varepsilon)^{-1} \prod_{\varepsilon} \prod_{j \text{ odd}} (1 - a_{\varepsilon,j} t_\varepsilon),$$

which is exactly (6.9). Let $\rho = \max_{\varepsilon,j} |a_{\varepsilon,j}| < 1$. Because W is finite-dimensional, there are only finitely many variables and eigenvalues. Choose R with $1 < R < \rho^{-1}$; the strict inequality $R > 1$ is what places the evaluation point $\mathbf{t} = \mathbf{1}$ in the interior of the convergence domain. On the closed polydisc $|t_\varepsilon| \le R$, every even geometric series converges absolutely and uniformly, while the odd factors are polynomials. Their finite product therefore gives absolute and locally uniform convergence of the multi-index series. Applying a finite product of Euler operators only multiplies its coefficients by a fixed polynomial in the integers N_ε; the resulting geometric series still converge uniformly on every smaller closed polydisc containing $\mathbf{1}$. □

Lemma 6.7 (Actual-eigenvalue differential identity)**.** *Let W be a finite-dimensional graded super vector space, let F be an invertible degree-preserving endomorphism, and let $A_1, \dots, A_r$ be commuting degree-preserving endomorphisms commuting with F. Write $W = \bigoplus_\varepsilon W[\varepsilon]$ for their simultaneous generalized eigenspaces, with actual generalized eigenvalue ε_i for A_i. For scalars c_i, one has the coefficientwise formal identity*

$$\sum_{(N_\varepsilon)} \mathrm{STr}\left(F^{-1} \prod_{i=1}^{r} (c_i \,\mathrm{id} + \nabla_{A_i}) \middle| \bigotimes_{\varepsilon} \mathrm{Sym}^{N_\varepsilon} W[\varepsilon] \right) \prod_{\varepsilon} t_\varepsilon^{N_\varepsilon}$$

$$= \prod_{i=1}^{r} \left(c_i + \sum_{\varepsilon} \varepsilon_i t_\varepsilon \partial_{t_\varepsilon} \right) \Lambda(\mathbf{t}). \tag{6.10}$$

If every eigenvalue of F^{-1} on W has absolute value < 1, then both sides are analytic near $\mathbf{1}$ and

$$\mathrm{STr}\left(F^{-1} \prod_{i=1}^{r} (c_i \,\mathrm{id} + \nabla_{A_i}) \mid \mathrm{Sym}(W) \right)$$

$$= \prod_{i=1}^{r} \left(c_i + \sum_{\varepsilon} \varepsilon_i t_\varepsilon \partial_{t_\varepsilon} \right) \Lambda(\mathbf{t}) \Big|_{\mathbf{t}=\mathbf{1}}. \tag{6.11}$$

Nilpotent parts of the A_i do not contribute to the supertrace.

Proof. After a finite scalar extension, simultaneously upper-triangularize the commuting family consisting of F and the A_i on each graded piece. On $\bigotimes_\varepsilon \mathrm{Sym}^{N_\varepsilon} W[\varepsilon]$, the diagonal part of ∇_{A_i} is the scalar $\sum_\varepsilon N_\varepsilon \varepsilon_i$. Any summand containing a strictly upper-triangular part is itself strictly upper triangular and has supertrace zero. Applying the Euler operators $t_\varepsilon \partial_{t_\varepsilon}$ to (6.8) therefore proves the coefficientwise identity (6.10). Under the spectral-radius hypothesis, Lemma 6.6 permits termwise differentiation and evaluation at $\mathbf{t} = \mathbf{1}$, giving (6.11). □

For $W = H_{>0}(X, \mathbb{V})$ and $F = \mathrm{Frob}$, Lemma 6.3 shows that every eigenvalue of F^{-1} has absolute value at most $q^{-1/2} < 1$. Hence Lemma 6.6 applies. We now record the normalized Poincaré-duality pairing with all twists displayed. If $V_0^{2d} \cong K_\ell(-d)^{\oplus m_d}$ and $z \in H_j(X')$, then the corresponding positive generator has total degree $n = 2d - j$ and belongs to

$$H_j(X', K_\ell(-d)).$$

Its dual negative-degree summand is $H^j(X', K_\ell(d))$, of total degree $j - 2d = -n$. Curve Poincaré duality and coefficient evaluation give

$$H_j(X', K_\ell(-d)) \otimes H^j(X', K_\ell(d)) \longrightarrow K_\ell.$$

After descent and decomposition into simultaneous generalized eigenspaces this becomes a Frobenius-equivariant perfect pairing

$$W^n[\varepsilon] \otimes H^{-n}(X, \mathbb{V}^\vee)[\varepsilon] \longrightarrow K_\ell. \tag{6.12}$$

If a is a Frobenius eigenvalue on $W^n[\varepsilon]$, then a^{-1} is the Frobenius eigenvalue on the paired negative-degree summand and also the eigenvalue of F^{-1} on $W^n[\varepsilon]$. Parity is preserved, so multiplication over all generalized eigenvalues yields

$$\Lambda((q^{-s_\varepsilon})_\varepsilon) = L_X^*(\mathbf{s}, \mathbb{V}^\vee). \tag{6.13}$$

Since

$$t_\varepsilon \partial_{t_\varepsilon}\big|_{t_\varepsilon = q^{-s_\varepsilon}} = -(\log q)^{-1} \partial_{s_\varepsilon},$$

Lemma 6.7 gives the differential operators (1.4) with the actual generalized eigenvalues of the derivations. In one dimension this sign can be seen without any formalism: if A acts on a generator by ε, then ∇_A acts on its Nth symmetric power by $N\varepsilon$, which is represented by $\varepsilon t\partial_t$ on the generating series. Substituting $t = q^{-s}$ gives precisely $-(\log q)^{-1}\varepsilon\partial_s$.

6.4. **Compact and ordinary iteration.** Let

$$\mathcal{F}_0 \dashrightarrow_{\lambda_1} \mathcal{F}_1 \dashrightarrow_{\lambda_2} \cdots \dashrightarrow_{\lambda_r} \mathcal{F}_r$$

be the geometric order of the iterated Hecke correspondence, and put $\omega_i = \omega + \bar{\lambda}_1 + \cdots + \bar{\lambda}_i$. The compactly supported one-leg operator has direction

$$\Gamma_{c,\lambda_i}^{\omega_{i-1}} : H_c^*(\mathrm{Bun}_G^{\omega_{i-1}}) \longrightarrow H_c^*(\mathrm{Bun}_G^{\omega_i}),$$

whereas its graded dual is the ordinary operator

$$\Gamma_{\lambda_i}^{\omega_{i-1}} : H^*(\mathrm{Bun}_G^{\omega_i}) \longrightarrow H^*(\mathrm{Bun}_G^{\omega_{i-1}}).$$

Consequently the compact and ordinary iterated operators are, respectively,

$$\Gamma_{c,I}^{\omega} = \Gamma_{c,\lambda_r}^{\omega_{r-1}} \circ \cdots \circ \Gamma_{c,\lambda_1}^{\omega_0}, \tag{6.14}$$

$$\Gamma_I^{\omega} = \Gamma_{\lambda_1}^{\omega_0} \circ \cdots \circ \Gamma_{\lambda_r}^{\omega_{r-1}}. \tag{6.15}$$

The rightmost compact factor acts first on $H_c^*(\mathrm{Bun}_G^{\omega_0})$, and the rightmost ordinary factor acts first on $H^*(\mathrm{Bun}_G^{\omega_r})$. Thus (6.15) is exactly the graded dual of (6.14). This is the same compact/ordinary reversal as [WW, (2.24)–(2.25)]; the displayed component superscripts here are derived directly from the geometric chain $\mathcal{F}_0 \dashrightarrow \cdots \dashrightarrow \mathcal{F}_r$. By (2.9), $\omega_r = \omega$.

Lemma 6.8 (Duality normalization)**.** *Let $\mathcal{T}_{c,I}^{\omega}$ be an iterated compactly supported correspondence whose one-leg factors have total relative degree two over the leg curve, and let T_I^{ω} be its graded-dual ordinary operator. This includes both an ordinary degree-two local datum and the sum of such a datum with a point-decorated degree-zero datum. For a Frobenius-stable component and zero total component shift,*

$$\mathrm{vol}(\mathcal{T}_{c,I}^{\omega}) = q^{\dim \mathrm{Bun}_G} \mathrm{STr}\big(\mathrm{Frob}^{-1} \circ T_I^{\omega} \mid H^*(\mathrm{Bun}_G^{\omega})\big)\,. \tag{6.16}$$

Proof. Degreewise, Wang–Wei's construction identifies the ordinary one-leg operator with the graded dual of the compactly supported one, and graded duality reverses the composition, giving (6.14) and (6.15). The Beck–Chevalley maps in Proposition 3.8 were chosen compatibly with Verdier duality, so the same statement holds after representative-leg globalisation.

Feng–Yun–Zhang's Poincaré–Verdier normalization on a component of Bun_G converts the compact Frobenius trace into the ordinary supertrace with geometric Frobenius inverted and

contributes the factor $q^{\dim \mathrm{Bun}_G}$ [FYZ, §7.5.2]. Each one-leg factor has total relative degree two before integration: either it comes from a degree-two local datum or from a degree-zero datum multiplied by a normalized degree-two class on the leg. Integration over the one-dimensional leg therefore makes the absolute operator degree zero, with no further shift or twist. The equality holds in each cohomological degree; Proposition 6.4 permits summation and proves (6.16). □

6.5. **The volume theorem.** The trace calculation depends only on a scalar-plus-derivation description of the one-leg associated graded. We first state this formal principle and then apply it to undecorated and decorated local volume data. The proof is linear-algebraic once an Atiyah–Bott model Sym(W), a Frobenius action, and degreewise trace comparison are available; consequently the same statement applies to the connected-reductive Atiyah–Bott model used by Feng–Yun–Zhang.

Theorem 6.9 (Scalar-plus-derivation master formula)**.** *Let $T_i^{\omega_{i-1}} : H^*(\mathrm{Bun}_G^{\omega_i}) \to H^*(\mathrm{Bun}_G^{\omega_{i-1}})$ be Frobenius-equivariant, filtration-preserving one-leg operators in the geometric order of* (6.15)*. Assume that each T_i is graded-dual to a compactly supported one-leg correspondence of total relative degree two over the leg curve, so that Lemma 6.8 applies. Suppose that, under the componentwise Atiyah–Bott identifications,*

$$\mathrm{Gr}_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,T_i^{\omega_{i-1}} = c_i^{\omega_{i-1}}\,\mathrm{id} + \nabla_{A_i}, \tag{6.17}$$

where the $c_i^{\omega_{i-1}}$ are scalars and the adjoints $A_i^\vee$ commute on $H^{<0}(X, \mathbb{V}^\vee)$. Let $\mathcal{T}_{c,I}^\omega$ denote the iterated compactly supported correspondence graded-dual to $T_I^\omega = T_1 \circ \cdots \circ T_r$. Then

$$\mathrm{vol}(\mathcal{T}_{c,I}^\omega) = q^{\dim \mathrm{Bun}_G} \prod_{i=1}^{r} \left(c_i^{\omega_{i-1}} - (\log q)^{-1} \sum_{\varepsilon} \varepsilon_i \partial_{s_\varepsilon} \right) L_X^*(\mathbf{s}, \mathbb{V}^\vee)\Big|_{\mathbf{s}=0}. \tag{6.18}$$

Proof. By Lemma 6.8, the volume is the ordinary-cohomology supertrace. The hypotheses (6.17) allow Proposition 6.4 to be applied to T_I^ω, so this supertrace is absolutely convergent and may be computed on the doubly associated graded. Substitute (6.17), identify all component associated gradeds by Lemma 2.1, and apply Lemma 6.7 and Equation (6.13). □

Theorem 6.10 (General graph-labelled local-volume-data formula)**.** *For each i, let v_{λ_i} be a degree-two local volume datum with compatible Weil structure, equipped with its graph-labelled pinned orbit. Assume* (2.9) *and* (6.1)*. Then, for every Frobenius-stable component Bun_G^ω, the arithmetic volume satisfies*

$$\mathrm{vol}\big({}^\omega \mathrm{Sht}_{G,I}, v_I^{\mathrm{rep}}\big) = q^{\dim \mathrm{Bun}_G} \prod_{i=1}^{r} \left(\delta_{\lambda_i,v_i}^{\omega_{i-1}} + \kappa(X') b_{\lambda_i,v_i} - (\log q)^{-1} \sum_{\varepsilon} \varepsilon_i \partial_{s_\varepsilon} \right) L_X^*(\mathbf{s}, \mathbb{V}^\vee)\Big|_{\mathbf{s}=0}. \tag{6.19}$$

The product is taken in the composition order of (6.15)*; under* (6.1) *its factors commute.*

Proof. Apply Theorem 6.9 to the compact and ordinary correspondences attached to v_I^{rep}, with $T_i = \Gamma_{\lambda_i,v_i}$, and use Theorem 4.9, for which $c_i^{\omega_{i-1}} = \delta_{\lambda_i,v_i}^{\omega_{i-1}} + \kappa(X') b_{\lambda_i,v_i}$ and $A_i = A_{\lambda_i,v_i}$. □

Corollary 6.11 (Decorated graph-labelled local-volume-data formula)**.** *Let $\mathfrak{v}_i = (v_{\lambda_i}; w_{\lambda_i}, \zeta_i)$ be globally decorated graph-labelled data as in Definition 5.2, with compatible Weil structures, zero total component shift, and commuting graph-transfer adjoints $A_{\lambda_i,v_i}^\vee$. Put*

$$D_i^{\mathfrak{v}} = \delta_{\lambda_i,v_i}^{\omega_{i-1}} + \kappa(X') b_{\lambda_i,v_i} + e_{\lambda_i,\zeta_i,w_i}^{\omega_{i-1}} - (\log q)^{-1} \sum_{\varepsilon} \varepsilon_i \partial_{s_\varepsilon}. \tag{6.20}$$

Then

$$\mathrm{vol}\big({}^\omega \mathrm{Sht}_{G,I}, \mathfrak{v}_I^{\mathrm{rep}}\big) = q^{\dim \mathrm{Bun}_G} \left(\prod_{i=1}^{r} D_i^{\mathfrak{v}} \right) L_X^*(\mathbf{s}, \mathbb{V}^\vee)\Big|_{\mathbf{s}=0}. \tag{6.21}$$

Proof. By Lemma 5.3, the direct-sum compact correspondence is graded-dual to the ordinary decorated operator. Apply Theorem 6.9 and use (5.12). □

6.6. **Recovery of the source theorems.** The source comparisons require two extensions of the standing semisimple, coweight-indexed notation. First, Feng–Yun–Zhang work with connected reductive groups and retain the horizontal degree-two class in $H^2(BG_0)$. Second, their class at a conjugate graph is obtained by transporting the chosen representative class; it need not be invariant under the stabilizer of the coweight. The connected-reductive trace package and the graph-labelled formalism needed for these two points are recorded explicitly below.

Proposition 6.12 (Connected-reductive scalar-plus-derivation trace package)**.** *Work in the quasisplit connected-reductive setup of* [FYZ, §7]*. Let*

$$W_G := H_{>0}(X, \mathbb{V}_G), \qquad M_G := H^{<0}(X, \mathbb{V}_G^\vee),$$

where $\mathbb{V}_G$, the negative-degree normalization, and $L_X^*(\mathbf{s}, \mathbb{V}_G^\vee)$ are those of [FYZ, §7.4.5–§7.4.6]*. Suppose that*

$$T_i^{\omega_{i-1}} : H^*(\mathrm{Bun}_G^{\omega_i}) \longrightarrow H^*(\mathrm{Bun}_G^{\omega_{i-1}})$$

are Frobenius-equivariant, filtration-preserving one-leg operators, graded-dual to compactly supported correspondences of total relative degree two, and that under the componentwise reductive Atiyah–Bott identifications one has

$$\mathrm{Gr}_{\mathrm{aug}}\,\mathrm{Gr}_{\mathrm{Ran}}\,T_i^{\omega_{i-1}} = c_i^{\omega_{i-1}}\,\mathrm{id} + \nabla_{A_i} \qquad \text{on } \mathrm{Sym}(W_G). \tag{6.22}$$

Assume that the adjoints $A_i^\vee$ commute on M_G. After a finite scalar extension, write

$$M_G = \bigoplus_{\varepsilon} M_G[\varepsilon]$$

for their simultaneous generalized eigenspace decomposition, where $A_i^\vee$ has actual generalized eigenvalue ε_i on $M_G[\varepsilon]$. Frobenius equivariance of (6.22) *makes every $M_G[\varepsilon]$ Frobenius-stable, and the variables s_ε in the normalized L-function are indexed by these summands. Then the iterated compact correspondence satisfies*

$$\mathrm{vol}(\mathcal{T}_{c,I}^{\omega}) = q^{\dim \mathrm{Bun}_G} \prod_{i=1}^{r} \left(c_i^{\omega_{i-1}} - (\log q)^{-1} \sum_{\varepsilon} \varepsilon_i \partial_{s_\varepsilon} \right) L_X^*(\mathbf{s}, \mathbb{V}_G^\vee)\Big|_{\mathbf{s}=0}. \tag{6.23}$$

Proof. This proposition is deliberately conditional on the one-leg scalar-plus-derivation calculation; it does not extend Wang–Wei's arbitrary-Schubert local-volume-data theorem from semisimple to reductive groups. Apply Feng–Yun–Zhang's componentwise reductive Atiyah–Bott theorem [FYZ, Theorem 4.2.8]. Their normalization removes from the negative Gross module the split-central summand responsible for the pole at the origin. Thus the normalized positive generator space W_G is finite-dimensional and positively graded, and the normalized negative module M_G is its Poincaré dual.

We spell out convergence because the quasisplit section of FYZ states that it is analogous to the split case and omits the estimate. In cohomological degree n, only finitely many Ran and augmentation degrees occur: a monomial of degree n contains at most $n/d_{\min}$ positive-degree generators. Trace additivity therefore replaces the filtered one-leg operators by the product

$$B_n = \prod_{i=1}^{r} \left(c_i^{\omega_{i-1}}\,\mathrm{id} + \nabla_{A_i} \right) \quad \text{on } \mathrm{Sym}(W_G)^n.$$

The normalized reductive purity statement of FYZ gives Frobenius absolute value $q^{n/2}$ on degree n, hence $q^{-n/2}$ for inverse Frobenius. A fixed derivation has norm $O(1+n)$ on $\mathrm{Sym}(W_G)^n$, and

the Hilbert function of the finite-dimensional positively graded super vector space W_G is polynomially bounded. Since every A_i commutes with Frobenius, simultaneous upper triangularization, separately on even and odd parts, gives

$$\left|\mathrm{STr}\big(\mathrm{Frob}^{-1}\, B_n \mid \mathrm{Sym}(W_G)^n\big)\right| \le C(1+n)^N q^{-n/2}$$

for constants independent of n. Hence the regularized trace converges absolutely. This is the quasisplit reductive version of the estimate proved in [FYZ, Proposition 5.5.3].

Finally, decompose M_G into the simultaneous generalized eigenspaces of the commuting $A_i^\vee$. The analytic superdeterminant argument of Lemmas 6.6 and 6.7 applies verbatim to W_G and its normalized L-function. It sends ∇_{A_i} to $-(\log q)^{-1}\sum_\varepsilon \varepsilon_i \partial_{s_\varepsilon}$ and leaves the scalar $c_i^{\omega_{i-1}}$ unchanged. Poincaré–Verdier duality contributes $q^{\dim \mathrm{Bun}_G}$, giving (6.23). □

Corollary 6.13 (Wang–Wei)**.** *Take $X' = X$, $\Gamma = 1$, and no point decoration. For any sequence of split semisimple degree-two local volume data satisfying Wang–Wei's component-shift, rationality, and commutativity assumptions, Theorem 6.10 recovers the relative-Hecke-operator and symmetric-algebra trace identity underlying* [WW, Theorem 2.13]*; with the actual-eigenvalue convention of* [WW, (2.21)]*, it is the same differential formula. If one inserts the determinant IC data used by Wang–Wei, summing over the split semisimple components gives* [WW, Theorem 1.5]*. In the almost-simple case these determinant data are literally the data used in the determinant application of this article; for a general semisimple product our separate determinant theorem uses the factorwise-normalized datum of* (7.2)*.*

Proof. In the split semisimple case, $\delta_{\lambda_i,v_i}^{\omega_{i-1}} = 0$, $\kappa(X') = 2g(X) - 2$, the graph-transfer is the local operator E_{v_i}, and the multivariable superdeterminant is Wang–Wei's L-function. Thus the factor in (6.19) is

$$(2g-2)b_{v_i} - (\log q)^{-1}\sum_\varepsilon \varepsilon_i \partial_{s_\varepsilon},$$

and the associated-graded operator is $(2g-2)b_{v_i}\,\mathrm{id} + \nabla_{E_{v_i}}$, exactly [WW, Proposition 3.7]. Their number-field hypothesis implies convergence for every chosen complex realization; the convergence theorem here gives the same value after choosing the same embedding. The component sum is $|\pi_1(G)|$ in the split semisimple case. □

Remark 6.14 (The ordered eigenweight symbols in Wang–Wei). Wang–Wei first define the eigenweights in [WW, (2.21)] as the eigenvalues of E_v. Later, in [WW, §2.4.2], their ordered symbols are introduced by writing $E_v|_{V_j} = -\epsilon_{v,j}^{\mathrm{ord}}\,\mathrm{id}$, while the displayed differential operator [WW, (2.28)] retains a minus sign in front of $\epsilon_{v,j}^{\mathrm{ord}}\partial_{s_j}$. These printed conventions are not simultaneously compatible with the associated-graded formula [WW, Proposition 3.7]. The present article uses actual eigenvalues throughout; comparison with the later ordered symbols therefore requires $\epsilon^{\mathrm{ord}} = -\epsilon^{\mathrm{actual}}$. The specialization above is literal at the relative-Hecke-operator and symmetric-algebra-trace levels, which are convention-independent.

Corollary 6.15 (Feng–Yun–Zhang: connected splitting cover)**.** *Work in the quasisplit connected reductive setup of* [FYZ, §7]*, and assume in this corollary that X' is geometrically connected. Let $\mu = (\mu_1, \dots, \mu_r)$ be minuscule and satisfy FYZ's component-shift condition. For each j, let*

$$\eta_j \in H^{2D_{\mu_j}+2}(BL_{\mu_j}), \qquad \eta_j' \in H^{2D_{\mu_j}}(BL_{\mu_j}),$$

and assume [FYZ, Assumption 7.4.9]*. If $D_{\mu_j}^{\eta_j}$ has generalized eigenvalue ε_j on the simultaneous summand indexed by ε, put*

$$\mathscr{D}_j = d_{\mu_j}^{\omega_{j-1}}(\eta_j) + d_{\mu_j}(\eta_j') - (\log q)^{-1}\sum_\varepsilon \varepsilon_j \partial_{s_\varepsilon}. \tag{6.24}$$

Then

$$\mathrm{vol}\big({}^{\omega}\,\mathrm{Sht}^{\mu}_{G},(\eta_1+\xi'\eta'_1,\dots,\eta_r+\xi'\eta'_r)\big)=q^{\dim \mathrm{Bun}_G}\left(\prod_{j=1}^{r}\mathscr{D}_j\right)L_X^*(\mathbf{s},V^*)\Big|_{\mathbf{s}=0}. \tag{6.25}$$

This is [FYZ, Theorem 7.4.12] *in the geometrically connected full-leg case.*

Proof. Because μ_j is minuscule, the local Schubert stack is smooth of relative dimension D_{μ_j} and

$$\mathrm{IC}_{\mu_j}\simeq \mathbf{1}\langle D_{\mu_j}\rangle.$$

Let

$$\pi^{\ell}_{\mu_j}:\mathcal{H}^{\ell}_{\mu_j}\longrightarrow BP_{\mu_j}$$

be the canonical parabolic-reduction map. The global minuscule Hecke stack has its canonical parabolic-reduction map $\mathrm{ev}_{\mu_j}: S_{\mu_j}\to BP_{\mu_j}$, and the local-model construction gives a 2-commutative triangle

$$\begin{array}{ccc} S_{\mu_j} & \xrightarrow{f_{\mathrm{Hk}}} & \mathcal{H}^{\ell}_{\mu_j} \\ {\scriptstyle \mathrm{ev}_{\mu_j}}\big\downarrow & & \big\downarrow{\scriptstyle \pi^{\ell}_{\mu_j}} \\ BP_{\mu_j} & = & BP_{\mu_j}. \end{array}$$

Via $H^*(BP_{\mu_j})\simeq H^*(BL_{\mu_j})$, put

$$\widetilde{\eta}_j=(\pi^{\ell}_{\mu_j})^*\eta_j,\qquad \widetilde{\eta}'_j=(\pi^{\ell}_{\mu_j})^*\eta'_j.$$

Define

$$v_j^{(2)}:=\big(\mathrm{IC}_{\mu_j},\widetilde{\eta}_j\cap[\mathcal{H}^{\ell}_{\mu_j}]_{\mathrm{BM}},[\mathcal{H}^{\ell}_{\mu_j}]\big), \tag{6.26}$$

$$w_j^{(0)}:=\big(\mathrm{IC}_{\mu_j},\widetilde{\eta}'_j\cap[\mathcal{H}^{\ell}_{\mu_j}]_{\mathrm{BM}},[\mathcal{H}^{\ell}_{\mu_j}]\big). \tag{6.27}$$

Their degree pairs are respectively $(D_{\mu_j}+2,-D_{\mu_j})$ and $(D_{\mu_j},-D_{\mu_j})$. After multiplication of the second datum by the point class ξ', (5.1) raises its c-degree by two, so both global summands have the common pair $(D_{\mu_j}+2,-D_{\mu_j})$. They have the same coefficient object and the same fundamental-class d-map. Hence the compressed model of Definition 5.2 has c-map

$$c_{\mathfrak{v}_j}=\Big(\mathrm{ev}^*_{\mu_j}\,\eta_j+p^*\xi'\cup\mathrm{ev}^*_{\mu_j}\,\eta'_j\Big)\cap[S_{\mu_j}]_{\mathrm{BM}} \tag{6.28}$$

and the common fundamental-class d-map. Therefore its ordinary one-leg operator is

$$x\longmapsto h_{0,!}\Big((\mathrm{ev}^*_{\mu_j}\,\eta_j+p^*\xi'\cup\mathrm{ev}^*_{\mu_j}\,\eta'_j)\cup h_1^*x\Big), \tag{6.29}$$

and its compact operator is the adjoint push–pull expression with h_0 and h_1 interchanged. These are exactly the FYZ cohomological correspondences for the class $\eta_j+\xi'\eta'_j$; the identification is made before taking associated gradeds or traces.

No stabilizer-invariance condition is imposed on η_j or η'_j. We use the graph-labelled pinned orbits of Definition 3.3: the datum on D_γ is obtained by pulling (6.26)–(6.27) back by $a_{\gamma^{-1}}$. The explicit c- and d-cocycle diagrams in the proof of Proposition 3.4 show that these graph-labelled local maps are compatible with the exceptional-pullback identifications used in (6.28). Thus, even if $\gamma\mu_j=\mu_j$, the graph label is retained and no invariance under the coweight stabilizer is forced, exactly as in [FYZ, Remark 7.4.3].

Let $\mathfrak{v}_j=(v_j^{(2)};w_j^{(0)},\xi')$. The preceding construction proves at the correspondence level—not only numerically—that $\mathfrak{v}_j$ is FYZ's one-leg datum. The graph-transport argument in Proposition 3.4 then applies to the minuscule reductive multi-graph correspondence. We do not invoke an arbitrary-Schubert reductive extension of Wang–Wei; the required one-leg scalar and derivation are the minuscule calculations already proved by FYZ.

FYZ's one-leg calculation gives

$$\delta^{\omega_{j-1}}_{\mu_j,v_j} + \kappa(X')b_{\mu_j,v_j} = d^{\omega_{j-1}}_{\mu_j}(\eta_j)$$

and identifies the graph-transfer with $D^{\eta_j}_{\mu_j}$; see [FYZ, Definition 5.6.5, Lemma 5.6.13, Proposition 5.6.17, Proposition 7.5.1, and §7.5.2]. The cited part of §7.5.2 is the one-leg associated-graded calculation preceding the final trace step, not an invocation of Theorem 7.4.12. The local unit of $w_j^{(0)}$ is $d_{\mu_j}(\eta'_j)$. Because X' is geometrically connected and ξ' is the class of one point, $\int_{X'} \xi' = 1$, so

$$e^{\omega_{j-1}}_{\mu_j,\xi',w_j} = d_{\mu_j}(\eta'_j).$$

Moreover, apply the cap-with-ξ' argument of Propositions 5.4 and 5.5 to FYZ's minuscule multigraph commutator [FYZ, Proposition 7.5.1]. It proves directly that every non-unit $\xi'\eta'_j$ commutator term lies two Ran steps lower; this step uses no arbitrary-Schubert reductive extension. Hence the one-leg top associated graded is

$$\big(d^{\omega_{j-1}}_{\mu_j}(\eta_j) + d_{\mu_j}(\eta'_j)\big)\,\mathrm{id} + \nabla_{D^{\eta_j}_{\mu_j}}.$$

Apply the connected-reductive scalar-plus-derivation trace package Proposition 6.12. This gives (6.25) term for term, using FYZ's normalized Gross local system and L-function, without invoking their Theorem 7.4.12 in the proof. □

Proposition 6.16 (Disconnected-cover point normalization)**.** *Let $N = |\pi_0(X'_{\bar{k}})|$, and let ξ' restrict to the class of one point on every geometric connected component. For the ordinary total leg pushforward used in Equation* (5.4)*, the FYZ degree-zero datum satisfies*

$$e^{\omega}_{\mu,\xi',w} = N\, d_\mu(\eta'). \tag{6.30}$$

If instead one uses the normalized leg trace

$$\mathrm{pr}^{\mathrm{norm}}_{1,!} := N^{-1}\,\mathrm{pr}_{1,!} \tag{6.31}$$

for the point-decorated summand, then its scalar is $d_\mu(\eta')$. With this explicitly normalized point-decorated pushforward, the scalar in the printed differential operator of [FYZ, Theorem 7.4.12] *is reproduced for a geometrically disconnected cover. This is a normalization statement; it does not assert that FYZ v1 explicitly defines the normalized cohomological correspondence.*

Proof. Ordinary proper pushforward is additive over the open-and-closed geometric components. Since $\int_{X'_a} \xi'|_{X'_a} = 1$ for every component, $\int_{X'} \xi' = N$, proving (6.30). Multiplication by N^{-1} gives the second assertion. The graph commutator, graph transfer, and Ran-filtration argument are unaffected because the normalization changes only the final scalar pushforward of the decorated summand. □

Remark 6.17 (The public disconnected-cover convention in FYZ). The public v1 of [FYZ] allows geometrically disconnected splitting covers and says that ξ' is Poincaré dual to a point in each connected component of X', but it does not spell out how that phrase is to be combined with the geometric-component decomposition and the final leg trace. Under the interpretation used in Proposition 6.16—namely, that ξ' restricts to one point class on every geometric component—ordinary total pushforward gives (6.30), while the scalar printed in their theorem is reproduced by the normalized trace (6.31). The geometrically connected case has $N = 1$ and is unambiguous. We record both normalizations and do not attribute either unresolved disconnected convention to FYZ beyond what is explicitly printed in v1.

Remark 6.18 (What is reconstructed in the FYZ specialization). The point-decoration theorem gives an independent proof that the $\xi'\eta'$ summand contributes only its unit scalar on the top Ran associated graded. The horizontal scalar and the one-bracket derivation are the FYZ one-leg

calculations cited in the proof above. Thus the master theorem reconstructs FYZ Theorem 7.4.12 from their local scalar and derivation inputs, while extending the point-decoration and graph-labelled mechanisms to the general local-volume-data setting.

7. The determinant volume datum

We now apply the general theorem to determinant intersection-complex data. For an almost-simple group this is precisely the local datum of Wang–Wei. For a semisimple product we use the factorwise-normalized rational determinant line specified below. The role of the determinant line is to manufacture a degree-two local correspondence from an arbitrary Schubert closure. The intersection complex controls the singular coefficient theory, the high Chern power supplies the required Borel–Moore degree, and the fundamental class on the opposite side completes the correspondence. Once its degree and pinned equivariance are verified, the general theorem applies without another global calculation.

7.1. **The determinant line and intersection-complex datum.** Let A index the connected components of the pinned Dynkin diagram of G_0. For $a \in A$, let $G_{0,a} \subset G_0$ be the canonical minimal nontrivial smooth connected normal subgroup generated by the root subgroups in that component. The subgroups $G_{0,a}$ pairwise centralize one another, and multiplication is a central isogeny

$$m : \prod_{a\in A} G_{0,a} \longrightarrow G_0. \tag{7.1}$$

This is the split case of the canonical decomposition of a semisimple group into its minimal normal factors [Con, Theorem 5.1.19]. Put

$$\mathfrak{g}_a := \mathrm{Lie}(G_{0,a}).$$

Normality of $G_{0,a}$ makes conjugation by G_0 into a representation

$$\rho_a : G_0 \longrightarrow \mathrm{GL}(\mathfrak{g}_a).$$

No simplicity of the Lie algebra $\mathfrak{g}_a$, and no separability of the central isogeny (7.1), is used. Let $\mathcal{L}_{\rho_a}$ be Wang–Wei's determinant-of-modification line attached to ρ_a, and define

$$\mathcal{L}_{\det} := \bigotimes_{a\in A} \mathcal{L}_{\rho_a}^{1/(2h_a^\vee)} \in \mathrm{Pic}(\mathcal{H}^\ell_{G_0})_{\mathbf{Q}}. \tag{7.2}$$

For an almost-simple group, $G_{0,a} = G_0$, $\rho_a = \mathrm{Ad}$, and this is exactly the rational determinant line of [WW, §4.1.1]. For a product, the tensor product in (7.2) is a single rational line on the full local Hecke stack. Its Chern-class powers therefore include the mixed terms dictated by the tensor product; the datum is not defined as an exterior product of separate almost-simple volume data.

Let $d_\lambda = \langle 2\rho, \lambda\rangle$, and let IC_λ be the intersection complex of the closed local Schubert stack $\mathcal{H}^\ell_{\le\lambda}$, normalized as in [WW, §4.1.2]. Let

$$[\mathcal{H}^\ell_{\le\lambda}]_{\mathrm{BM}} \in \mathrm{Hom}^0\Big(\overleftarrow{h}^\ell_{\lambda,!}\,\mathrm{IC}_\lambda\langle d_\lambda\rangle, \mathbf{1}_{\mathcal{B}_0}\Big)$$

and

$$[\mathcal{H}^\ell_{\le\lambda}] \in \mathrm{Hom}^0\Big(\mathbf{1}_{\mathcal{B}_0}, \overrightarrow{h}^\ell_{\lambda,!}\,\mathrm{IC}_\lambda\langle -d_\lambda\rangle\Big)$$

be the relative Borel–Moore and cohomological fundamental classes. The determinant datum is

$$v_{\lambda,\det} = \Big(\mathrm{IC}_\lambda, c_1(\mathcal{L}_{\det})^{d_\lambda+1}\cdot[\mathcal{H}^\ell_{\le\lambda}]_{\mathrm{BM}}, [\mathcal{H}^\ell_{\le\lambda}]\Big). \tag{7.3}$$

Here $d_c = d_\lambda + 2$ and $d_d = -d_\lambda$, so the datum has degree two. More explicitly, the Borel–Moore fundamental class by itself has $d_c = -d_\lambda$; multiplication by the class $c_1(\mathcal{L}_{\det})^{d_\lambda+1}$ of normalized

degree $2d_\lambda + 2$ changes this to $d_c = d_\lambda + 2$. Adding $d_d = -d_\lambda$ gives total degree 2. For a general semisimple group, (7.3) is the factorwise-normalized determinant datum fixed in this article.

Proposition 7.1 (Semisimple normalization)**.** *The rational line* (7.2) *is canonical and functorial for pinned automorphisms. Under the multiplication central isogeny* (7.1)*, one has*

$$\phi_m^*\mathcal{L}_{\det} = \bigotimes_{a\in A} \mathrm{pr}_a^*\,\mathcal{L}_{\det,G_{0,a}} \qquad \text{in } \mathrm{Pic}\Big(\mathcal{H}^\ell_{\prod_a G_{0,a}}\Big)_{\mathbf{Q}}. \tag{7.4}$$

Consequently (7.3) *is a well-defined degree-two local volume datum in arbitrary characteristic.*

Proof. For a representation $r : G_0 \to \mathrm{GL}(U)$, Wang–Wei's determinant-of-modification construction [WW, §4.1.1] is functorial for group homomorphisms: if $\varphi : H \to G_0$, then

$$\phi_\varphi^*\mathcal{L}_U = \mathcal{L}_{U|_H}. \tag{7.5}$$

Both sides assign to an H-modification the determinant of the relative lattice position after applying U.

Distinct normal factors centralize one another. Hence the restriction of ρ_a along m is the adjoint action of $G_{0,a}$ on $\mathfrak{g}_a$ and is trivial on every $G_{0,b}$, $b \neq a$. Applying (7.5) gives

$$\phi_m^*\mathcal{L}_{\rho_a} = \mathrm{pr}_a^*\,\mathcal{L}_{\mathrm{Ad},G_{0,a}}.$$

Taking the rational powers and tensoring proves (7.4). A pinned automorphism permutes the connected Dynkin components, carries $G_{0,a}$ to the corresponding normal factor, intertwines the representations ρ_a, and preserves $h_a^\vee$; therefore it preserves $\mathcal{L}_{\det}$.

The Chern class of a rational line is a rational cohomology class. The fundamental classes in (7.3) have shifts d_λ and $-d_\lambda$, while the Chern-class power adds degree $2d_\lambda + 2$. Thus $d_c = d_\lambda + 2$, $d_d = -d_\lambda$, and the datum has degree two. The power of the sum $c_1(\mathcal{L}_{\det})$ may contain mixed terms; these are part of the single local datum (7.3). □

7.2. **Pinned equivariance of determinant data.**

Proposition 7.2 (Equivariance of the determinant datum)**.** *The factorwise-normalized determinant IC volume data form a coherently pinned-equivariant family in the Weil category.*

Proof. The pinned automorphism a_γ identifies the Schubert closure of type λ with that of type $\gamma\lambda$, preserves $\langle 2\rho, \lambda\rangle$, and identifies their open Schubert strata. Writing the normalized intersection complex as the intermediate extension of the normalized constant sheaf on the open stratum gives a canonical isomorphism

$$a_\gamma^*\,\mathrm{IC}_{\gamma\lambda} \xrightarrow{\sim} \mathrm{IC}_\lambda\,.$$

Normalize this isomorphism by requiring that its restriction to the open Schubert stratum be the identity on the normalized constant sheaf. For γ, δ, the direct isomorphism attached to $\gamma\delta$ and the composite

$$a_\delta^* a_\gamma^*\,\mathrm{IC}_{\gamma\delta\lambda} \xrightarrow{a_\delta^*\theta_{\gamma,\delta\lambda}} a_\delta^*\,\mathrm{IC}_{\delta\lambda} \xrightarrow{\theta_{\delta,\lambda}} \mathrm{IC}_\lambda$$

have the same restriction to the open stratum. Full faithfulness of intermediate extension therefore identifies them, proving the cocycle (3.10) without choosing a scalar. All constructions are defined over k, so the normalized isomorphisms commute with the canonical Weil structures.

By Proposition 7.1, a pinned automorphism permutes the canonical normal almost-simple factors and carries $\mathcal{L}_{\det}$ to itself. Thus $a_\gamma^* c_1(\mathcal{L}_{\det}) = c_1(\mathcal{L}_{\det})$. The cohomological fundamental class d_λ^ℓ is the element corresponding to 1 under the canonical Hom identification displayed immediately before [WW, Definition 4.1]. The relative Borel–Moore fundamental class entering $c^\ell_{\lambda,\det}$ is functorial under isomorphisms, and cap product with the algebraic class $c_1(\mathcal{L}_{\det})^{d_\lambda+1}$ is likewise functorial. Hence pullback by a_γ carries both correspondence morphisms for $\gamma\lambda$ to those

for λ. The resulting isomorphisms of triples are compatible with the cocycle and with Frobenius, proving coherent pinned equivariance in the Weil category. □

7.3. **The determinant volume formula and specializations.**

Proof of Theorem 1.1. Apply Theorem 6.10 to the factorwise-normalized determinant IC volume data. Their coherent pinned equivariance from Proposition 7.2 induces the graph-labelled pinned orbits required by Theorem 6.10, and their compatible Weil structures are part of the same proposition; Corollary 6.5 and Proposition 6.4 show that no common number-field model is needed. In the almost-simple case their constants and local operators are Wang–Wei's b_λ and E_λ; for a semisimple product they are the invariants of the explicitly specified factorwise-normalized local datum. The descended operator is the graph-transfer of Definition 4.7 and Proposition 4.8, so (6.19) is (1.5). □

Corollary 7.3 (Almost-simple determinant application)**.** *Assume G_0 is almost simple. The local datum in Theorem 1.1 is Wang–Wei's determinant IC datum. If $X' = X$ and $\Gamma = 1$, the associated-graded operator is* [WW, Proposition 3.7]*; the differential formula is written with the actual eigenvalues of $E_\lambda^\vee$.*

Corollary 7.4 (Semisimple-product application)**.** *For a general split semisimple G_0, the same formula holds for the factorwise-normalized determinant datum* (7.3)*, under the commutativity hypothesis* (6.1)*. Its coherent Weil structure is supplied by Proposition 7.2, and Proposition 6.4 gives the trace without a number-field-model hypothesis.*

Appendix A. Foundations for the local-model construction

This appendix states the precise geometric and six-functor inputs used in Section 3 and verifies their hypotheses for the bounded local and global Hecke stacks. It proves the two local-model squares at the level of stacks and gives the mate calculations needed for relative Borel–Moore compatibility. Readers interested only in the geometric computation may treat Propositions 3.8 and 3.10 as the output of the appendix. The reason for the length of the foundations is that ordinary proper Beck–Chevalley supplies the $f_!$ comparison, whereas relative Borel–Moore cohomology also requires a compatible comparison of relative dualizing objects; the latter is the role of the product-duality argument below. The logical order is: stack-level Beauville–Laszlo gluing gives the two Cartesian local-model squares; Laszlo–Olsson supplies the finite-stage adic duality and Künneth identities; Liu–Wang supplies the right-Kan-extended ind-constructible categories and the locally pro-smooth $f_!$-base change; Heyer–Mann supplies only coherent mate and adjunction identities; and Proposition A.8 passes the finite-stage exceptional product-duality exchange to the elementary pro-smooth limit. These inputs give the relative Borel–Moore pullback formulas used in the exact product calculation.

A.1. **Étale stacks and ind-constructible sheaves.** Let Sch denote the category of classical schemes over k, endowed with the étale topology, and let Stack denote the category of étale sheaves of spaces on Sch. A morphism $u : \mathcal{X} \to \mathcal{Y}$ is *representable by algebraic spaces* if, for every scheme $T \to \mathcal{Y}$, the fiber product $T \times_{\mathcal{Y}} \mathcal{X}$ is an algebraic space. A representable morphism is said to be quasi-compact, separated, of finite presentation, smooth, or proper when every such base change has the indicated property. All representable morphisms used below are in fact representable by schemes after passage to the explicit atlases constructed in Section A.4.

For a finite-type k-scheme T, let $\mathrm{Shv}_c(T) := \mathrm{Shv}_c(T_{\bar{k}}, K_\ell)$ be the stable infinity-category of constructible geometric adic étale complexes, and put

$$\mathrm{Shv}(T) = \mathrm{Ind}(\mathrm{Shv}_c(T)).$$

All operations in this appendix are performed after geometric base change to $\bar{k}$. If an object, morphism, or correspondence descends to k, its Weil structure and Frobenius action are recorded separately; they are not built into the mapping spaces defining geometric cohomology. For a qcqs scheme written as an inverse limit $T \simeq \varprojlim_i T_i$ of finite-type schemes with affine transition maps, we use the canonical equivalence

$$\mathrm{Shv}(T) \simeq \mathrm{Ind}\left(\varinjlim_i \mathrm{Shv}_c(T_i)\right), \tag{A.1}$$

where the transition functors are pullbacks. For an étale stack $\mathcal{X}$, define

$$\mathrm{Shv}(\mathcal{X}) := \varprojlim_{(T\to\mathcal{X})\in(\mathrm{Sch}_{\mathrm{qcqs}}/\mathcal{X})^{\mathrm{op}}} \mathrm{Shv}(T), \tag{A.2}$$

again with transition functors given by pullback. This is the right-Kan extension construction of [LW, §4.1.1].

Proposition A.1 (Foundational input used in this article)**.** *We use only the following parts of the right-Kan-extended ind-constructible étale formalism.*

(i) *Liu–Wang's construction supplies the sheaf categories, symmetric monoidal pullback, the bounded extraordinary pushforwards used below, and filtered-open gluing.*

(ii) *For the bounded local Hecke projections and their Cartesian global base changes, which are representable and proper of finite presentation, there are adjunctions*

$$u_! \dashv u^!,$$

proper Beck–Chevalley equivalences, and the projection formula

$$u_!(M \otimes u^*N) \xrightarrow{\sim} u_!M \otimes N. \tag{A.3}$$

No comparison between $u_!$ and u_* is used in the proof.

(iii) *Liu–Wang's locally pro-smooth theorem gives the $u_!$-exchange stated in Proposition A.5. The exceptional exchange needed for relative Borel–Moore cohomology is proved separately in Propositions A.8 and A.14.*

(iv) *Liu–Zheng and Laszlo–Olsson supply finite-stage constructible duality, constructibility of the six operations, smooth purity, and the Künneth formula for dualizing complexes.*

(v) *Heyer–Mann is used only for coherent functoriality of adjunctions, units, counits, horizontal mates, and pasting of mate squares.*

All these constructions commute with extension of scalars from $\mathbf{Q}_\ell$ to K_ℓ. No six-operation assertion for maps outside this list is used in the article.

Proof. The right-Kan-extended categories and filtered-open gluing are constructed in [LW, §4.1.1 and Proposition 4.2]; the proper-map compatibilities used in (ii) are the proper case recorded in [LW, §4.1.1, item (1)], and locally pro-smooth $u_!$-base change is [LW, Proposition 4.10]. Preservation of constructibility by finite-presentation $f_!$ and by $f^!$ is [LZ, Propositions 7.4.1–7.4.2], and the compatibility with the Laszlo–Olsson adic formalism is [LZ, Proposition 7.7.1 and Remark 7.7.2]. Smooth purity and the Künneth identity for dualizing complexes are [LO, Lemma 9.3 and Lemma 11.2]. We use [HM, Proposition 3.2.2] only for coherence of adjoints and mates. The one right-adjointability statement not supplied directly by these citations is the constant-proper-factor comparison proved in Proposition A.8. □

Proposition A.2 (Filtered open gluing)**.** *Let $\mathcal{X} = \bigcup_\beta \mathcal{X}_\beta$ be a filtered union of quasi-compact open substacks. Then extension by zero induces*

$$\operatorname*{colim}_\beta \mathrm{Shv}(\mathcal{X}_\beta) \xrightarrow{\sim} \mathrm{Shv}(\mathcal{X}).$$

Proof. This is [LW, Proposition 4.2]. We use it for the Harder–Narasimhan exhaustions of Bun_G, $\mathrm{Bun}_G \times X'$, and the bounded global Hecke stacks. The global bounded functors are supplied independently by the formalism in Proposition A.1. We use the displayed equivalence only to assemble compatible objects and natural transformations from the quasi-compact opens, and to detect that such a transformation is an equivalence after restriction to every open. No assertion about arbitrary six operations on an arbitrary filtered union is intended. □

A.2. **Pro-smooth morphisms and base change.**

Definition A.3 (Elementary and locally pro-smooth morphisms)**.** A morphism $u : T \to S$ of qcqs schemes is *elementary pro-smooth* if there is a cofiltered presentation

$$T \simeq \varprojlim_i T_i$$

by qcqs schemes such that every $T_i \to S$ is smooth and every transition $T_j \to T_i$ is affine, smooth, and surjective. A morphism of stacks is elementary pro-smooth if it is representable and every base change to a qcqs scheme is elementary pro-smooth. A morphism $g : \mathcal{Y} \to \mathcal{Z}$ is *locally pro-smooth* if there is a pro-smooth surjection $P \to \mathcal{Y}$ from a qcqs scheme such that the composite $P \to \mathcal{Z}$ is also a pro-smooth surjection.

Lemma A.4 (Base change, smooth composition, and conservativity)**.** *Elementary pro-smooth morphisms are stable under base change. If $T \to S$ is elementary pro-smooth and $S \to R$ is smooth, then $T \to R$ is elementary pro-smooth. If $P \to \mathcal{Y}$ is a representable pro-smooth surjection, then pullback* $\mathrm{Shv}(\mathcal{Y}) \to \mathrm{Shv}(P)$ *is conservative.*

Proof. For base change, pull back every term of an elementary pro-smooth presentation; smoothness, affineness, and surjectivity are stable under base change, and inverse limits commute with the resulting fiber products. For the second assertion, if $T \simeq \varprojlim_i T_i$ over S, then every composite $T_i \to S \to R$ is smooth and the same transition maps present $T \to R$ as elementary pro-smooth. We make no general composition claim for two arbitrary pro-smooth inverse systems.

The conservativity assertion is [LW, Lemma 4.7], applied to the representable pro-smooth surjection $P \to \mathcal{Y}$. In the right-Kan-extended description (A.2), a morphism in $\mathrm{Shv}(\mathcal{Y})$ is therefore an equivalence as soon as its pullback to P is an equivalence. □

Proposition A.5 (Locally pro-smooth base change)**.** *Consider a Cartesian square*

$$\begin{array}{ccc} \mathcal{W} & \xrightarrow{g'} & \mathcal{X} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{Y} & \xrightarrow{g} & \mathcal{Z}. \end{array}$$

Assume that f is representable of finite presentation and that there is a pro-smooth surjection $P \to \mathcal{Y}$ from a qcqs scheme such that the composite $P \to \mathcal{Z}$ is also a pro-smooth surjection. Then the Beck–Chevalley transformation

$$f'_!(g')^* \xrightarrow{\sim} g^* f_! \tag{A.4}$$

is an equivalence. Its horizontal right mate is a canonical exceptional exchange transformation

$$(g')^* f^! \longrightarrow f'^! g^*. \tag{A.5}$$

The theorem of Liu–Wang asserts the first equivalence; invertibility of the mate is a separate right-adjointability condition and will be verified for the local-model squares in Proposition A.14.

Proof. The canonical Beck–Chevalley equivalence of [LW, Proposition 4.10] is $g^* f_! \xrightarrow{\sim} f'_!(g')^*$. The displayed map (A.4) is its inverse. Taking the horizontal right mate of that inverse under $f_! \dashv f^!$ and $f'_! \dashv f'^!$ gives (A.5); functoriality of mates is [HM, Proposition 3.2.2]. We do not infer that this mate is invertible merely from invertibility of (A.4). □

The proof of the exceptional product-duality exchange has three steps. We first record that finite-stage constructible objects generate the sheaf category on an elementary pro-smooth limit and that their mapping spaces are computed at a sufficiently large finite stage. We next identify the finite-stage exceptional exchange with the usual dualizing-complex Künneth isomorphism. Finally, we test the global canonical mate on the compact stage generators and pass the finite-stage equivalences to the inverse limit.

Lemma A.6 (Compact stage generation and mapping continuity)**.** *Let*

$$P \simeq \varprojlim_n P_n$$

be an elementary pro-smooth presentation as in Proposition A.8, let Z be a finite-type k-scheme, and write $r'_n : P \times Z \to P_n \times Z$. Then:

(i) *smooth pullback along each transition $P_m \times Z \to P_n \times Z$ preserves constructible objects;*

(ii) *the objects*

$$(r'_n)^* F_n, \qquad F_n \in \mathrm{Shv}_c(P_n \times Z),$$

are compact and form a set of generators of $\mathrm{Shv}(P \times Z)$;

(iii) *for $F_n \in \mathrm{Shv}_c(P_n \times Z)$ and $G_a \in \mathrm{Shv}_c(P_a \times Z)$, one has a natural equivalence*

$$\mathrm{Map}_{P\times Z}\big((r'_n)^* F_n, (r'_a)^* G_a\big) \simeq \varinjlim_{m\geq n,a} \mathrm{Map}_{P_m\times Z}\big((r'_{mn})^* F_n, (r'_{ma})^* G_a\big); \tag{A.6}$$

(iv) *if Z is proper over k and $\pi_n : P_n \times Z \to P_n$, then $\pi_{n,!}$ carries constructible objects to constructible objects.*

The analogous assertions hold with $Z = *$. The compact subcategory of the Ind-completion can be larger than the literal union of the stage categories by idempotent completion; only the stated compact generating family is used below.

Proof. Smooth pullback preserves constructibility in the finite-stage adic formalism. Put

$$\mathcal{C}_0 := \varinjlim_n \mathrm{Shv}_c(P_n \times Z).$$

By (A.1), $\mathrm{Shv}(P \times Z) \simeq \mathrm{Ind}(\mathcal{C}_0)$. The Yoneda image of every object of $\mathcal{C}_0$ is compact in the Ind-completion, and those images generate under colimits. This proves (ii).

We prove (iii) directly. Filtered colimits in Cat_∞ are left exact, and the maximal-subgroupoid functor $(-)^{\simeq} : \mathrm{Cat}_\infty \to \mathcal{S}$ preserves filtered colimits; see [LurGC, Remark 1.4.15]. Moreover $\mathrm{Fun}(\Delta^1, -)$ preserves filtered colimits because Δ^1 is compact in Cat_∞. The mapping space between two objects is the fiber, in spaces, of

$$\mathrm{Fun}(\Delta^1, \mathcal{C}_0)^{\simeq} \longrightarrow (\mathcal{C}_0^{\simeq})^2$$

over the corresponding ordered pair of objects. Hence filtered colimits commute with these mapping-space fibers. For objects represented at stages n and a, this gives

$$\mathrm{Map}_{\mathcal{C}_0}(F_n, G_a) \simeq \varinjlim_{m\geq n,a} \mathrm{Map}_{\mathrm{Shv}_c(P_m\times Z)}\big((r'_{mn})^* F_n, (r'_{ma})^* G_a\big),$$

which is also seen concretely by representing every finite simplex at one sufficiently large common stage. The fully faithful Yoneda embedding $\mathcal{C}_0 \hookrightarrow \mathrm{Ind}(\mathcal{C}_0)$ preserves these mapping spaces, proving

(A.6). Finally, constructibility of proper finite-presentation pushforward follows from [LZ, Proposition 7.4.1] (equivalently, from the Laszlo–Olsson finiteness theorem, cf. [LO, Proposition 8.2]). The proof for $Z = *$ is identical. □

Lemma A.7 (Finite-stage product duality and naturality)**.** *Let S be a smooth finite-type k-scheme and let Z be a proper finite-type k-scheme. Write*

$$p_S : S \to *, \qquad q : Z \to *, \qquad \pi_S : S \times Z \to S, \qquad \rho_S : S \times Z \to Z.$$

Then $q^!\mathbf{1}_*$ is constructible and the canonical horizontal right mate

$$\vartheta_S : \rho_S^* q^!\mathbf{1}_* \xrightarrow{\sim} \pi_S^!\mathbf{1}_S \tag{A.7}$$

is an equivalence. For every smooth morphism $u : S' \to S$, with $u' = u \times \mathrm{id}_Z$, the square

$$\begin{array}{ccc} (u')^*\rho_S^* q^!\mathbf{1}_* & \xrightarrow{(u')^*\vartheta_S} & (u')^*\pi_S^!\mathbf{1}_S \\ \cong\downarrow & & \downarrow\cong \\ \rho_{S'}^* q^!\mathbf{1}_* & \xrightarrow{\vartheta_{S'}} & \pi_{S'}^!\mathbf{1}_{S'} \end{array} \tag{A.8}$$

commutes, where the right vertical arrow is the smooth exceptional-pullback exchange.

Proof. Define ϑ_S in the enhanced Liu–Zheng category as the horizontal right mate of the inverse proper Beck–Chevalley equivalence for the product square. Proposition 7.7.1 and Remark 7.7.2 of [LZ] identify the enhanced adic operations, adjunctions, and base-change transformations with the Laszlo–Olsson formalism after passage to homotopy categories. Hence it suffices to show that the image of ϑ_S is an isomorphism there.

Put $\omega_S = p_S^!\mathbf{1}_*$ and $\omega_Z = q^!\mathbf{1}_*$. Constructibility of $q^!\mathbf{1}_*$ follows from [LZ, Proposition 7.4.2] and [LO, §9.1]. Since S is smooth, ω_S is invertible by [LO, Lemma 9.3]. The Künneth formula for dualizing complexes gives

$$\omega_{S\times Z} \simeq \pi_S^*\omega_S \otimes \rho_S^*\omega_Z$$

by [LO, Lemma 11.2]. Using $f^! = D_X f^* D_Y$ from [LO, §9.1], one obtains

$$\begin{aligned} \pi_S^!\mathbf{1}_S &= \mathrm{RHom}\big(\pi_S^*\omega_S, \omega_{S\times Z}\big) \\ &\simeq \mathrm{RHom}\big(\pi_S^*\omega_S, \pi_S^*\omega_S \otimes \rho_S^*\omega_Z\big) \\ &\simeq \rho_S^*\omega_Z. \end{aligned}$$

Under $\pi_{S,!} \dashv \pi_S^!$, the inverse of this isomorphism corresponds to

$$\pi_{S,!}\rho_S^* q^!\mathbf{1}_* \xrightarrow[\text{BC+PF}]{\sim} \mathbf{1}_S \otimes q_! q^!\mathbf{1}_* \xrightarrow{\mathrm{id}\otimes\epsilon_q} \mathbf{1}_S,$$

where the first arrow is proper Beck–Chevalley for the product square followed by the projection formula. This is exactly the counit composite defining the image of the canonical horizontal mate ϑ_S. Therefore ϑ_S is an equivalence in the enhanced category.

The transformation ϑ_S was defined before passing to homotopy categories. Naturality under smooth u is consequently the pasting identity for horizontal right mates, using the smooth exceptional-pullback exchange of [LZ, Proposition 7.2.3] for the right vertical arrow and the functoriality of mates in [HM, Proposition 3.2.2]. It gives the commutative square (A.8). □

Proposition A.8 (Relative duality for a constant proper factor)**.** *Let*

$$P \simeq \varprojlim_n P_n$$

be a qcqs elementary pro-smooth k-scheme, where every P_n is a smooth finite-type k-scheme and every transition $r_{mn} : P_m \to P_n$ is smooth, affine, and surjective. Let Z be a proper finite-type k-scheme, and write

$$q : Z \to *, \qquad \pi : P \times Z \to P, \qquad \rho : P \times Z \to Z.$$

Then the horizontal right mate of proper base change is an equivalence

$$\rho^* q^! \mathbf{1}_* \xrightarrow{\sim} \pi^! \mathbf{1}_P. \tag{A.9}$$

Proof. Let $r_n : P \to P_n$ and $r'_n : P \times Z \to P_n \times Z$ be the projections, put $r'_{mn} = r_{mn} \times \mathrm{id}_Z$, and let π_n and ρ_n denote the finite-stage product projections. By Lemma A.6, the objects

$$F = (r'_n)^* F_n, \qquad F_n \in \mathrm{Shv}_c(P_n \times Z),$$

form compact generators. The finite-stage mates

$$\vartheta_m : \rho_m^* q^! \mathbf{1}_* \xrightarrow{\sim} \pi_m^! \mathbf{1}_{P_m}$$

are compatible with every transition r_{mn} by the commutative square (A.8). Moreover $q^! \mathbf{1}_*$ and $\pi_{n,!} F_n$ are constructible by Lemmas A.6 and A.7.

For a compact generator F, mapping out of F and using the compatible finite-stage mates gives

$$\begin{aligned}
\mathrm{Map}_{P\times Z}\Big(F, \rho^* q^! \mathbf{1}_*\Big) &\simeq \varinjlim_{m \geq n} \mathrm{Map}_{P_m \times Z}\Big((r'_{mn})^* F_n, \rho_m^* q^! \mathbf{1}_*\Big) \\
&\xrightarrow[\vartheta_m]{\sim} \varinjlim_{m \geq n} \mathrm{Map}_{P_m}\big(\pi_{m,!}(r'_{mn})^* F_n, \mathbf{1}_{P_m}\big) \\
&\simeq \varinjlim_{m \geq n} \mathrm{Map}_{P_m}(r_{mn}^* \pi_{n,!} F_n, \mathbf{1}_{P_m}) \\
&\simeq \mathrm{Map}_P(r_n^* \pi_{n,!} F_n, \mathbf{1}_P) \\
&\simeq \mathrm{Map}_P(\pi_! F, \mathbf{1}_P) \\
&\simeq \mathrm{Map}_{P\times Z}\Big(F, \pi^! \mathbf{1}_P\Big).
\end{aligned} \tag{A.10}$$

The third line is elementary pro-smooth base change for $\pi_!$ at finite stages, and the fifth is the same theorem for $r_n : P \to P_n$; see [LW, Lemma 4.8(1)]. The first and fourth lines are Lemma A.6. Because the squares (A.8) commute, the displayed equivalence of mapping spaces is induced by the global canonical mate (A.9), not merely by an abstract objectwise identification. The compact generators detect equivalences, so Yoneda proves the proposition. □

A.3. **Jet groups and the classifying stack.** We now prove Lemma 3.5. Put $A_n = k[t]/(t^{n+1})$. For a k-algebra R,

$$L_n^+ G_0(R) = G_0(R[t]/(t^{n+1})),$$

and $\mathrm{Aut}_n(\mathcal{D})(R)$ is the group of substitutions

$$t \longmapsto a_1 t + a_2 t^2 + \cdots + a_n t^n, \qquad a_1 \in R^\times.$$

Substitution acts on $L_n^+ G_0$, and $H_n = L_n^+ G_0 \rtimes \mathrm{Aut}_n(\mathcal{D})$.

Proof of Lemma 3.5. Because A_n/k is finite locally free and $G_0 \times_k A_n$ is smooth affine over A_n, its Weil restriction $L_n^+ G_0$ is a smooth affine k-group. The displayed coordinates identify $\mathrm{Aut}_n(\mathcal{D})$ with $\mathbf{G}_m \times \mathbf{A}^{n-1}$ as a scheme, with a polynomial group law; it is therefore smooth and affine. Hence H_n is smooth and affine.

The kernel of $\mathrm{Aut}_{n+1}(\mathcal{D}) \to \mathrm{Aut}_n(\mathcal{D})$ consists of substitutions $t \mapsto t + a t^{n+1}$ after a chosen lift at level n; it is a torsor under $\mathbf{G}_a$. The kernel of $L_{n+1}^+ G_0 \to L_n^+ G_0$ is, after fpqc localization and translation to the identity, the vector group $\mathrm{Lie}(G_0) \otimes_k (t^{n+1})/(t^{n+2})$. Smoothness of G_0 therefore

makes both transition maps smooth and surjective; they are affine because the groups are affine. The same holds for $H_{n+1} \to H_n$.

For every R, compatibility of truncated points gives

$$\varprojlim_n L_n^+ G_0(R) = G_0(R[[t]]), \qquad \varprojlim_n \mathrm{Aut}_n(\mathcal{D})(R) = \mathrm{Aut}_R(R[[t]],(t)).$$

Thus $H \simeq \varprojlim_n H_n$, so $H \to *$ is elementary pro-smooth. This is the semidirect-product combination of the positive-loop and formal-coordinate examples in [LW, Examples 4.5–4.6].

We verify the classifying-stack atlas on an arbitrary test scheme. Let $T \to BH$ classify an H-torsor $Q \to T$, and put $K_n = \ker(H \to H_n)$. The contracted product

$$Q_n := Q \times^H H_n = Q/K_n$$

is represented by an affine H_n-torsor over T: fpqc-locally on T it is $H_n \times T$, and representability and affineness descend. The transition $Q_{n+1} \to Q_n$ is a torsor under $\ker(H_{n+1} \to H_n)$, hence is smooth, affine, and surjective. The canonical map

$$Q \longrightarrow \varprojlim_n Q_n$$

is an isomorphism of fpqc sheaves, since it becomes the evident isomorphism $H \times T \simeq \varprojlim_n (H_n \times T)$ after an fpqc trivialization of Q. Therefore $Q \to T$ is elementary pro-smooth. Since this holds after every base change $T \to BH$, the atlas $* \to BH$ is representable elementary pro-smooth; it is surjective by the definition of BH. □

A.4. **Coordinate-and-trivialization torsors.** We next prove Lemma 3.6. The purpose of the construction is not to choose coordinates globally. Rather, it produces a qcqs scheme on which the local-model map becomes an honest map to the Schubert variety and on which Liu–Wang's pro-smooth base-change theorem can be applied. Passing through all jet levels records exactly the data of a formal coordinate and a formal bundle trivialization.

Let $Y = \mathrm{Bun}_G \times X'$. Choose a filtered exhaustion $Y = \bigcup_\beta Y_\beta$ by finite-type qcqs open substacks, discarding empty terms. As explained in Lemma 3.6, Y is smooth over k, so we choose a smooth surjective atlas $U_\beta \to Y_\beta$ with U_β a nonempty smooth finite-type k-scheme. Let $D_\beta \subset X' \times U_\beta$ be the graph of the selected point, let I_β be its ideal, and put

$$D_{\beta,n} := \mathrm{Spec}_{X' \times U_\beta}(\mathcal{O}/I_\beta^{n+1}).$$

Definition A.9. The functor $\mathrm{Coor}_{\beta,n}$ assigns to $T \to U_\beta$ the set of isomorphisms of pointed T-schemes

$$D_{\beta,n} \times_{U_\beta} T \xrightarrow{\sim} \mathrm{Spec}_T(\mathcal{O}_T[t]/(t^{n+1})).$$

After a coordinate is chosen, the functor $\mathrm{Triv}_{\beta,n}$ assigns a trivialization of the restricted G_0-bundle on $D_{\beta,n}$. Their combined functor is denoted $P_{\beta,n}$.

Lemma A.10 (Lifting coordinates and trivializations)**.** *For every $n \geq 1$, the restriction morphism*

$$P_{\beta,n+1} \longrightarrow P_{\beta,n}$$

is an fpqc-locally nonempty torsor under $\ker(H_{n+1} \to H_n)$*. In particular it is smooth, affine, and surjective.*

Proof. Fix a test scheme $T \to U_\beta$ and an n-jet coordinate together with an n-th order trivialization. Étale-locally on T, smoothness of the relative curve supplies a formal parameter u along the graph. The given n-jet coordinate differs from the truncation of u by an element of $\mathrm{Aut}_n(\mathcal{D})$. Since $\mathrm{Aut}_{n+1}(\mathcal{D}) \to \mathrm{Aut}_n(\mathcal{D})$ is smooth and surjective, this element lifts fpqc-locally, and composing with the truncation of u gives an $n+1$-jet coordinate. Two such lifts differ uniquely by $\ker(\mathrm{Aut}_{n+1}(\mathcal{D}) \to \mathrm{Aut}_n(\mathcal{D}))$.

After a coordinate lift is fixed, the functor of extensions of the given bundle trivialization across the square-zero immersion $D_{\beta,n} \hookrightarrow D_{\beta,n+1}$ is the fiber of the smooth isomorphism torsor under G_0. Smoothness of G_0 makes this fiber fpqc-locally nonempty, and two lifts differ uniquely by an element of $\ker(L^+_{n+1}G_0 \to L^+_nG_0)$. Coordinate change acts on the loop-group kernel by substitution, so the combined set of lifts is a simply transitive torsor under the semidirect-product kernel $\ker(H_{n+1} \to H_n)$. The latter is smooth affine by Lemma 3.5, proving the final assertion. □

Proof of Lemma 3.6. The graph of a section of the smooth relative curve $X' \times U_\beta \to U_\beta$ is an effective relative Cartier divisor. Étale-locally on U_β its completed local ring is $\mathcal{O}_{U_\beta}[[t]]$. Hence $\mathrm{Coor}_{\beta,n}$ is an $\mathrm{Aut}_n(\mathcal{D})$-torsor. After choosing a coordinate, the restriction of the pulled-back G_0-bundle to the finite flat scheme $D_{\beta,n}$ is fpqc locally trivial, and the functor of its trivializations is an $L^+_nG_0$-torsor. Coordinate change acts by substitution, so the combined functor is an H_n-torsor.

An H_n-torsor over a scheme is represented by an affine scheme: fpqc locally it is $H_n \times U_\beta$, and affineness descends. It is smooth and surjective because H_n is smooth. By Lemma A.10, the restriction $P_{\beta,n+1} \to P_{\beta,n}$ is a torsor under the smooth affine kernel of $H_{n+1} \to H_n$, and is therefore smooth, affine, and surjective.

Since the transition maps are affine, the inverse limit $P_\beta = \varprojlim_n P_{\beta,n}$ exists as an affine scheme over U_β. It is qcqs because U_β is qcqs. A T-point of P_β is precisely a compatible system of truncated coordinates and truncated trivializations, hence a formal coordinate and a formal trivialization. The map $U_\beta \to BH$ classifies the H-torsor of such choices. Consequently the preceding moduli description gives a canonical equivalence

$$P_\beta \xrightarrow{\sim} U_\beta \times_{BH} *.$$

This proves effectivity directly; no general assertion about inverse limits of nonabelian torsors is being used.

For $P_\beta \to Y_\beta$ no general composition theorem is needed: the same inverse system presents it, because every composite

$$P_{\beta,n} \longrightarrow U_\beta \longrightarrow Y_\beta$$

is smooth and the transition maps remain smooth, affine, and surjective. Hence $P_\beta \to Y_\beta$ is an elementary pro-smooth surjection.

We verify the second covering condition directly after arbitrary base change. Let $T \to BH$ classify an H-torsor $Q \to T$, and let $Q_m = Q \times^H H_m$ be the finite-level torsors constructed in the proof of Lemma 3.5. Because a point of P_β includes a full trivialization, the map $P_\beta \to BH$ is canonically 2-isomorphic to the composite $P_\beta \to * \to BH$. Since $* \times_{BH} T$ is the torsor Q, there is a canonical identification

$$P_\beta \times_{BH} T \simeq P_\beta \times_k Q.$$

The right side is the inverse limit, over the cofiltered product of the two indexing sets, of

$$P_{\beta,n} \times_k Q_m \longrightarrow T.$$

Each such map is smooth: it factors through the smooth map $P_{\beta,n} \times Q_m \to Q_m$ and the smooth H_m-torsor $Q_m \to T$. It is surjective because $P_{\beta,n} \to U_\beta$ is a torsor over the nonempty scheme U_β, hence $P_{\beta,n} \to \operatorname{Spec} k$ is surjective, and $Q_m \to T$ is surjective. All transition maps are smooth, affine, and surjective. Thus $P_\beta \times_{BH} T \to T$ is elementary pro-smooth and surjective. This proves directly that $P_\beta \to BH$ is a pro-smooth surjection, without using closure of arbitrary pro-smooth morphisms under composition. The same assertions persist after base change to either bounded global Hecke projection. □

A.5. **The bounded local quotient stack.** We prove Lemma 3.7. Put

$$Z_\lambda = \mathcal{H}^\ell_{\leq\lambda} \times_{BH, \overleftarrow{h}^\ell_\lambda} *.$$

Target trivialization identifies Z_λ with the projective Schubert variety $\overline{\mathrm{Gr}}_\lambda$.

Proposition A.11. *There is a canonical equivalence*

$$\mathcal{H}^\ell_{\leq\lambda} \simeq [Z_\lambda/H].$$

The two local projections $\overleftarrow{h}^\ell_\lambda, \overrightarrow{h}^\ell_\lambda : \mathcal{H}^\ell_{\leq\lambda} \to BH$ *are representable and proper of finite presentation.*

Proof. Fpqc descent along $* \to BH$ gives the quotient description. Explicitly, a T-point of $[Z_\lambda/H]$ is an H-torsor $Q \to T$ together with an H-equivariant map $Q \to Z_\lambda$, equivalently a bounded local modification with its target formal bundle but without a chosen coordinate or trivialization.

For a morphism $T \to BH$ classified by Q, the base change of the target projection is

$$Q \times^H Z_\lambda \longrightarrow T.$$

It is fpqc-locally $Z_\lambda \times T \to T$, hence representable and proper of finite presentation. These properties descend fpqc-locally. Put

$$\lambda^\vee = -w_0\lambda.$$

Inversion of a local modification exchanges source and target and identifies the source-trivialized fiber of the type-λ correspondence with the projective Schubert variety $\overline{\mathrm{Gr}}_{\lambda^\vee}$. Hence the same argument proves representability and properness of finite presentation for $\overrightarrow{h}^\ell_\lambda$. □

The stack BH is covered by the qcqs pro-smooth atlas $* \to BH$, while the global stacks are treated by their filtered qc-open presentations and Proposition A.2. The explicit atlases of Section A.4 verify the stronger two-cover hypothesis required by Proposition A.5.

A.6. **The two local-model squares.** We give the stack-level proof of the Cartesian assertion in Proposition 3.8. Let T be a test scheme, let $x' \in X'(T)$, and write $D_\gamma \subset X' \times T$ for the graph of $\gamma x'$. Since $X' \to X$ is a Galois torsor, the D_γ are pairwise disjoint relative Cartier divisors.

We first record the affine-section and principal-bundle forms of the gluing theorem that are used below.

Lemma A.12 (Beauville–Laszlo equalizer for affine targets)**.** *Let C be a scheme, let $D \subset C$ be an effective Cartier divisor, and put $U = C \setminus D$. Write $\widehat{C}_D$ for the formal completion and $\widehat{U}_D = \widehat{C}_D \setminus D$ for the punctured completion. For every affine C-scheme Y, restriction induces an equivalence*

$$\mathrm{Hom}_C(C, Y) \xrightarrow{\sim} \mathrm{Hom}_U(U, Y) \times_{\mathrm{Hom}_{\widehat{U}_D}(\widehat{U}_D, Y)} \mathrm{Hom}_{\widehat{C}_D}(\widehat{C}_D, Y). \tag{A.11}$$

The equivalence is functorial in (C, D, Y). It is compatible with the base changes used below, for which D remains an effective Cartier divisor and the infinitesimal neighborhoods of D commute with base change.

Proof. The statement is Zariski-local on C. Write $C = \operatorname{Spec} A$, $D = V(f)$, and $Y = \operatorname{Spec} B$ with B an A-algebra. Since D is effective Cartier, f is a nonzerodivisor. Multiplication by f is also injective on $\widehat{A}$: if $fx = 0$ in $\widehat{A} = \varprojlim_n A/f^n$, then at level $n+1$ regularity of f forces a representative of x to lie in $f^n A$, so its image in A/f^n is zero. Thus both $A[f^\infty]$ and $\widehat{A}[f^\infty]$ vanish. By [Sta, Tag 0BNR], (A, f) is a Beauville–Laszlo gluing pair and the ring square is Cartesian,

$$A \xrightarrow{\sim} A_f \times_{\widehat{A}_f} \widehat{A}. \tag{A.12}$$

Since $\mathrm{Hom}_{A\text{-alg}}(B,-)$ preserves limits, (A.12) gives

$$\mathrm{Hom}_{A\text{-alg}}(B,A) \xrightarrow{\sim} \mathrm{Hom}_{A\text{-alg}}(B,A_f) \times_{\mathrm{Hom}_{A\text{-alg}}(B,\widehat{A}_f)} \mathrm{Hom}_{A\text{-alg}}(B,\widehat{A}),$$

which is (A.11). These local equivalences agree on overlaps by uniqueness and therefore glue. The construction is functorial in the gluing square. In the applications below the divisors are graphs in a smooth relative curve; their infinitesimal neighborhoods are finite locally free over the base and commute with the base changes under consideration. This gives the final assertion. □

Proposition A.13 (Multi-disc Beauville–Laszlo gluing for torsors)**.** *Let $C \to T$ be a relative smooth curve, and let $D = \coprod_i D_i$ be a finite disjoint union of graphs of sections of C/T; put $U = C \setminus D$. For the smooth affine reductive group G_0, restriction is an equivalence between*

(i) *G_0-torsors on C, and*
(ii) *a G_0-torsor on U, a G_0-torsor on each formal completion $\widehat{C}_{D_i}$, and compatible identifications on the punctured completions.*

The equivalence is fully faithful on morphisms and is compatible with the base changes in T for which the divisors remain effective Cartier and their infinitesimal thickenings commute with base change. In particular, this holds for the graph divisors used in this article. The same proof applies more generally to finite locally free relative effective Cartier divisors with this base-change property.

Proof. We first treat one divisor. Choose a faithful closed immersion $G_0 \hookrightarrow \mathrm{GL}_N$. The arbitrary-characteristic Matsushima theorem implies that GL_N/G_0 is affine because both groups are reductive [Alp, Theorem 9.4.1 and Corollary 9.7.7]. Beauville–Laszlo gluing for vector bundles [BL] reconstructs the associated rank-N vector bundle, hence its frame GL_N-torsor P.

A reduction of P to G_0 is a section of

$$P \times^{\mathrm{GL}_N} (\mathrm{GL}_N/G_0) \longrightarrow C.$$

This associated bundle is affine over C, because affineness is fpqc-local on the base. The reductions given on U and $\widehat{C}_D$ agree over $\widehat{U}_D$ and therefore glue uniquely by Lemma A.12. The resulting reduction is a G_0-torsor, a property which is fpqc-local. For two glued torsors P_1, P_2, the sheaf $\underline{\mathrm{Isom}}_{G_0}(P_1,P_2)$ is affine over C; applying Lemma A.12 to it glues morphisms and proves full faithfulness. For a graph divisor in a smooth relative curve, every infinitesimal thickening is finite locally free over T and its formation commutes with the base changes used in the local-model argument. Hence every construction above is functorial for those base changes. This also agrees with the stack-theoretic formal-gluing theorem [HPV, Theorem 5.3 and Remark 5.4]; in the finite-étale divisor setting, compare [Fed, Proposition 4.4].

For pairwise disjoint D_i, apply the one-divisor equivalence successively. Completion along one D_i leaves the complement and formal data at every D_j, $j \neq i$, unchanged. Uniqueness in the fully faithful gluing equivalence shows that the result is independent of the order. This proves the multi-disc statement, including compatibility with morphisms and base change. □

Stack-level proof of Proposition 3.8. Put $Y = \mathrm{Bun}_G \times X'$. The target square defines a canonical map

$$\Phi_1 : S_\lambda \longrightarrow Y \times_{BH,\overleftarrow{h}^{\ell}_{\lambda}} \mathcal{H}^{\ell}_{\leq\lambda}.$$

An object of the target fiber product over T consists of a target G-bundle $\mathcal{F}_1$ on $X \times T$, a selected point x', a bounded local modification at the representative formal disc, and an identification of its target formal bundle with the formal restriction of $\mathcal{F}_1$.

Pull $\mathcal{F}_1$ to a G_0-bundle $\mathcal{F}_1'$ on $X' \times T$. The descent presentation of G supplies semilinear maps

$$\psi_{\delta,1} : \delta^*\mathcal{F}_1' \xrightarrow{\sim} a_\delta(\mathcal{F}_1')$$

satisfying the cocycle

$$\psi_{\delta_1\delta_2,1} = a_{\delta_1}(\psi_{\delta_2,1}) \circ \delta_2^*(\psi_{\delta_1,1}).$$

Let $\mathcal{E}_{0,1}$ be the source lattice in the representative local modification. On the completion along D_γ define

$$\mathcal{E}_{0,\gamma} := a_\gamma\big((\gamma^{-1})^*\mathcal{E}_{0,1}\big). \tag{A.13}$$

It has relative position $\leq \gamma\lambda$. For $\delta, \gamma \in \Gamma$, deck pullback commutes with the constant pinned automorphisms, and

$$\delta^*(\delta\gamma)^{-1*} = \gamma^{-1*}.$$

Together with $a_{\delta\gamma} = a_\delta a_\gamma$, this gives canonical isomorphisms

$$\begin{aligned}\delta^*\mathcal{E}_{0,\delta\gamma} &= \delta^* a_{\delta\gamma}\big((\delta\gamma)^{-1}\big)^*\mathcal{E}_{0,1} \\ &\cong a_\delta a_\gamma(\gamma^{-1})^*\mathcal{E}_{0,1} = a_\delta(\mathcal{E}_{0,\gamma}).\end{aligned}$$

Let

$$\eta_1 : \mathcal{E}_{0,1}|_{\widehat{D}_1^\times} \xrightarrow{\sim} \mathcal{F}_1'|_{\widehat{D}_1^\times}$$

be the punctured-disc isomorphism in the representative local modification. The target descent datum defines

$$\chi_{\gamma,1} := \big(a_\gamma(\psi_{\gamma^{-1},1})\big)|_{\widehat{D}_\gamma} : \big(a_\gamma((\gamma^{-1})^*\mathcal{F}_1')\big)|_{\widehat{D}_\gamma} \xrightarrow{\sim} \mathcal{F}_1'|_{\widehat{D}_\gamma},$$

and we put

$$\eta_\gamma := \chi_{\gamma,1} \circ \big(a_\gamma((\gamma^{-1})^*\eta_1)\big)|_{\widehat{D}_\gamma^\times} : \mathcal{E}_{0,\gamma}|_{\widehat{D}_\gamma^\times} \xrightarrow{\sim} \mathcal{F}_1'|_{\widehat{D}_\gamma^\times}.$$

For every $\delta, \gamma \in \Gamma$, the square

$$\begin{array}{ccc} \delta^*\mathcal{E}_{0,\delta\gamma}|_{\widehat{D}_\gamma^\times} & \xrightarrow{\delta^*\eta_{\delta\gamma}} & \delta^*\mathcal{F}_1'|_{\widehat{D}_\gamma^\times} \\ {\scriptstyle\vartheta_{\delta,\gamma}}\downarrow \cong & & {\scriptstyle\psi_{\delta,1}}\downarrow \cong \\ a_\delta\mathcal{E}_{0,\gamma}|_{\widehat{D}_\gamma^\times} & \xrightarrow{a_\delta\eta_\gamma} & a_\delta\mathcal{F}_1'|_{\widehat{D}_\gamma^\times}. \end{array}$$

commutes. Indeed, substituting the definition of η_γ reduces the assertion to the cocycle for $\psi_{\delta,1}$ and the equality $a_{\delta\gamma} = a_\delta a_\gamma$. The maps $\vartheta_{\delta,\gamma}$ therefore satisfy the cocycle and are compatible with the punctured-disc gluing data.

Let $U = (X' \times T) \setminus \coprod_\gamma D_\gamma$. Applying Proposition A.13 to $\mathcal{F}_1'|_U$, the formal bundles $\mathcal{E}_{0,\gamma}$, and the punctured-disc isomorphisms produces a G_0-bundle $\mathcal{F}_0'$ and a modification $\mathcal{F}_0' \dashrightarrow \mathcal{F}_1'$. On U use $\psi_{\delta,1}$, and on the completion along D_γ use the transported local descent map. Full faithfulness in Beauville–Laszlo gluing produces a unique global isomorphism

$$\psi_{\delta,0} : \delta^*\mathcal{F}_0' \xrightarrow{\sim} a_\delta(\mathcal{F}_0').$$

The cocycle holds on U and on every formal completion, hence globally by full faithfulness. By finite étale descent, G_0-bundles on $X' \times T$ equipped with semilinear isomorphisms $\delta^*\mathcal{E} \simeq a_\delta(\mathcal{E})$ satisfying this cocycle are equivalent to G-bundles on $X \times T$, where $G = (\mathrm{Res}_{X'/X} G_0)^\Gamma$. Hence $\mathcal{F}_0'$ descends to a source G-bundle $\mathcal{F}_0$, and the equivariant modification descends to $\mathcal{F}_0 \dashrightarrow \mathcal{F}_1$. This construction is functorial in T and defines a quasi-inverse to Φ_1.

Formal restriction of the reconstructed modification recovers the original local object. Conversely, gluing the formal restrictions of a global modification recovers its source bundle by uniqueness. The morphism part of Beauville–Laszlo gluing and finite étale descent give the same statement on automorphism groupoids. Thus $\Phi_1(T)$ is an equivalence for every T, and the 2-Yoneda lemma proves that the target square is 2-Cartesian. For the source square put $\lambda^\vee = -w_0\lambda$. Inversion identifies the local correspondence of type $\leq \lambda$, with source trivialized, with the target-trivialized local correspondence of type $\leq \lambda^\vee$, and it interchanges $\overleftarrow{h}^\ell$ and $\overrightarrow{h}^\ell$. Starting from the prescribed global source bundle, apply the preceding construction to the inverse local modification of type

$\lambda^\vee$ to reconstruct the global target bundle, and then invert the resulting modification. Pinned automorphisms commute with inversion and carry $\lambda^\vee$ to $(\gamma\lambda)^\vee$, so the same semilinear descent and cocycle calculation applies. This gives a quasi-inverse on objects and automorphisms and proves that the source square is also 2-Cartesian.

By Lemma 3.7, the local horizontal arrows are representable and proper of finite presentation. The global horizontal arrows are their Cartesian base changes, so they have the same properties. This completes the geometric part of Proposition 3.8. □

A.7. **Beck–Chevalley and relative dualizing objects.** For the target square write the proper base-change transformation as

$$\overleftarrow{h}_!f^*_{\mathrm{Hk}}K \longrightarrow f^*\overleftarrow{h}^{\ell}_{\lambda,!}K. \tag{A.14}$$

It is the inverse of the usual transformation $f^*\overleftarrow{h}^{\ell}_{\lambda,!} \to \overleftarrow{h}_!f^*_{\mathrm{Hk}}$; we use this orientation to match (3.20). Define the source transformation similarly.

Proposition A.14. *The transformations in* (3.20) *are equivalences. There are canonical exceptional exchanges*

$$f^*_{\mathrm{Hk}}(\overleftarrow{h}^{\ell}_{\lambda})^!\mathbf{1}_{BH} \xrightarrow{\sim} \overleftarrow{h}^!\mathbf{1}_Y, \qquad f^*_{\mathrm{Hk}}(\overrightarrow{h}^{\ell}_{\lambda})^!\mathbf{1}_{BH} \xrightarrow{\sim} \overrightarrow{h}^!\mathbf{1}_Y.$$

They are compatible with composition, tensor products, the projection formula, and passage to right-adjoint mates.

Proof. The proper base-change maps are equivalences by Proposition A.1(ii) and the properness proved in Sections A.5 and A.6.

For the exceptional exchanges, restrict first to a qcqs open $Y_\beta \subset Y$. By Section A.4, there is a qcqs elementary pro-smooth scheme $P_\beta \simeq \varprojlim_n P_{\beta,n}$ with pro-smooth surjections $P_\beta \to Y_\beta$ and $P_\beta \to BH$, where the $P_{\beta,n}$ are smooth finite-type k-schemes and the transitions are smooth, affine, and surjective. Proposition A.5 supplies the canonical right-mate transformation for the target square. Pull it back along the conservative morphism $P_\beta \to Y_\beta$.

The chosen full formal coordinate and target trivialization identify the pulled-back square with a product because the bounded local modification is then represented solely by a point of the projective Schubert variety; all dependence on the global bundle has been removed by the trivialization. Concretely, the pulled-back square is

$$\begin{array}{ccc} P_\beta \times Z_\lambda & \xrightarrow{\mathrm{pr}_2} & Z_\lambda \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle q_\lambda} \\ P_\beta & \longrightarrow & *, \end{array}$$

where $q_\lambda : Z_\lambda \to *$ is the projective Schubert variety. Under this identification, the pullback of the exceptional exchange is the horizontal right mate for the product square. It is an equivalence by Proposition A.8. Pullback along $P_\beta \to Y_\beta$ is conservative by Lemma A.4; hence the target exceptional exchange is an equivalence on Y_β.

For the source square the same argument uses the source-trivialized fiber $Z_{\lambda^\vee}$, with $\lambda^\vee = -w_0\lambda$. The equivalences on the Y_β are compatible under restriction because they are the canonical mates of the same Cartesian 2-cells. They glue by Proposition A.2.

Compatibility with composition and mates is the functoriality of passing to adjoints in the six-functor 2-category [HM, Proposition 3.2.2]. Compatibility with multiplication by cohomology classes follows from the projection formula (A.3). This proves the sheaf-theoretic part of Proposition 3.8. □

Let

$$\delta_\lambda : (\overleftarrow{h}^{\ell}_{\lambda})^!\mathbf{1}_{BH} \xrightarrow{\sim} (\overrightarrow{h}^{\ell}_{\lambda})^!\mathbf{1}_{BH}$$

be the canonical local relative-dualizing identification used in [WW, (2.8)], and let

$$\beta_{\mathrm{tar}} : f_{\mathrm{Hk}}^{*}(\overleftarrow{h}_{\lambda}^{\ell})^{!}\mathbf{1}_{BH} \xrightarrow{\sim} \overleftarrow{h}^{!}\mathbf{1}_{Y}, \qquad \beta_{\mathrm{src}} : f_{\mathrm{Hk}}^{*}(\overrightarrow{h}_{\lambda}^{\ell})^{!}\mathbf{1}_{BH} \xrightarrow{\sim} \overrightarrow{h}^{!}\mathbf{1}_{Y}$$

be the exceptional exchanges of Proposition A.14. The global relative-dualizing identification is the explicit composite

$$\overleftarrow{h}^{!}\mathbf{1}_{Y} \xrightarrow{\beta_{\mathrm{tar}}^{-1}} f_{\mathrm{Hk}}^{*}(\overleftarrow{h}_{\lambda}^{\ell})^{!}\mathbf{1}_{BH} \xrightarrow{f_{\mathrm{Hk}}^{*}\delta_{\lambda}} f_{\mathrm{Hk}}^{*}(\overrightarrow{h}_{\lambda}^{\ell})^{!}\mathbf{1}_{BH} \xrightarrow{\beta_{\mathrm{src}}} \overrightarrow{h}^{!}\mathbf{1}_{Y}. \tag{A.15}$$

We denote the local and global objects by $\omega_{\mathcal{H}^{\ell}/BH}$ and $\omega_{S_{\lambda}/Y}$ respectively.

A.8. **Relative Borel–Moore cohomology and module structures.** Relative Borel–Moore cohomology is the natural receptacle for the first half of a volume datum. The map c becomes a morphism from the coefficient object to a relative dualizing object, ordinary cohomology acts on that morphism, and the map d evaluates it by the counit of the opposite Hecke projection. The bigrading below records cohomological shift and Tate twist separately so that the normalized diagonal used in the body is obtained only when the relevant class is genuinely Tate-normalized.

Definition A.15 (Bigraded relative Borel–Moore cohomology)**.** Let $p : \mathcal{C} \to \mathcal{Z}$ be representable of finite presentation and let $K \in \mathrm{Shv}(\mathcal{C})$. Put

$$\omega_{\mathcal{C}/\mathcal{Z}} = p^{!}\mathbf{1}_{\mathcal{Z}}.$$

For $r \in \mathbf{Z}$ and $s \in \frac{1}{2}\mathbf{Z}$, define

$$H_{\mathrm{BM}}^{r,s}(\mathcal{C}/\mathcal{Z}, K) := \mathrm{Hom}_{\mathrm{Shv}(\mathcal{C})}^{0}\big(K, \omega_{\mathcal{C}/\mathcal{Z}}[r](s)\big).$$

We write $H_{\mathrm{BM}}^{r} = H_{\mathrm{BM}}^{r,0}$ for the ordinary grading and

$$H_{\mathrm{BM,Tate}}^{m} := H_{\mathrm{BM}}^{m,m/2} = \mathrm{Hom}^{0}\big(K, \omega_{\mathcal{C}/\mathcal{Z}}\langle m\rangle\big)$$

for the normalized diagonal. The notation H_{-*}^{BM} in the body suppresses the Tate index and uses the homological reindexing of [WW, (2.8)].

The same convention gives

$$H^{a,u}(\mathcal{C}) := \mathrm{Hom}^{0}\big(\mathbf{1}_{\mathcal{C}}, \mathbf{1}_{\mathcal{C}}[a](u)\big).$$

There is a bigraded module action

$$H^{a,u}(\mathcal{C}) \otimes H_{\mathrm{BM}}^{b,v}(\mathcal{C}/\mathcal{Z}, K) \longrightarrow H_{\mathrm{BM}}^{a+b,u+v}(\mathcal{C}/\mathcal{Z}, K).$$

Indeed, if $\theta : \mathbf{1} \to \mathbf{1}[a](u)$ and $m : K \to \omega_{\mathcal{C}/\mathcal{Z}}[b](v)$, define

$$\theta \cdot m : K \cong K \otimes \mathbf{1} \xrightarrow{\mathrm{id}\otimes\theta} K[a](u) \xrightarrow{m[a](u)} \omega_{\mathcal{C}/\mathcal{Z}}[a+b](u+v),$$

with the usual Koszul convention. Ordinary cohomology acts by taking $u = 0$, while the normalized multipliers of (3.1) lie on the diagonal $u = a/2$.

Let

$$\mathcal{Z} \xleftarrow{p} \mathcal{C} \xrightarrow{q} \mathcal{Z}$$

be a proper correspondence equipped with an identification $p^{!}\mathbf{1} \simeq q^{!}\mathbf{1}$. For a volume datum $v = (K, c, d)$, the correspondence morphisms induce maps on the preceding bigraded groups; after suppressing the Tate index these are the maps

$$H^{*}(c) : H^{*}(\mathcal{Z}) \longrightarrow H_{-*}^{\mathrm{BM}}(\mathcal{C}/\mathcal{Z}, K), \qquad H^{*}(d) : H_{-*}^{\mathrm{BM}}(\mathcal{C}/\mathcal{Z}, K) \longrightarrow H^{*+d_v}(\mathcal{Z})$$

used in the body. Under the adjunction

$$\mathrm{Hom}(p_{!}K\langle -d_c\rangle, \mathbf{1}) \simeq \mathrm{Hom}(K\langle -d_c\rangle, p^{!}\mathbf{1}),$$

the morphism c corresponds to the normalized class $u_c = H^*(c)(1) \in H_{\mathrm{BM}}^{d_c, d_c/2}$. The projection formula gives source-linearity

$$H^*(c)(\alpha) = p^*\alpha \cdot u_c, \tag{A.16}$$

and $H^*(d)$ is target-linear:

$$H^*(d)(q^*\alpha \cdot m) = \alpha \cdot H^*(d)(m). \tag{A.17}$$

Both formulas are homogeneous for the full bidegree.

Proof. The first formula is the adjoint form of the composite

$$p_!K\langle -d_c\rangle \otimes \alpha \xrightarrow{\text{projection formula}} p_!(K\langle -d_c\rangle \otimes p^*\alpha) \xrightarrow{c\otimes \mathrm{id}_\alpha} \alpha.$$

For the second formula, represent $m \in H_{\mathrm{BM}}^{b,v}(\mathcal{C}/\mathcal{Z}, K)$ by $m : K \to q^!\mathbf{1}[b](v)$. By definition, $H^*(d)(m)$ is

$$\mathbf{1} \xrightarrow{d} q_!K\langle d_d\rangle \xrightarrow{q_!m} q_!q^!\mathbf{1}[b+d_d](v+d_d/2) \xrightarrow{\epsilon_q} \mathbf{1}[b+d_d](v+d_d/2).$$

For $\alpha \in H^{a,u}(\mathcal{Z})$, the class $q^*\alpha \cdot m$ is obtained by tensoring m with $q^*\alpha$. The projection formula identifies the middle part of the resulting composite with α tensored with $q_!m$, and naturality of the counit identifies the last arrow with $\alpha \otimes \epsilon_q$. The result is $\alpha \cdot H^*(d)(m)$, with the Koszul sign dictated by the displayed order. These are the module identities used in [WW, (3.7)–(3.8)]. □

A.9. **Proof of relative Borel–Moore compatibility.** We finish by proving Proposition 3.10. Write $Y = \mathrm{Bun}_G \times X'$, $K^{\mathrm{rep}} = f_{\mathrm{Hk}}^*K$, and let

$$u^\ell \in \mathrm{Hom}(K\langle -d_c\rangle, (\overleftarrow{h}{}^\ell_\lambda)^!\mathbf{1}_{BH})$$

be the adjoint of c^ℓ.

Proof of Proposition 3.10. The global morphism c^{rep} is the composite

$$\overleftarrow{h}{}_!K^{\mathrm{rep}}\langle -d_c\rangle \xrightarrow{\sim} f^*\overleftarrow{h}{}^\ell_{\lambda,!}K\langle -d_c\rangle \xrightarrow{f^*c^\ell} \mathbf{1}_Y.$$

Passing to the adjoint and using the compatibility of Beck–Chevalley with mates gives

$$K^{\mathrm{rep}}\langle -d_c\rangle \xrightarrow{f_{\mathrm{Hk}}^*u^\ell} f_{\mathrm{Hk}}^*(\overleftarrow{h}{}^\ell_\lambda)^!\mathbf{1}_{BH} \xrightarrow{\sim} \overleftarrow{h}{}^!\mathbf{1}_Y.$$

Thus

$$H^*(c^{\mathrm{rep}})(1) = f_{\mathrm{Hk}}^*H^*(c^\ell)(1).$$

Applying the same argument to an arbitrary class on BH, together with (A.16), proves commutativity of the left square of (3.22).

Write $q = \overrightarrow{h}{}^\ell_\lambda$ and $q' = \overrightarrow{h}$. Let

$$m : K \longrightarrow q^!\mathbf{1}_{BH}[r](s)$$

be a relative Borel–Moore class, and let

$$m^{\mathrm{rep}} : K^{\mathrm{rep}} = f_{\mathrm{Hk}}^*K \longrightarrow q'^!\mathbf{1}_Y[r](s)$$

be its pullback through the exceptional exchange of Proposition A.14. By definition, $H^*(d^\ell)(m)$ is the composite

$$\mathbf{1}_{BH} \xrightarrow{d^\ell} q_!K\langle d_d\rangle \xrightarrow{q_!m} q_!q^!\mathbf{1}_{BH}[r+d_d](s+d_d/2) \xrightarrow{\epsilon_q} \mathbf{1}_{BH}[r+d_d](s+d_d/2). \tag{A.18}$$

Pulling (A.18) back along f and inserting the source Beck–Chevalley equivalence gives

$$\mathbf{1}_Y \xrightarrow{d^{\mathrm{rep}}} q'_!K^{\mathrm{rep}}\langle d_d\rangle \xrightarrow{q'_!m^{\mathrm{rep}}} q'_!q'^!\mathbf{1}_Y[r+d_d](s+d_d/2) \xrightarrow{\epsilon_{q'}} \mathbf{1}_Y[r+d_d](s+d_d/2). \tag{A.19}$$

The square comparing the last arrows is the naturality square for the counit under the right-mate correspondence: it follows directly by expanding the mate as the unit–Beck–Chevalley–counit composite in Proposition A.5. Thus (A.19) equals $f^*H^*(d^\ell)(m)$, proving commutativity of the right square of (3.22).

Let $\theta \in H^*_{\mathrm{Tate}}(\mathcal{H}^\ell_{\leq\lambda})$ be homogeneous. Pullback is symmetric monoidal, so

$$f^*_{\mathrm{Hk}}(\theta \cdot u^\ell) = f^*_{\mathrm{Hk}}\theta \cdot f^*_{\mathrm{Hk}}u^\ell.$$

Apply the right square of (3.22) and target-linearity (A.17). This gives

$$H^*(d^{\mathrm{rep}})\big(f^*_{\mathrm{Hk}}\theta \cdot H^*(c^{\mathrm{rep}})(1)\big) = f^*H^*(d^\ell)\big(\theta \cdot H^*(c^\ell)(1)\big).$$

By definition, the last term is $f^*\Gamma^\ell_{\theta\cdot v_\lambda}(1)$, proving (3.23). The class $\theta \cdot v_\lambda$ has degree $d_v + |\theta|$, so every term has that degree. This completes the proof. □

Open-and-closed restriction to a source component commutes with every construction above because it is ordinary Cartesian base change. The formulas on $\mathrm{Bun}_G \times X'$ therefore restrict to the componentwise correspondences used in the main theorem.

References

[Alp] J. Alper, *Adequate moduli spaces and geometrically reductive group schemes*, Algebr. Geom. 1 (2014), no. 4, 489–531.

[BL] A. Beauville and Y. Laszlo, *Un lemme de descente*, C. R. Acad. Sci. Paris Sér. I Math. 320 (1995), no. 3, 335–340.

[Con] B. Conrad, *Reductive group schemes*, in *Autour des schémas en groupes. Vol. I*, Panoramas et Synthèses 42/43, Société Mathématique de France, Paris, 2014, 93–444.

[Fed] R. Fedorov, *Affine Grassmannians of group schemes and exotic principal bundles over* $\mathbf{A}^1$, Amer. J. Math. 138 (2016), no. 4, 879–906; arXiv:1308.3078.

[FYZ] T. Feng, Z. Yun, and W. Zhang, *Arithmetic volumes of moduli stacks of Shtukas*, arXiv:2601.18557v1, 2026.

[HM] C. Heyer and L. Mann, *6-Functor Formalisms and Smooth Representations*, arXiv:2410.13038v1, 2024.

[HPV] B. Hennion, M. Porta, and G. Vezzosi, *Formal glueing along non-linear flags*, arXiv:1607.04503v2, 2016.

[LurGC] J. Lurie, $(\infty, 2)$*-Categories and the Goodwillie Calculus I*, arXiv:0905.0462, 2009.

[LZ] Y. Liu and W. Zheng, *Enhanced six operations and base change theorem for higher Artin stacks*, arXiv:1211.5948v4, 2024.

[LO] Y. Laszlo and M. Olsson, *The six operations for sheaves on Artin stacks II: Adic coefficients*, Publ. Math. Inst. Hautes Études Sci. 107 (2008), 169–210.

[Sta] The Stacks Project Authors, *The Stacks Project*, Tag 0BNR.

[LW] S. Liu and Z. Wang, *Higher period integrals and derivatives of L-functions*, arXiv:2504.00275v3, 2026.

[WW] Z. Wang and W. Wei, *Arithmetic volume of Shtukas and Langlands duality*, arXiv:2604.03763v1, 2026.

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